

**An Extended Finite Element Method (XFEM) approach to
hydraulic fractures:
Modelling of oriented perforations**

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Abstract

In the current context of energy markets global dynamics, production of Shale Reservoirs has been a change in the energy paradigm, with the unconventional reservoirs now seen as a potential "game changer".

The Hydraulic Fracturing (HF) technique is used to maximize their economic potential. Due to the high cost of hydraulic fracturing operations, is essential to build reliable tools to predict the formations behavior, and for this purpose, computer modeling of hydraulically induced fractures is an important method to study fracture parameters, such as length, width or fracture efficiency (fluid loss), amongst others.

In general, software used in the industry for fracture modelling allows very few independent input parameters. In contrast, recent advances in available numerical methods – in particular the extended finite element method (XFEM) – have increased the fracture modelling capabilities. The XFEM (extended Finite Element Method), is a new method for discontinuities/fractures modelling, based on the concept of local nodal enrichment functions and phantom nodes, which reduces the convergence problems and increases the results accuracy, and is used in the study presented herein.

To validate the numerical tools, numerical simulations of a series of laboratory tests that reproduce hydraulic fracturing by oriented perforations on rectangular blocks of gypsum cement by Abass H. et al. (1994) were carried out. The numerical results provide a good match to the experimental observations.

Following that, a parametric study was carried out on the effect of a series of parameters on the outcome of hydraulic fracturing operations, in terms of fracture initiation, propagation and reorientation, when using 180°-phased oriented perforations.

It was found that various variables influence the fracture behavior; of those considered in this study, flow rate, stress anisotropy, rock permeability and phasing were found to introduce major changes in fracture initiation, propagation, reorientation and width.

Keywords: Hydraulic Fracturing, XFEM, oriented perforations, hydro-geomechanical model, fracture propagation

Resumo

No contexto das atuais dinâmicas globais do mercado da energia, a exploração e produção de reservatórios de Argilitos laminados introduziu uma mudança no paradigma energético, sendo os reservatórios não convencionais hoje vistos como um potencial “game changer”.

A fracturação hidráulica (FH) é utilizada para maximizar o seu potencial económico. No entanto, dados os elevados custos destas operações, torna-se essencial a construção de ferramentas para prever o comportamento das formações, e nesse sentido, a modelação numérica de fracturas induzidas hidraulicamente é um método importante para estudar diversos parâmetros, como o comprimento, abertura ou eficiência da fractura.

Em geral, os softwares utilizados na indústria para a modelação de fracturas permitem a introdução de poucos parâmetros de entrada independentes. Por oposição, avanços recentes em métodos numéricos disponíveis - em particular o Método dos Elementos Finitos Alargado/Extendido (XFEM) - aumenta a capacidade de modelação de fracturas. O XFEM, é um novo método para a modelação de descontinuidades/singularidades, baseado nos conceitos de enriquecimento dos nós e nós-fantasma, que permitem reduzir problemas de convergência e aumentar a precisão dos resultados, sendo usado no presente estudo.

Por forma a validar a ferramenta de cálculo, procedeu-se à simulação de uma série de ensaios experimentais descritos por Abass et al. (1994), efetuados em laboratório sob provetes de gesso consolidados em câmara de triaxial verdadeiro. Os resultados numéricos apresentam boa concordância com os dados experimentais.

Em seguida, procedeu-se à execução de um estudo paramétrico sobre o efeito de um conjunto de parâmetros nos resultados de uma operação de fracturação hidráulica, em termos da iniciação, propagação e reorientação da fractura, considerando perfurações orientadas com fase 180° .

Verificou-se que diferentes variáveis influenciam o comportamento da fractura; das consideradas, o caudal de injeção, anisotropia de tensões, permeabilidade e fase das perfurações verificou-se introduzirem alterações significativas na iniciação, propagação, reorientação e abertura da fractura.

Palavras-chave: Fracturação hidráulica, XFEM, perfurações orientadas, modelo hidro-geomecânico, propagação da fractura.

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III - Index of Symbols

$\overline{N_i(x)}$ – New set of shape functions

σ'_{00} – Effective tangential stress

$\sigma'_{effective}$ – Effective stresses

Π_0 – Initial elastic potential energy

Ω_e – Element volume

A_{com} – Communication area between the fracture and the perforation tunnel

A_m – Amplitude dimensionless function dependent on θ

A_{perf} – Perforation cross – sectional area

B_i – Global matrix of displacement – strain

C_f – Friction coefficient

C_{ij}^{lm} – Constitutive tensor

$C_{leak-off}$ – Leak – off coefficient

$D^{(e)}$ – Material stress – strain matrix – constitutive matrix

F_c – Critical load

G_I – Energy release rate for loading mode I

G_{IC} – Critical energy release rate

G_{II} – Energy release rate for loading mode II

G_{III} – Energy release rate for loading mode III

G_f – Fracture energy

$J^{(e)}$ – Element jacobian matrix

K_D – Set of enriched nodes associated with discontinuities

K_I – Fracture toughness for loading mode I

K_{IC} – Critical fracture toughness

K_{II} – Fracture toughness for loading mode II

K_{III} – Fracture toughness for loading mode III

K_T – Set of enriched nodes associated with the crack tip

K_e – Element stiffness matrix

N_i – Shape function

P_{BK} – Breakdown pressure

P_f – Fracture pressure

P_p – Pore pressure
 Q_0 – Fluid injected in the fracture
 W_s – Work required to form the crack
 a_{ij} – Additional degrees of freedom
 f_{ij} – Dimensionless function dependent on θ
 $p_j(x)$ – Enrichment function
 r_w – Wellbore radius
 $u^{Enrichment}$ – Finite element displacement due to enriched features
 u^{FEM} – Traditional finite element displacement
 u_f – Total damage fracture displacement
 u_x – Displacement in direction xx
 u_y – Displacement in direction yy
 w_f – Fracture width
 γ_P – Plastic energy of the material
 γ_S – Surface material energy
 ε_{ij} – Strain tensor
 η_i – Constant shape function
 ρ_w – Water density
 σ_h – Minimum horizontal stress
 σ_H – Maximum horizontal stress
 σ_T – Tensile strength
 σ_c – Compressive strength
 σ_f – Fracture stress
 σ_{ij} – Stress tensor
 σ_{rr} – Radial stress around the wellbore
 $\sigma_{y0} = \sigma_T$
 $\sigma_{\theta\theta}$ – Tangential stress around the wellbore
 ∂_{ij} – Delta dirac
 $2B$ – Fracture toughness determination sample width
 $2L$ – Fracture toughness determination crack length
 Γ – Outward normal

$\Delta\sigma'_{00}$ – Tangential stress variation between direction 0° and 90°

Π – Elastic potential energy

Ψ – Total system energy

B – Plate thickness

B – Pre – logarithmic energy factor tensor

D – Half – length of the sample in a principal direction

E – Elastic modulus

F – External load

G – Energy release rate

G – Shear modulus

$H(x)$ – Heaviside function

J – J integral

$K(x)$ – Near – tip asymptotic enrichment function

K – Bulk modulus

K – Permeability

M – Stiffness matrix

T – Traction vector

$W(\varepsilon)$ – Strain energy density

$a = r$

l – Fracture length

$q(x, t)$ – Flow

r – Distance from crack tip

r – Distance from wellbore center

t – Time

u – Displacement vector

w – Width

θ – Angle of perforation from the PFP

θ – Fracture propagation direction

λ – Lamé constant

μ – Fluid viscosity

ν – Poisson ratio

$\psi(x)$ – Arbitrary function

$\psi(x)$ – Tangential function

$\phi(x)$ – Normal function

ϕ – Porosity

IV - Index of Abbreviations

CAE – Computer aided engineering

CSM – Cohesive segment method

CZM – Cohesive zone method

EFEM – Embedded finite element method

EPFM – Elasto-plastic fracture mechanics

FEM – Finite element method

GF – Geometry factor

GFEM – Generalized finite element method

HC – Hydro-carbon

HF – Hydraulic fracturing

ISIP – Instantaneous shut-in pressure

LEFM – Linear elastic fracture mechanics

MAXPS – Maximum principal stress

PFP – Preferred fracture plane

PORPRES – Pore pressure in XFEM

PUFEM – Partition unit finite element method

SIF- Stress intensity factor

StatusXFEM – Measure of fracture local damage in XFEM

VCCT – Virtual crack closing technique

XFEM – eXtended finite element method

1. Introduction

1.1 Context

In the current context of energy markets global dynamics, exploitation of Shale Gas and Shale Oil Reservoirs has been a change in the energy paradigm, with the unconventional reservoirs now seen as a potential "game changer", with an extremely fast growth/expansion in E&P operations in the Oil and Gas industry.

Due to the extremely low permeability and porosity characteristics of this type of reservoirs, the Hydraulic Fracturing operations are essential to render these fields/reservoirs economically viable.

Hydraulic fracturing consists of a high pressure fluid injection into the reservoir, so that the tensile strength of the rock mass is exceeded and a fracture is formed (breakdown pressure) which constitute a preferential flow path for the hydrocarbons. In the absence of any discontinuity fracturing occurs along the direction of the maximum principal stress direction

This technique was and is still applied with an extremely empiricist base; however due to the high costs of such operations (including drilling costs, fluids injection and proppants), it becomes essential to build reliable tools to predict the formations behavior. For this purpose, computer modeling of hydraulically induced fractures is an important tool to control fracture parameters, such as its length, width or fracture efficiency (fluid loss), among others.

Understand the fracture initiation and propagation mechanisms becomes essential to ensure the efficiency of a hydraulic fracturing operation. In the last three decades, computational numerical modeling using finite difference and finite element methods had a key role in improving our understanding of the complex non-linear effects when coupling fluid, rock and fracture material response during hydraulic fracturing operations.

The finite element method allowed the fracture numerical simulation, a field extremely studied by industries such as aerospace (modeling micro-fractures), civil (modeling fractures in concrete) or the oil industry. Progresses in 2D and 3D modeling of fractures were made due to the necessity to predict the behavior of various materials.

For hydraulically induced fractures, various methods and techniques were used to investigate fracture initiation and propagation in homogeneous semi-infinite elastic mediums, for which there are analytical solutions.

Despite the advances, the traditional finite element method (FEM) has limitations in the modelling of singularities and discontinuities (such as fractures) e.g., it requires the reconfiguration of the finite element mesh at all time-steps during fracture propagation. The remeshing is necessary to ensure that the mesh conforms with the fracture geometry, which makes the method heavy computationally, introduces convergence problems and accuracy loss.

The XFEM (eXtended Finite Element Method), is a new method for discontinuities (strong and weak) modelling, based on the concept of partition of unity, by using local nodal enrichment shape functions nodal together with the introduction of additional degrees of freedom. This allows to overcome the limitations of the traditional FEM, through a complete independence of the fracture and its geometry in relation to the adopted mesh, without re-meshing needs. This gives improvements in solutions convergence and decreases the computational modeling heaviness. Since the introduction of XFEM, studies based on different formulations and applications have been widely developed by the scientific community to investigate the hydro-geomechanical behavior of the induced fractures.

Since the accurate determination of the in-situ stresses in rock masses is extremely complex, pre-design of the operations aims to improve the results through the control of other parameters or procedures.

Oriented perforations is a technology that consists in perforating the rock from the wellbore with pre-defined distances/lengths, widths and directions, to ensure that at least one of the perforations is a few angles of the preferred fracture plane (PFP) in an attempt to reduce the breakdown pressure. Since the excavation of the wellbore introduces a redistribution of stresses near the wellbore, it is essential to study the interaction of perforations with the stress state, in terms of breakdown pressure, fracture geometry and reorientation.

This study presents an overview of a computational approach to model hydraulically induced fractures for oriented perforation with XFEM. The software chosen to perform the study is the Dassault Systèmes™ commercial software, Abaqus ©, which since its version 6.9 introduced the XFEM as a functionality.

1.2 Objectives

The general objectives of this study are:

- Conceptualize the hydro-geomechanical behavior of induced fractures
- Transmit the basic principles of XFEM and its specific features and advantages
- Implement in Abaqus© a hydro-geomechanical model to simulate the induced fracture behavior based on the principles of XFEM

- Explore the capabilities of a numerical simulation tool that mimics fracture initiation and propagation
- Evaluate how the in-situ conditions may influence the fracture initiation and propagation
- Investigate how oriented perforations and parameters may affect the fracture initiation and propagation

1.3 Structure

In order to comply with the pre-established aims, the study is divided into four distinct parts:

- Theoretical framework (Chapters 1 to 4)
- Modeling – Validation (Chapter 5)
- Modeling – Parametric study (Chapter 6)
- Conclusions (Chapter 7)

Theoretical framework

Chapter 1 provides the context for the work and sets its objectives. In addition the structure of the thesis is presented.

Chapter 2 describes briefly hydraulic fracturing operations and its relevant phases, with main emphasis given to the design stage of hydraulic fracturing operation and oriented perforations. In addition it provides a literature review on modelling hydraulic induced fractures.

Chapter 3 frames the problem from a theoretical point of view. It presents the basic concepts of rock, fluid and fracture mechanics, which are essential to understand, from a conceptual point of view, the phenomena involved and aid in the interpretation of the numerical results.

Chapter 4 describes the basis of the main numerical methods used for the study of fractures. This chapter includes a description of the foundations of the finite element method and the underlying concepts of extended finite element method XFEM, highlighting their main capabilities and disadvantages.

Modeling - Validation

Chapter 5 aims to evaluate the ability of XFEM functionality to model fracture behavior. In order to ensure the quality of the results, an initial study is performed for the fracture toughness determination in an infinite plate under tension. This study provides information on both stationary and propagating fractures.

The study is then focused on modelling oriented perforations, and a set of analysis is carried out to simulate the laboratory experiments done by Abass H. et al. (1994) that mimic hydraulic

fracturing using oriented perforations on samples of gypsum cement consolidated in a true-triaxial test apparatus. Some of the input parameters were not given in the Abass H. et al (1994) experiments, reason why several simplifications and assumption were done and these are extensively explained through the chapter. The focus of the analysis is the control/prediction of fracture initiation/breakdown pressure, fracture propagation, reorientation, pressure and fracture opening mechanisms.

Modelling – Parametric study

Chapter 6 describes a parametric study on the role of oriented perforations on the initiation and propagation of induced fractures, investigating the effect of a set of parameters, including, permeability, porosity, flow injection rate, fluid leak-off, fluid viscosity, fracture surface friction, stress anisotropy and perforation phasing on breakdown pressure, fracture geometry, reorientation and width.

Conclusions

Chapter 7 presents a synthesis of the results presented in the two previous chapters and the conclusions that can be drawn from them. In addition, it is proposed future research work to clarify aspects raised by this study and improve our understanding regarding the formation of induced fractures by oriented perforations.

2. Hydraulic Fracturing

2.1 Introduction

In addition to horizontal drilling, hydraulic fracturing is proven as a key technology to increase the economic feasibility of unconventional reservoirs (e.g. shale). Hydraulic fracturing (HF) is a formation stimulation practice used to create additional permeability of a producing formation, for hydrocarbons to flow more easily toward the wellbore (Veatch R. W. J. et al, 1989).

Hydraulic fracturing consists in applying a pressure that induces stresses higher than the formation tensile strength (breakdown pressure) that causes the formation of a fracture. Then, a specified fluid volume is pumped and propagated through the opened cracks, creating high flow channels for HC extraction.

This technique presents a high success rate and financial payback, being commonly used in unconventional reservoirs. After undergoing the first application of HF, wells that show a decline in production, and are no longer economically viable, may be refractured, in order to continue its operation.

Fracturing, or refracturing, is still a challenge for engineers. Significant research work has been conducted in the last decade using new planning software, geomechanical analysis in finite element and finite differences or artificial intelligence techniques, aiming to map existing data and build predictive systems that can maximize the results of a particular operation.

When considering a hydraulic fracturing treatment, four stages must be well defined and projected: well selection, treatment design, operation planning and execution. Each of these stages has equal importance to the operations outcome, and appropriate attention should be given to each in order to carry out an efficient job. For the purpose of this study, treatment design and planning is detailed, due to its importance for the global understanding of the numerical analysis presented in Chapter 5 of this thesis.

Over the years, the scientific community has devoted much attention to the development of the technique. Due to the evolution of mathematical models, fluids, materials and equipment, it has become common practice in the industry today and stands out as one of most effective methods for formation stimulation and has increased the volume of exploitable oil and gas reserves.

2.2 Technique History

Hydraulic fracturing operation has been performed since the early days the petroleum industry. The first experimental test was done in 1947, on a gas well operated by the company Stanolind Oil in the Hugoton field in Grant County, Kansas, USA (Holditch S. A., 2007).

In 1949, the company HOWCO (Halliburton Oil Well cementing Company), the exclusive patent holder, performed a total of 332 wells stimulation, with an average production increase of 75%.

It is estimated that nearly 2.5 million fracturing operations have already been performed around the world, and approximately 60% of wells drilled today are fractured ((Montgomery C. T. et Smith M. B., 2010) and (Valkó P. et Economides M. J., 1995)).

The application of fracturing goes beyond increasing well's productivity, it also provides increased reserves, making possible the exploration of new fields - only in the United States the growth in oil reserves may have been at least 30% and in the natural gas, 90% (Holditch S. A., 2007).

Hydraulic fracturing is a common technique not just for enhancing hydrocarbon production but also geothermal energy extraction (Sasaki S., 1998). It is widely used for other purposes like hazardous solid waste disposal (Hainey B.W. et al, 1999), measurement of in-situ stresses (Raaen A. M. et al, 2001), fault reactivation in mining and remediation of soil and ground water aquifers (Murdoch L. C. et Slack W., 2002).

2.3 Hydraulic fracturing operation

The main objective of a hydraulic fracturing operation is to create a path for hydrocarbons migration from the reservoir to the wellbore.

The hydraulic fracturing procedure comprises two steps (Weijers, 1995); First, following casing perforation, a viscous fluid called “pad” is pumped into the formation through completed areas. When the downhole pressure exceeds the breakdown pressure, the fracture is initiated and propagates through the reservoir. The fluid pumped at pressures up to 50 MPa is able to form hundred meters long fractures in a cohesive rock in each direction around the well, with widths between 3,175 mm to 6,35 mm ((Frantz H. J. et Jochen V., October 2005) and (Economides M. J. et al., 1993)). In a second phase, slurry made of a fluid mixed with sand, typically named as proppant, is injected. This slurry has the function of extending the fracture that was initially created, transport the proppant deep into the fracture and delay or prevent the fracture from closing due to the overburden pressure. With these two phases, a highly conductive, narrow and long propped path is created, increasing the local permeability and the flow of HC to the wellbore (Veatch R. W. J. et al, 1989).

Hydraulic fracturing is a very quick operation. The process for the execution of a single horizontal well typically includes 4 to 8 weeks for preparing the site for the drilling, 4 or 5 additional weeks for drilling, including casing and cementing and 2 to 5 days for HF operation. Figure 2.1 shows

the evolution of the injection rate and pressure of the injected fluid and the proppant concentration during HF operation (Daneshy A., 2010).

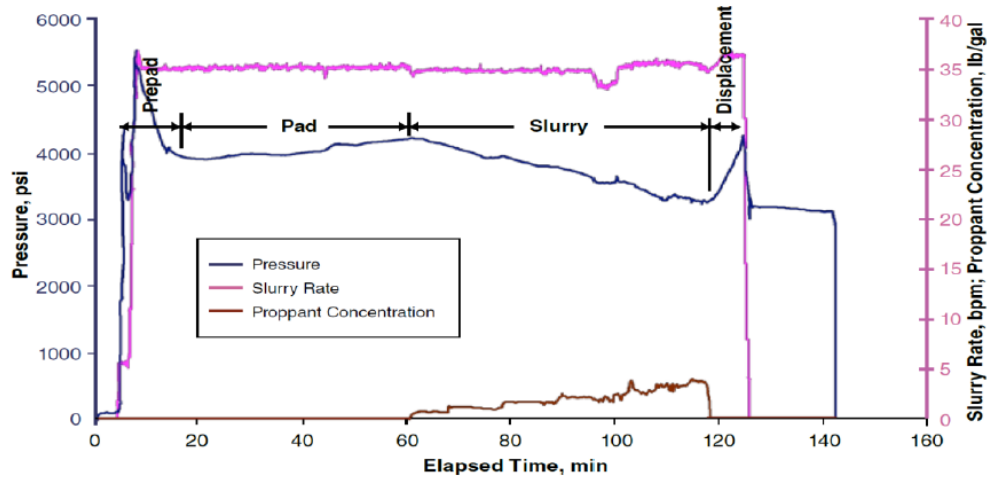


Figure 2.1 - A typical two-phases fracturing chart with discretization of time steps (Daneshy A., 2010)

2.4 Perforations

Execution of perforations is the last task to be performed before a HF operation. The perforations serve as a singularity to help fracture initiation and control the propagation direction.

Perforations play an important role in the complex fracture geometries around wellbore. Initiation of a single wide fracture from a wellbore is one of the main objectives of using the perforation as a means to avoid multiple T-shaped and reoriented fractures, increasing the possibility of predicting the fracture behaviour in an operation (Sepehri J., 2014).

The parameters that can be varied in the design of perforation in a HF operation are: 1) perforation phasing; 2) perforation density (shots per meter); 3) perforation length; 4) perforation diameter and 5) stimulation type.

Regarding perforation phasing, several arrangements are possible, some of which are shown in Figure 2.2. Different designs are adequate for specific situations, to maximize the well productivity.

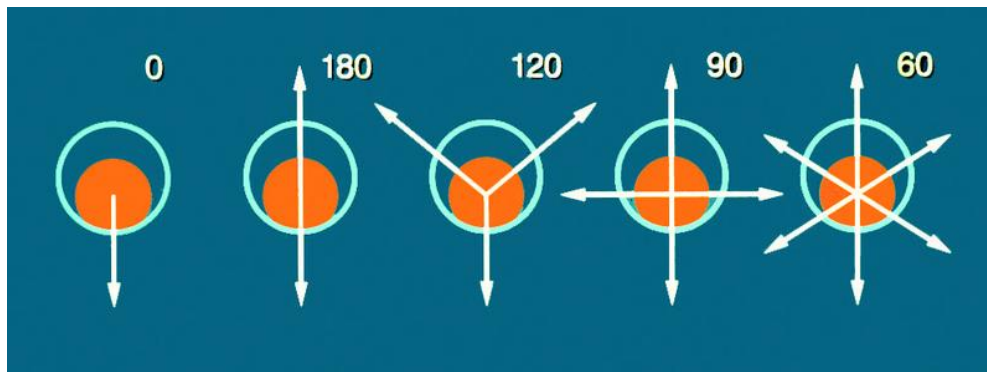


Figure 2.2 - Perforations phasing designs (Petrowiki, 2015)

In a hydraulic fracturing stimulation operation, the perforation should ensure the initiation of a single wide fracture from the wellbore, with minimum tortuosity, ensuring fracture propagation with minimal injection pressure and having control over fracture propagation direction (Behrmann L. A. et Nolte K. G., 1999).

During completion and production phase is important to minimize the near-wellbore effects associated with the stress redistribution caused by wellbore opening e.g. perforation friction, micro-annulus pinch points from gun phasing misalignment, multiple competing fractures and fracture tortuosity caused by a curved fracture path (Romero J. et al, 1995).

Phasing of 60°, 90°, 120° and 180° are usually the most efficient options for hydraulic fracture treatment because in these directions, the perforation angle and the preferable fracture plane are not very dissimilar, and with such angles the use of several perforation wings reduces the probability of screen out (Aud W. et al, 1994).

2.5 Numerical studies on hydraulic fracturing – state of the art

Analysis and modelling of fracture behaviour is a subjected studied since the beginning of the XIX century. Initially, the subject was studied empirically.

Inglis (1913) quantified the effects of stress concentration through the analysis of elliptical holes in plates. Inglis (1913) obtained an expression of the stresses at the tip of the major axis of the ellipse and found that the effect of the stress concentration becomes higher with the reduction of the curvature radius of the ellipse. However, when the distance approaches zero, the stress tends to infinity.

Griffith (1920) suggested that internal faults act as "stress intensifiers" affecting the strength of solids. In addition, Griffith (1920) formulated the thermodynamic (based on the system total energy changes during the fracturing process) criterion for fracture initiation.

Westergaard (1939) formulated the expression for the stress field near the crack tip. This is the transition moment where fracture mechanics passed from a purely empirical science to an analytical problem.

After the Second World War (1939-1945), driven by the fracture problems encountered in airplanes, Irwin (1948) using the ideas of Griffith (1920) proposed the fundamentals of fracture mechanics. Irwin (1948) extended the Griffith theory for metals, and altered the general solution of Westergaard (1939), introducing the concept of Stress Intensity Factor (SIF). Irwin (1948) also found and proved the concept of energy release rate G and studied the relations between K (SIF) and G , the basis of Linear Elastic Fracture Mechanics (LEFM). An overview of the basis of the linear elastic fracture mechanics is presented in section 3.3.

Rice (1968) introduced the J-integral concept, a contour line integral that is found to be independent of the route/path taken, and corresponds to the rate of change in the potential energy, for linear elastic or nonlinear elastic solids during crack extension.

In the last three decades, with the development of numerical methods for the analysis of cracked rocks/solids, many studies have been published in the literature on the subject. Numerical methods are used in Fracture Mechanics to calculate the stress intensity factors and simulate crack initiation and propagation in materials. Initially such studies used the finite differences method and later the finite element method. Several modifications need to be introduced in traditional FEM in order to allow the modelling of fractures within a continuous media and their interaction. As important modifications, it is possible to name the GFEM (generalized finite element method), EFEM (embedded finite element method) and XFEM (extended finite element method). A discussion on the various numerical methods available to model discontinuities in an otherwise continuous media, including their potentialities and shortcomings, is presented in Chapter 4.

Modelling discontinuities has always been a challenge in the field of computational mechanics. When modelling cracks with the standard finite element method (FEM), the FEM mesh is required to conform to the geometry of the crack, and this creates the necessity of a re-meshing for each time-step, increasing the computational memory consumption, convergence problems and loss of accuracy.

Belytschko (1999) developed the extended finite element method. It is able to incorporate the local enrichment into the finite elements space. The resulting enriched space is then capable of capturing the non-smooth solutions with optimal convergence rate, without the necessity of

remeshing in each time step. This becomes possible due to the notion of partition of unity as identified by Melenk (1996) and Duarte (1996).

Several studies have been published in the literature on the analysis and modelling of hydraulic induced fractures, using the finite element and the finite differences methods. These introduce new variables and interactions, as:

Atukorala (1983) developed a finite element model to simulate vertical or horizontal hydraulic fracturing in oil sands. The solutions for fluid flow and mechanic response was obtained separately. These two equations were solved iteratively imposing a compatibility condition on the fluid volume in the fracture. The linear elastic fracture mechanics (LEFM) was used to analyze the mechanical response of the model.

Settari (1989) investigated the effects of soils deformation in the fracture initiation with a partially coupled hydromechanic model. The effect of leak-off on the fracture dimensions was emphasized. A Mohr Coulomb shear failure criterion was considered. Later, Settari extended this work to incorporate the effects of temperature (heat flux) in the formulation.

Advani (1990) developed a finite element method based software to simulate three-dimensional hydraulic fractures in multilayer reservoirs, with emphasis to the propagation of planar tensile hydraulic fractures in layered elastic reservoirs.

Frydman (1997) simulated the process of pressurization in the well, the same mechanism as that employed in hydraulic fracturing treatments, by means of coupled hydromechanic numerical analyses. The model considered the effect of a cohesive zone in fracture analysis, with the fracture propagation direction previously predefined.

Itaoka (2002) studied the behaviour of crack growth under high tectonic stress. A finite element model for hydraulic fracturing, accounting for the fracture mixed mode and the possible effect on the crack growth was presented.

Yang (2002) defined numerically a numerical model to account for the effect of heterogeneity and rock permeability on the initiation and propagation of hydraulic fractures.

Lu (2004) developed a three-dimensional hydraulic fracturing model admitting the occurrence of radial flow, which increased the quality of the fracture height predictions.

Garcia (2005) developed a hydromechanic fully coupled model to study the effect of pore pressure and stress distribution in hydraulically fractured reservoirs. The fully implicit finite difference model takes into account 3D nonlinear poroelastic deformation of the reservoir rock. A local grid refinement around the wells was considered. The equations governing fluid flow are

coupled with the equations governing the deformation of rock fracture and reservoir, and solved numerically under different reservoir-fracture conditions.

Pak (2008) developed a hydraulic fracture finite element model under isothermal and non-isothermal conditions. With this model various boundary conditions can be simulated, allowing the specification of pore pressure, temperature or traction loads. The model was used to simulate a laboratory experiment on the propagation of hydraulic fracturing in oil sands.

In 2009, Abaqus 6.9 software introduced the possibility of carrying out analysis using the XFEM. With the high diffusion of the *software*, a great amount of studies were and are carried out on hydraulically induced fractures.

Arlanoglu (2011) using the XFEM approach simulated numerically the smearing mechanism of drilling solids into the wellbore wall and the effect on the stress distribution around the wellbore. Fracture propagation was not considered in this model, but the evolution of pressures and fracture width is analyzed for stationary fractures, assuming a damage softening law (cohesive behaviour).

Keswani (2012) compared the theoretical solutions for a single edge notch specimen and panel with the numerical simulation obtained using the XFEM approach. This study was based on the analysis of the stress intensity factor and how it controls the crack growth/propagation. Several other similar studies comparing fracture analytical solutions with numerical predictions are found in the literature (McNary, 2009 and Oliveira, 2013)

Shin (2013) modelled with the XFEM the simultaneous propagation of multiple fractures in a 3D geomechanical model to understand the effect of competing fractures on their propagation characteristics. The effect of parameters as fluid viscosity and flow rate was found to be important.

With Abaqus 6.14 version it becomes possible to carry out the analysis of fracture propagation (with the XFEM) with full hydromechanic coupling. (Zielonka, 2014) developed 2D and 3D fully-coupled hydromechanic models in XFEM for the prediction of both fracture initiation and propagation of penny-shaped and KGD (Kristonovich-Geertsma-de Klerk) fractures. The aim of the study was to validate the coupled hydromechanic XFEM formulation, compare the numerical predictions with available analytical solutions and analyse the XFEM mesh dependency.

3. Material Mechanics

3.1 Introduction

It is an important aspect of rock mechanics, and solid mechanics in general, to determine the relationship between stresses and strains, which are often referred to as constitutive equations. Various constitutive models for rock masses have been proposed and described in the literature, the simplest one being the linear elastic model, which assumes a reversible and linear correspondence between stress and strains. This is usually the adopted constitutive model for the rock mass in the simulation analyses of hydraulic fracturing, at least to describe the behavior of the rock mass up to failure. Other, more complex, constitutive models have been developed, to take into account different aspects of rock behavior; for example plasticity based constitutive models, are found to be particularly useful to predict the stress concentration around a wellbore or the behavior of soft materials during reservoir depletion (Economides M. J. et al, 2000).

Fracture Mechanics is the area of mechanics that studies the behavior of cracks, and it is an important tool to improve the knowledge about the mechanical performance materials, as rocks.

The Linear Elastic Fracture Mechanics (LEFM) assumes that the material is isotropic and linear elastic. On that basis, the stress field near the crack tip is calculated using the theory of elasticity. When the stresses near the crack tip exceeds the resistance limit of the material, the crack grows. In Linear Elastic Fracture Mechanics, most formulas are defined to plane stress and strain states, associated with one of the three modes of relative movements of the crack surfaces (Economides M. J. et al, 2000).

3.2 Rock Mechanics

3.2.1 Constitutive laws

Linear Elastic model

The linear elastic model, also called the Hooke's law, is characterized by the occurrence of instantaneous elastic deformation due to the application of load. The behavior of an isotropic linear elastic material is fully described by the following constitutive equation:

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij} \quad (3.1)$$

where $\lambda = \frac{\nu E}{(1+\nu)(1-2\nu)}$ and $\mu = G = \frac{E}{2(1+\nu)}$ are Lamé constants, E is the Young's modulus, ν is the Poisson's ratio, G is the shear modulus and δ is the Kronecker delta. Alternatively, Equation 3.1 can be written in function of any other two elastic parameters from the following list: Young's modulus, Shear modulus, Poisson ratio, Bulk modulus or Lamé constants. The

relationship amongst these various parameters can be found in the literature (Gross D. et Seelig T., 2011).

The elastic model is frequently employed to describe rock behavior up to failure.

Poroelasticity and the influence of pore pressure

Terzaghi principle of effective stresses states that when a rock is subjected to a stress, it is opposed by the fluid pressure of pores in the rock. It accounts for the discrete nature of soils and considers the effect of the pore fluid pressure on the soil response, and is mathematically described by the following equation:

$$\sigma' = \sigma - P_p \quad (3.2)$$

Where σ' are the effective stresses, σ are the total stresses and P_p is the pore pressure.

It states that the soil response is controlled by the effective stresses (the stresses acting on the soil skeleton) and changes to the effective stresses (and thus changes to the soil state) can be achieved through changes to the applied stresses at the soil element boundaries or to the pore fluid pressure. The above equation is found to be suitable for saturated soils.

The principle of effective stress is found to be also applicable to rocks. If there is fluid within the rock pores the analysis should be done in terms of effective stresses.

The bases for applying the principle of effective stresses to rock formations are the following observations:

- An increase of pore pressure induces rock dilation.
- Compression of the rock produces a pore pressure increase if the fluid is prevented from escaping through the pores network.

The mechanical response of rock to pore pressure diffusion is a time dependent variable; i.e. the response is dependent on the loading rate and the capacity of the fluid to escape through the pores, with the response to be drained or undrained as consequence of the stated above.

If a load is applied instantaneously, the response will be undrained, because there is no time for pore pressure diffusion through rock mass. This effect is more important if the fluid is a relatively incompressible liquid rather than a relatively compressible gas.

Based on this relationship between pore pressure diffusion (and thus pore water pressure) and rock mass deformation, a new variable was introduced by Biot (1956), to take into account the modifications to the overall response of the rock due to the pore pressure – pore volume relation.

The poroelastic behaviour assumes that the effective stresses can be calculated as:

$$\sigma' = \sigma - \alpha P_p \quad (3.3)$$

Where α is the Biot constant. The poroelastic constant varies between 0 and 1 as it describes the efficiency of the fluid pressure in counteracting the total applied stress, and typically, for petroleum reservoirs, it is about 0.7, though its value changes over the life of the reservoir.

The general solution for Biot coefficient is:

$$\alpha = \frac{3(\nu_u - \nu)}{B(1 - 2\nu)(1 + \nu_u)} \quad (3.4)$$

Where B is the skempton pore pressure coefficient defined as:

$$B = \frac{\Delta P}{\Delta \sigma} \quad (3.5)$$

where ΔP represents the variation in pore pressure resulting from a change in the applied confining stress $\Delta \sigma$ under undrained conditions.

In the ideal case, where no porosity change occurs under an equal variation of pore and confining pressure, Equation 3.4 can be simplified to:

$$\alpha = 1 - \frac{K}{K_s} \quad (3.6)$$

Where K is the bulk modulus of the material and K_s is the bulk modulus of the solid constituents.

3.2.2 Failure criteria

A failure criterion is a relationship between the principal effective stresses corresponding to the stress states at which failure occurs. Stress states located outside the zone defined by the failure criterion cannot be reached. In a HF treatment the rock fails in tension, i.e. the tensile failure criterion is reached. The stress increments required to induce tensile rock failure are usually ten times lower than those required to achieve shear (compressive) failure. Nevertheless, shear failure criteria are presented due to its importance for the general overview of the geomechanical rock behavior.

Shear failure criteria

There are several compressional failure criteria defined in the literature. Among the most used criteria are: 1) Mohr-coulomb; 2) Hoek-Brown; 3) Modified Wiebols-Cook and 4) Drucker-Prager.

For the first two criteria, failure conditions are independent of the intermediate stress, because they are most often calibrated from compression triaxial test results, in which the intermediate

stress is equal to the minimum stress. The Hoek-Brown criterion has the advantage of considering the curvature of the failure surface at low stress levels. Experimental evidence shows that they often provide good approximations (Economides M. J. et al, 2000).

The other two shear failure criteria account for the effect of the intermediate principal stress magnitude and thus for full calibration true-triaxial tests are ideally required.

Tensile failure criteria

Simple analytical formulations are used for this purpose. The maximum tensile stress criterion maintains that failure initiates as soon as the minimum principal effective stress component reaches the tensile strength of the material:

$$\sigma'_h = \sigma_T \quad (3.7)$$

Where σ'_h is the minimum principal effective stress and σ_T is the rock tensile strength.

3.2.3 In-situ stresses

Vertical stresses

When estimating the in-situ stresses in rock masses, it is commonly assumed that the principal stresses are the vertical stress and the stresses in two directions within the horizontal plane.

The geostatic vertical total stress at any point can be estimated as the weight of the soils/rocks above that depth. Therefore, if the material unit weight is constant with depth, then:

$$\sigma_v = Z \gamma \quad (3.8)$$

Where σ_v is the vertical total stress, Z is the depth and γ is the soil/rock unit weight.

Soils density usually increases with depth due to the compression caused by the geostatic stresses, so that the specific gravity is not constant. If the weight of soil varies with the depth, the vertical forces can be calculated by an integral:

$$\sigma_v = \int \gamma dz \quad (3.9)$$

If the soil is stratified and the specific weight of each layer is known, vertical forces can be calculated by summing the contribution of each layer.

Horizontal stresses

The horizontal stresses occur as a result of vertical stress, the material behavior and tectonic stresses. Assuming that the rock is homogeneous, isotropic linear elastic, the two horizontal stresses are equal (isotropic), and defined by the following equation:

$$\sigma_H \approx \sigma_h = \frac{\nu}{(1 - \nu)} \sigma_v \quad (3.10)$$

Where σ_H is the maximum horizontal stress, σ_h is the minimum horizontal stress, σ_v is the vertical total stress and ν is the poisson ratio.

Due to material anisotropy, tectonic stresses or geological singularities, most of the time, the assumption of similar horizontal stresses in both direction is not realistic, and several relations are presented in the literature for their estimation (Zoback M., 2007).

The difference between the minimum horizontal stress and the maximum horizontal stress is difficult to determine from direct measurements and thus methods to estimate it from observations of fracture propagation in wellbore walls (through acoustic images records or resistivity measurements) have been developed.

Tectonic stress regimes

In general, the vertical stress σ_v can be considered to be a principal stress. There are two horizontal principal stresses, the maximum horizontal stress σ_H and the minimum horizontal stress σ_h . In 1905, Anderson found that the three tectonic stress regimes - normal, strike-slip and thrust faulting - can be characterized by different stress states patterns. For the three different tectonic stress regimes, the principal stresses are arranged as follows (Altmann J., 2010), and illustrated in Figure 3.1:

Normal faulting regime:
 $\sigma'_v > \sigma'_H > \sigma'_h$

Strike-slip regime:
 $\sigma'_H > \sigma'_v > \sigma'_h$

Thrust faulting regime:
 $\sigma'_H > \sigma'_h > \sigma'_v$

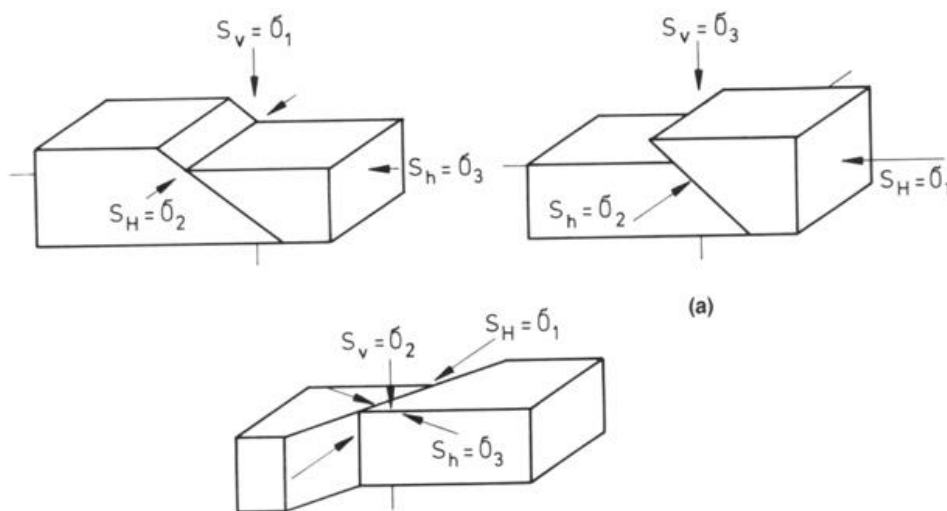


Figure 3.1 - Principal stresses in normal faulting (NN) (left), strike-slip (SS) (middle) and reverse faulting (RF) (right) regimes (Zoback M., 2007)

3.2.4 Stress changes near wellbore due to HF

The stress state near the wellbore is complex due to the stress equilibrium reached after wellbore drilling. Hydraulic fracturing operation, in particular the fluid injection and fracture propagation, causes the complexity of stress measurements in the near wellbore region to increase. Three major effects can be identified:

1. Wellbore excavation and stress re-equilibrium in the near-wellbore region
2. Minimum stress increase due to poroelastic rock behavior and fluid leak-off
3. Stress increase due to fracture opening and propagation

Assuming that a vertical wellbore is drilled in a linearly elastic semi-infinite, homogenous and isotropic medium, the stress state around the wellbore is given by (Jaeger J. C. et Cook N. G. W., 1971):

$$\sigma_{rr} = \frac{1}{2}(\sigma_H - \sigma_h) \left(1 - \frac{r_w^2}{r^2}\right) + \frac{1}{2}(\sigma_H - \sigma_h) \left(1 - \frac{4r_w^2}{r^2} + \frac{3r_w^4}{r^4}\right) \cos 2\theta \quad (3.11)$$

$$\sigma_{\theta\theta} = \frac{1}{2}(\sigma_H + \sigma_h) \left(1 + \frac{r_w^2}{r^2}\right) - \frac{1}{2}(\sigma_H - \sigma_h) \left(1 + \frac{3r_w^4}{r^4}\right) \cos 2\theta \quad (3.12)$$

Where σ_{rr} is the radial stress around the wellbore, $\sigma_{\theta\theta}$ is the tangential stress, r_w is the wellbore radius and σ_H and σ_h have been previously defined. The above expressions are written in polar coordinates, where r is the distance from the wellbore center and θ is the angle with the maximum horizontal stress direction.

Assuming a vertical well subject to two orthogonal horizontal stresses, the wellbore fluid pressure that initiates a fracture along σ_H direction, usually named as Breakdown Pressure, can be calculated as (Valkó P. et Economides M. J., 1995):

$$P_{bk} = 3\sigma_h - \sigma_H - P_p + \sigma_T \quad (3.13)$$

Where σ_T is the tensile strength of the rock. This expression was obtained from Equation 3.12 admitting that $r_w = r$ and $\theta = 0$ and invoking the tensile failure criteria expressed by Equation 3.7.

The equation is valid only in the case of no fluid leak-off assuming that a perfect non-penetrating mudcake has been formed on the wellbore wall by the fracturing fluid additives. The presence of other microfractures can retard the initiation of fracture by stress relieving. For this reason the assumption of homogeneous rock mass is done. The same effect may be felt if oriented perforations are drilled/completed.

Assuming fluid leak-off prior to the breakdown pressure, a more complex expression, assuming the poroelastic behaviour may be written:

$$P_{bk} = \frac{3\sigma_h - \sigma_H - 2\eta P_p + \sigma_T}{2(1 - \eta)} \quad (3.14)$$

Where η is called the poroelastic stress coefficient, which describes the in-situ stress change caused by injection and/or production, defined as (Detournay E. and Cheng A.H., 1993):

$$\eta = \alpha \frac{(1 - 2\nu)}{2(1 - \nu)} \quad (3.15)$$

Notice that an increase in the pore pressure corresponds to a decrease in the effective stresses, and hence a decrease in the breakdown pressure. Therefore, the use of high-viscosity fluids and low pumping rates may reduce the breakdown pressure (Valkó P. et Economides M. J., 1995). On the other hand, low viscosity fluids and low pumping rates reduced the capacity for a long sustentation of the overburden pressures, and the induced fracture tends to close rapidly. An equilibrium between both interactions should be achieved.

Regarding the second effect, during the fracturing process, fracturing fluids leak into the formation. This leakage induces a pore pressure increase around the fracture that results in dilation of the formation and, therefore, an increase of the minimum principal stress in this vicinity. For a 2D crack, the increase of minimum principal stress, as a function of time, is expressed (Detournay E. et Cheng A., 1991):

$$\Delta\sigma_h(T_c) = \eta(P_f - P_p)f(T_c) \quad (3.16)$$

Where P_f is the fracturing fluid pressure and T_c is the characteristic time, expressed as follow:

$$T_c = \frac{2tkG(1 - \nu)(\nu_u - \nu)}{\alpha^2\mu(1 - 2\nu)^2(1 - \nu_u)L^2} \quad (3.17)$$

Where t is the total injection time, k is the permeability, G is the shear modulus, ν_u is the undrained Poisson ratio, μ is the viscosity, ν is the Poisson ratio, α is the biot coefficient and L is the fracture half-length.

In typical reservoirs, the value of characteristic time at which poroelastic effects can start to influence the state of stress around the fracture is about 10^{-3} (adimensional). This effect decreases if a high-pressure drop occurs at the fracture face, which happens when a good fluid-loss control agent is used. Poroelastic effects eventually disappear when injection stops and the excess of pore pressure dissipates into the formation (Economides M. J. et al, 2000).

Regarding the third effect, the stress increase is consequence of the fracture being held open by proppant. Once injection has stopped, proppant is responsible for sustaining the fracture, and

due to the high concentration and proppant induced pressures, the stresses around the wellbore increase (Economides M. J. et al, 2000).

3.3 Linear elastic fracture mechanics

3.3.1 Stress Distribution around the fracture tip

According to the linear elastic fracture mechanics, the tangential stresses around a fracture are proportional to $\frac{1}{\sqrt{r}}$, which implies infinite stresses near by the crack tip (Figure 3.2). In this specific location, the cohesive compressive forces of Barenblatt, due to molecular attraction on the material structure, act on a small area around the tip of the fracture. These forces, in an equilibrium condition, counterbalance the infinite stresses at fracture tip, delaying the fracture propagation (Gidley J. et al, 1989), resulting in elasto-plastic behavior at the crack tip.

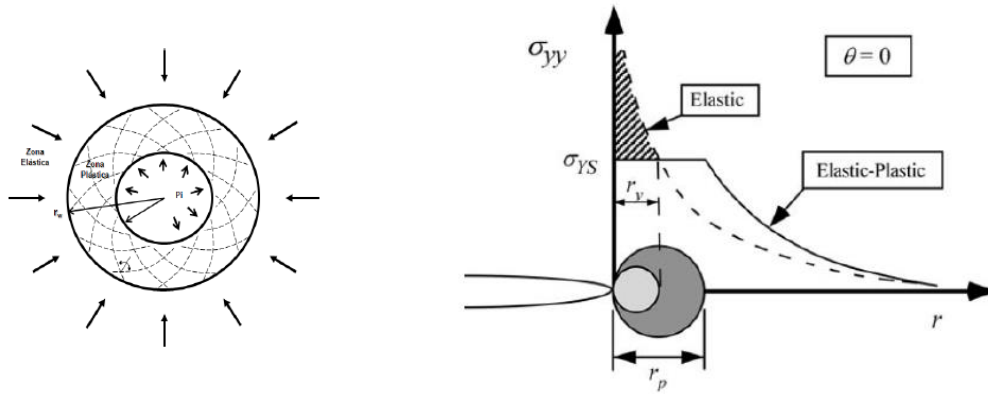


Figure 3.2 - Crack behavior in the near tip region (Abass H. et Neda J., 1988);

Notes: r is the distance to the crack tip; r_y is the cohesive behavior distance in LEFM and r_p is the same when considering the validity of the EPFM.

According to the Barenblatt theory (1962) of Cohesion Strength, the tip of the fracture should be smooth, so that the derivative of the thickness (du_y) to the fracture length (dx) is zero at the fracture tip, whose mathematical representation is:

$$\left(\frac{du_y}{dx} \right)_{x=L} = 0 \quad (3.18)$$

And graphical representation is:

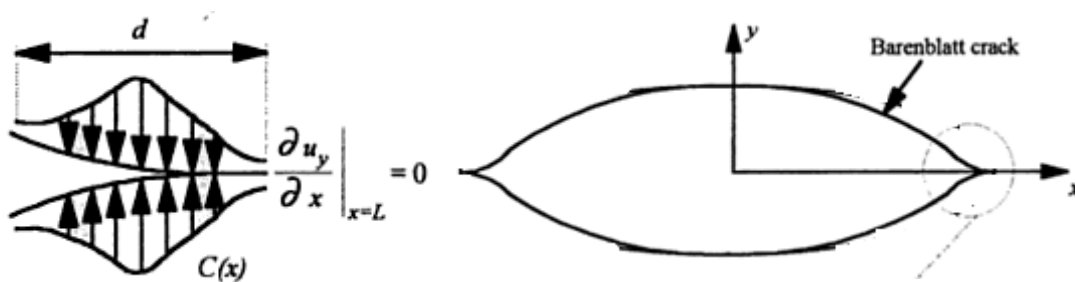


Figure 3.3 - Barenblatt theory for crack tip (Charlez A. Ph., 1997)

Each material has a cohesive force; the fracture width is dependent on the capacity of the acting force to exceed this cohesive force (Barenblatt G. I., 1962). This fact explains why cracked samples do not necessarily break when loaded. The cohesive forces are counteracting the infinite stresses generated at the fracture tip.

The LEFM is only valid when the plastic deformation is small compared to the size of the crack. If the plastic deformation zones are large compared with the size of the crack, the Elastic-plastic fracture mechanics (EPFM) should be used, as seen in figure 3.2. In the remaining of this thesis LEFM is used.

3.3.2 Crack Loading Modes

As the fluid pressure increases and becomes higher than the sum of the minimum principal stress and the tensile strength of the rock, failure occurs and fractures are opened. A fracture can be seen as some form of mechanical discontinuity in the material.

In fracture mechanics, materials can experience three different modes of fracture due to the applied loads, as seen in figure 3.4.

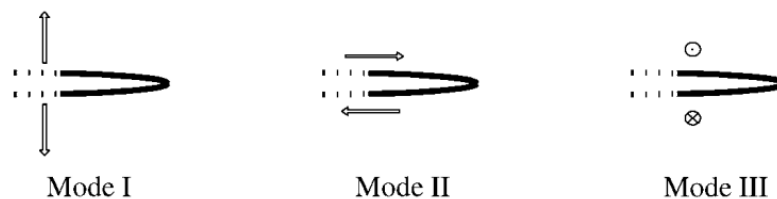


Figure 3.4 - Crack loading modes (Fjaer, 2008)

Mode I fracture occurs due to a principal applied load in the direction normal to the crack plane. Mode II fracture occurs when the loading is an in-plane shear loading and the crack faces are sliding. Mode III fracture occurs when the shear loading mode is a direction outside of the plan, introducing a torsional movement.

In hydraulic fractures analysis, mode I is primarily important, although the other modes can be important while modelling more complicated situations, as for example fracture turning from deviated wells (Economides M. J. et al, 2000). Notice that is possible to have more than one of these loading modes at a time, and if it happens, the linear superposition can be applied.

3.3.3 Stress Intensity factors

Stress intensity factor (SIF) is an important parameter to determine for crack initiation and the crack propagation (Its length, velocity and orientation). The stress intensity factor is the magnitude of the stress singularity at the crack tip, and is dependent on the crack itself (geometry, size and location) and the loads on the material (the magnitude and the mode)

(Taleghani A. D., 2009). SIF are used in fracture mechanics to increase the predictions accuracy of the near-tip stress state induced by fluid pressures and in-situ stresses.

For each crack loading mode, there is an associated stress intensity factor K_i . Knowing the stress intensity factor related to each loading mode, it is possible to estimate the stress field at the crack tip. This requires the definition of a polar coordinates system (r, θ) , with origin defined at the crack tip, as seen in figure 3.5.

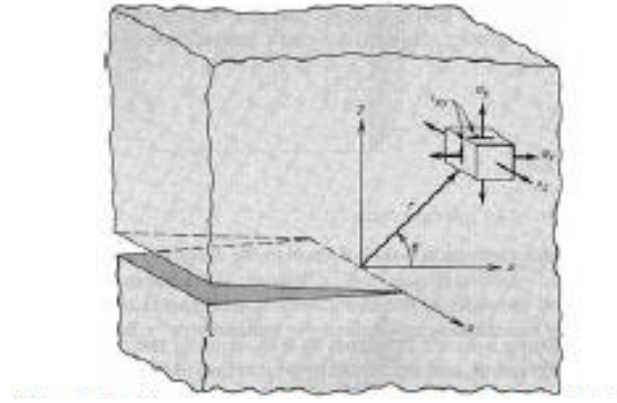


Figure 3.5 - Schematic representation of crack tip stresses defined in polar coordinates

In polar coordinates, for a linear elastic and cracked material, the stress field is:

$$\sigma_{ij} = \left(\frac{k}{\sqrt{r}}\right) f_{ij}(\theta) + \sum_{m=0}^{\infty} A_m \sqrt{r^m} g_{ij}^{(m)}(\theta) \quad (3.19)$$

Where σ_{ij} is the stress tensor, k is a constant related with SIF and f_{ij} is a dimensionless tensor function dependent on θ , where θ is the angle with the x axis in the polar coordinates system. For the higher-order terms, A_m is the amplitude and a dimensionless function dependent on θ . Equation 3.19 shows that independently of the fracture geometry, the stress state is asymptotic, tending to infinite near the crack tip ($r \approx 0$).

The stress field for Mode I, at a radial distance r of the crack tip is defined as:

$$\sigma_{xx} = \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \left(1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2}\right) \quad (3.20)$$

$$\sigma_{yy} = \frac{K_I}{\sqrt{2\pi r}} \cos \left(\frac{\theta}{2}\right) \left(1 + \sin \left(\frac{\theta}{2}\right) \sin \left(\frac{3\theta}{2}\right)\right) \quad (3.21)$$

$$\tau_{xy} = \frac{K_I}{\sqrt{2\pi r}} \cos \left(\frac{\theta}{2}\right) \sin \left(\frac{\theta}{2}\right) \cos \left(\frac{3\theta}{2}\right) \quad (3.22)$$

And the displacement field is expressed (Fleming M. et al, 1997):

$$u_x = \frac{K_I}{2\mu} \sqrt{\frac{r}{2\pi}} \cos\left(\frac{\theta}{2}\right) \left[k - 1 + 2 \left(\sin\left(\frac{\theta}{2}\right)^2 \right) \right] \quad (3.23)$$

$$u_y = \frac{K_I}{2\mu} \sqrt{\frac{r}{2\pi}} \cos\left(\frac{\theta}{2}\right) \left[k + 1 - 2 \left(\cos\left(\frac{\theta}{2}\right)^2 \right) \right] \quad (3.24)$$

Where u_i are the displacements and μ is the shear modulus.

3.3.4 Griffith energy balance equation

As stated before, the stress intensity in the near-tip region tends to infinite, however it is observed that cracked samples do not necessarily break when loaded. Griffith (1920) proposed a new theory based on the first law of thermodynamics, and on the postulate that when a system goes from a non-equilibrium state to equilibrium, there is a net decrease in energy. Griffith (1920) got satisfactory results for materials obeying the Hooke's law (linear elastic materials). Before Griffith, it was thought that the strength of a fracture was independent of its size. Griffith examined fractures of various sizes in a brittle material (glass), and found that as larger is the sample, the lower is the resistance. This fact became known as "size effect"; i.e. the larger the sample the more likely the material is to crack, hence lower resistance ((Anderson T. L., 2005) and (Ahmed A., 2009)).

Griffith's theory admits that the crack is represented by an elliptical shape, as seen in Figure 3.6. The crack propagates when the elastic potential energy supplied overcomes the crack growth resistance (the energy required to form a new surface cracking, i.e. the energy required to break the cohesion between the front atoms the crack).

The total energy of the system, Ψ , consists of two parts (Fischer-Cripps et Anthony C., 2007):

- The elastic potential energy supplied by the material strain and external loads applied to the system Π
- The work required to form the crack surface W_s

The total energy must decrease or remain constant for a crack to form or to enable its propagation. The formation of the crack (and the two new surfaces), when the surface energy of the material γ_s is exceeded, relaxes the stresses and the stored potential elastic strain energy, Π reduces near the crack faces.

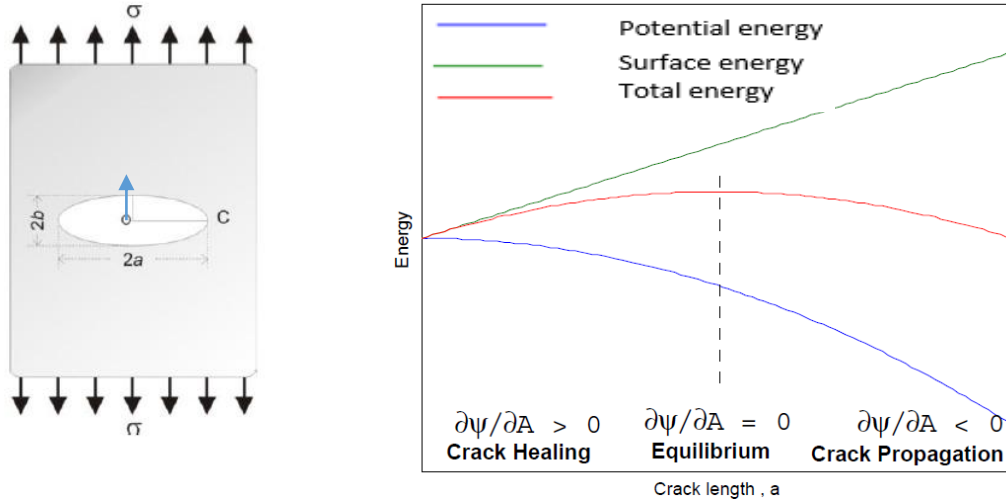


Figure 3.6 - Griffith energy balance for an elliptical shape crack

The Griffith energy balance for an incremental increase in the crack area ∂A , under equilibrium conditions, as seen in figure 3.6, can be expressed as:

$$\frac{\partial \Psi}{\partial A} = \frac{\partial \Pi}{\partial A} + \frac{\partial W_s}{\partial A} = 0 \quad (3.25)$$

For an elliptical crack, the following stress analysis can be done ((Ahmed A., 2009) and (Griffith A. A., 1921)):

$$\Pi = \Pi_0 - \frac{\pi \sigma^2 a^2 B}{E} \quad (3.26)$$

Where Π_0 is the potential energy of an uncracked plate and B is the plate thickness. Since the formation of a crack requires the creation of two surfaces, the work required to form the crack surface, W_s is given by the following expression:

$$W_s = 4aB\gamma_s \quad (3.27)$$

Where γ_s is the surface energy of the material. Thus,

$$\frac{\partial \Pi}{\partial A} = -\frac{\pi \sigma^2 a}{E} \quad (3.28)$$

And,

$$\frac{\partial W_s}{\partial A} = 2\gamma_s \quad (3.29)$$

Solving for the fracture stress,

$$\sigma_f = \sqrt{\frac{2E\gamma_s}{\pi a}} \quad (3.30)$$

Where σ_f is the critical fracture tangential stress to cause the fracture initiation. As mentioned before, the expression above was derived assuming that the fracture and the surrounding media

are isotropic linear elastic materials. This equation can be modified to account for the fracture elastic-plastic characteristic, as follows ((Orwan E., 1948) and (Anderson T. L., 2005)):

$$\sigma_f = \sqrt{\frac{2E(\gamma_s + \gamma_p)}{\pi a}} \quad (3.31)$$

Where γ_p is the plastic energy of the material.

3.3.5 The energy release rate – G

Irwin (1948) proposes an approach to describe the conditions of fracture initiation, which is essentially equivalent to the Griffith model. Irwin (1948) defined the energy release ratio, G , as a measure of the energy available for increasing the extent of the crack:

$$G = - \frac{\partial \Pi}{\partial A} \quad (3.32)$$

G is the rate of potential energy variation with crack area. Since G is obtained from the derivative of a potential, it is called the crack extension force or the crack driving force (Oliveira F., 2013).

For linear elastic material under loading mode I, this scalar entity can be expressed as:

$$G = \frac{K_I^2}{E'} \quad (3.33)$$

Where for a plane stress $E = E'$ and for a plane strain $E' = \frac{E}{1-\nu^2}$.

3.3.6 Failure Criteria

The critical value of the stress intensity factor, K_{IC} , also known as fracture toughness (Economides M. J. et al, 2000), determines the material resistance to the fracture initiation. At a microscopically and macroscopically level, the fracture toughness of a material exists due to the strength of connection bonds between constituent particles and the size and number of natural flaws in the material. Fracture toughness is widely used to describe fracture propagation through rock, and it appears to be a valid predictor of fracture propagation in cohesive soils. In rock, fracture toughness values are typically on the order of $0,5 - 3 \text{ MPa}\sqrt{m}$.

The propagation criterion, under mode I loading condition, assumes that the propagation will continue as far as the criterion for fracture advance is:

$$K_{IC} \leq K_I \quad (3.34)$$

On the other hand, the fracture will cease propagation when:

$$K_{IC} > K_I \quad (3.35)$$

Fracture toughness is not a material property, but depends on the problem geometry, as discussed later in Chapter 5. The failure criteria may also be written in terms of the energy release rate for LEFM, once there is a direct relationship between both.

3.3.7 J-Integral

To study crack growth, Rice (1977) developed an energy related method, based on a contour integral, to calculate the stress intensity factors. Even if there is plasticity, this method can be used in brittle and ductile materials. In LEFM, the value of the J-integral in a contour surrounding the crack tip is exactly the energy release rate, G and thus is directly related to the stress intensity factors of the crack (Li S. et Simkins Jr. D. C., 2002).

The J-Integral can be defined as a line integral around the crack tip, invariant of the adopted path since it starts on the underside and ends in the upper surface of the crack as seen in Figure 3.7. The path is represented by arrows counter-clockwise (Anderson T. L., 2005).

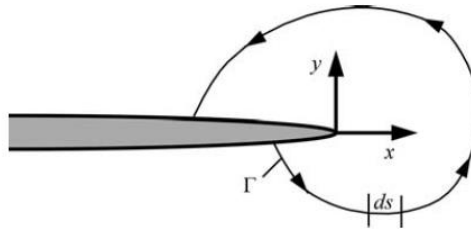


Figure 3.7 - Schematic representation of the 2D line J-Integral (Dassault Systèmes, 2015)

The J-integral thus provided an alternative approach to calculate the G or K (stress intensity factors). The Rice's integral in its original form can be expressed as:

$$J = \int_{\Gamma} \left(W \partial y - T \frac{\partial u}{\partial x} \partial s \right) \quad (3.36)$$

Where Γ is a curve surrounding the crack tip, T is the traction vector defined according to outward normal along Γ , $T_i = \sigma_{ij} n_j$, u is the displacement vector, and ∂s is an element of an arc with total length Γ .

W is the strain energy density, given by:

$$W(\varepsilon) = \int_0^{\varepsilon} \sigma_{ij} \delta \varepsilon_{ij} \quad (3.37)$$

As mentioned above, the J-integral is numerically equal to the energy release rate in LEFM, therefore it is also related to the stress intensity factor through the following expression:

$$J = \frac{1}{8\pi} K^T B^{-1} K \quad (3.38)$$

Where $K = [K_I, K_{II}, K_{III}]^T$ and B is the pre-logarithmic energy factor tensor. For homogeneous and isotropic materials, the expression can be simplified as:

$$J = \frac{1}{E'} (K_I^2 + K_{II}^2) + \frac{1}{2G} K_{III}^3 \quad (3.39)$$

3.4 Fluid Mechanics

3.4.1 Material behavior and constitutive equations

Basic concepts

Fluid mechanics is an important element in the application of fracturing techniques (Economides M. J. et al, 2000). The two basic fluid mechanics variables, injection rates (q_i) and fluid viscosity (μ) affect pressure, control the displacement rates of the proppant and have an important role when controlling the fluid loss to the formation.

Flow is usually defined as the relative sliding of parallel layers. The external forces can be originated from pressure and/or gravity differentials (Poiseuille flow) or from torque (Couette flow). As a reaction, in order to keep the equilibrium of the system, in the opposite direction, the shear stress has the magnitude:

$$\tau = \mu \times \dot{\gamma} \quad (3.40)$$

Where μ is the fluid apparent viscosity and $\dot{\gamma}$ is the shear rate. The shear rate is defined as:

$$\dot{\gamma} = \frac{\Delta u}{\Delta y} \quad (3.41)$$

The shear stress/shear rate relationship expressed in an algebraic form is the rheological constitutive equation and it is graphically represented as a rheological curve. The shear stress/shear rate relationship is a material property and it is independent of the flow's geometry. If the flow rates are extremely high, the application of the concept of parallel sliding layers is impracticable due to the appearance of more complex flow movements (turbulence) (Valkó P. et Economides M. J., 1995).

Rheological models

Fluids can be classified in terms of the shape of their rheological empirical curve, as shown in Figure 3.8.

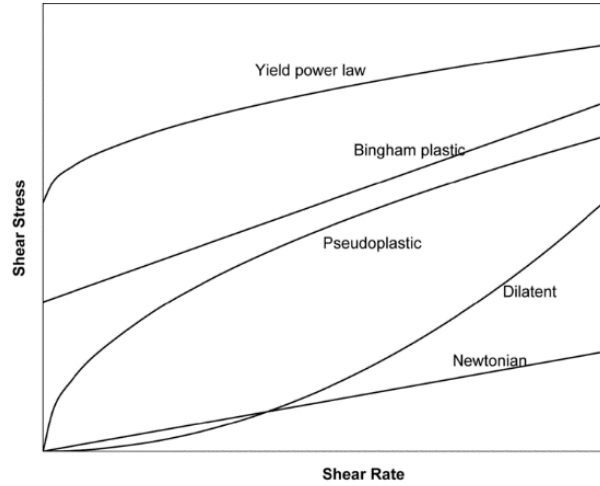


Figure 3.8 - Fluid types based on the rheological curves (Valkó P. et Economides M. J., 1995)

A fluid is Newtonian if the rheological curve is a straight line passing in the origin of the referential. The behavior of a Newtonian fluid can be mathematically described by the following equation:

$$\tau = \mu \dot{\gamma} \quad (3.42)$$

This is the simplest model, because due to the proportionality between stress rate and shear rate a single parameter – the viscosity μ – is required to fully characterize the fluid.

If a positive shear stress is necessary to initiate deformation this is called the yield stress and the fluid behavior is plastic or Bingham plastic. The behavior of a Bingham plastic fluid that exhibits a yield stress, τ_0 , is described by the following equation:

$$\tau - \tau_0 = \mu' \dot{\gamma} \quad (3.43)$$

The Bingham plastic model can be described by two parameters: τ_0 and μ' (the yield stress and the Bingham plastic viscosity), that correspond to the slope and the intercept of a straight line, respectively.

Pseudo-plastic or power law behavior consists on a fluid without a yield stress and a rheological curve with decreasing slope as the shear rate increases, but that plots as a line in double logarithmic space, being expressed by the following equation:

$$\tau = K \dot{\gamma}^n \quad (3.44)$$

Where, K is the consistency coefficient or consistency index and the exponent n is the flow behavior index or power law index, a dimensionless term that reflects the closeness to Newtonian flow. Equation 3.44 can describe the behaviour of both shear-thinning and shear thickening fluids. For the special case of a Newtonian fluid, n is equal to 1 and the consistency

index K corresponds then to the viscosity of the fluid, μ . When n is less than 1 the fluid is shear-thinning and when n is higher than 1 the fluid is shear-thickening in nature.

For power law fluids the apparent viscosity, which is defined as the ratio between applied shear stress and the shear rate, is given by the following equation:

$$\mu_a = K\dot{\gamma}^{n-1} \quad (3.45)$$

Polymeric solutions and melts are examples of power-law fluids. Some drilling fluids and cement slurries, depending on their formulation, may also exhibit shear thinning power-law behavior. When there is a yield stress and this is included in the power law model, the model is then known as the Herschel–Bulkley model, given by the following equation:

$$\tau - \tau_0 = K_H \dot{\gamma}^{n_H} \quad (3.46)$$

Where, n is the flow behavior index, K_H is the consistency index, and τ_0 is yield stress. The yield stress may be understood as a threshold for large scale yielding and thus an engineering reality and plays an important role.

Dilatant behavior is characterized by a monotonic increase in the slope of the rheological curve with shear rate (Montgomery C., 2013). Most fluids do not conform exactly to any of these models, but their behavior can be adequately described by one of them. Several other constitutive equations have been proposed and can be found in the literature to reproduce real fluid behavior ((Valkó P. et Economides M. J., 1995) and (Rao M. A., 2014)).

3.4.2 Fluid Flow – Hydraulic transport in rocks

Darcy law

Assuming Darcy's is valid, the relationship between the flow velocity vector q and the pore pressure P_p , under the assumption that the flow velocity vector depends only on the pressure gradient in one direction, is described by:

$$q = \frac{k}{\mu} i = \frac{k}{\mu} \frac{\Delta P_p}{\Delta l} = \frac{Q}{A} \quad (3.47)$$

Where Q is the flow rate, A is the cross section area of the fracture, k is the intrinsic permeability, μ is the viscosity and i is the pressure gradient along the distance Δl .

The following assumptions have to be met for the Darcy's law to apply:

- Inertial forces can be neglected.
- Steady state flow conditions are valid.
- The medium is isotropic and fully saturated with a single phase fluid.
- The fluid is homogeneous and contains only one phase.

If the medium is anisotropic (most usual situation), assuming a principal coordinate system $\{x', y', z'\}$ for the permeability tensor and neglecting the gravitational forces, Darcy's law can be rewritten in scalar form:

$$q_{x'} = \frac{k_{xx}}{\mu} \frac{\Delta P_p}{\Delta x'}, \quad q_{y'} = \frac{k_{yy}}{\mu} \frac{\Delta P_p}{\Delta y'}, \quad q_{z'} = \frac{k_{zz}}{\mu} \frac{\Delta P_p}{\Delta z'} \quad (3.48)$$

In unfractured sedimentary rocks, the permeability in the bedding planes is typically larger than the permeability in a plan perpendicular to the bedding. In fractured (hard) rocks the permeability is highest in the direction coincident with the direction of the fracture sets.

Flow regimes

The seepage properties of a rock depend on the size, shape and interconnections of the void spaces in the rock. Deformation of the pore space, closure or creation of micro cracks or deposition of particles will influence the seepage properties.

Through porous media can be classified into three different flow zones and regimes, depending on local fluid velocity within the void space, as seen in Figure 3.9 (Basak P., 1977).

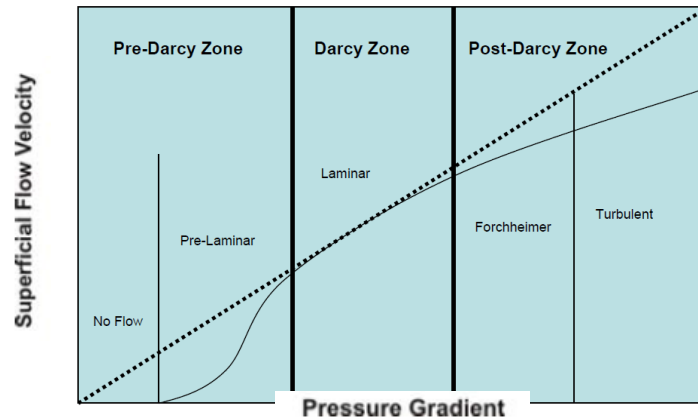


Figure 3.9 - Flow zones in porous media. Flow through porous media can be classified into three different flow zones, depending on local fluid velocity within the pore space (Basak P., 1977)

The flow zones are known as: Pre-Darcy zone where the increase in flow velocity is disproportionally high compared to the increase in pressure gradient; the Darcy zone where the flow is laminar and Darcy's law is valid; and the Post-Darcy zone where flow velocity increase is less than that proportional to the increase in pressure gradient. The transition between these flow zones is smooth. These different flow zones can be defined using the Reynolds number:

$$N_{Re} = \frac{\text{Inertial forces}}{\text{Viscous forces}} = \frac{\rho v D}{\mu} \quad (3.49)$$

Where ρ is the density of the fluid, μ is the viscosity of the fluid, v is the characteristic flow velocity and D the diameter of the pipe. Note that this is the simplest Reynolds number equation for Newtonian fluids flowing through a pipe. More complex expressions for pipes, parallel plates

and ellipses assuming other rheological models can be found in the literature ((Economides M. J. et al, 2000) and (Valkó P. et Economides M. J., 1995)).

The most typical N_{Re} threshold values are set in Table 3.1:

Table 3.1 - Reynolds number values associated with the different flow regimes (Amao A. M., 2007)

Zone	Flow Regime	Reynolds Number
Pre-Darcy	No flow	$N_{Re} < 1000$
	Pre laminar flow	
Darcy	Laminar flow	$N_{Re} < 2300$
Post-Darcy	Forcheimer flow	$2300 < N_{Re} < 4000$
	Turbulent flow	$N_{Re} > 4000$

A reasonable explanation why the flow does not follow Darcy's law for very small pressure gradients can be given by the boundary layer theory (Schlichting H. et Gersten K., 2000). Because of the liquid-solid interactions, the viscous forces at the interfaces/surface are much stronger than in the center of the pores. This effect leads to low to null velocities in the fracture surfaces, high velocities in the gravity center of the fracture and a subsequent creation of vortices according to the theory of laminar instability (Tani I., 1962).

With increasing pressure gradient (e.g. higher flow rate), the small pores start to take part in the flow leading to an apparent increase of permeability. In the Post-Darcy zone (high flow rates), inertial forces begin to control the flow velocity and the flow field is not laminar.

At the reservoir scale, the various flow regimes illustrated in Figure 3.9 can be found in the near field and the far field, around the borehole and hydraulic fracture, respectively. Near and within the wellbore, the fracture flow can be turbulent. In the rock mass surrounding the wellbore or the fracture, flow is likely to be laminar and described by the Darcy's law. At a sufficient distance from the borehole or fracture, the pressure gradient decreases potentially below the Pre-Darcy threshold resulting in a reduction in the apparent permeability (Basak P., 1977).

Forchheimer Equation and Non-Darcy Flow Correction

In 1901, Philippe Forchheimer discovered that, at sufficiently high velocity, the relationship between flow rate and potential gradient is non-linear, and that this non-linearity increases with flow rate. Initially this non-linearity was attributed to increased turbulence in the fluid flow; however it is now known that this non-linearity is due to inertial effects in the porous media. The pressure drop at high velocities, in the Forcheimer flow regime is given by Forchheimer empirical equation:

$$\frac{\partial p}{\partial x} = \frac{\mu}{k} q + \beta \rho v^2 \quad (3.50)$$

Where β is the inertial factor, ρ is the density of the fluid flowing through the medium, $\frac{\partial p}{\partial x}$ is the pressure gradient, μ is the apparent viscosity of the fluid and q is the flow rate.

The third term of the equation is introduced in the Forchheimer equation in situation where the Darcy's law is still valid. The term $(\beta \rho v^2)$ must be added to account for the increased pressure drop. The difference in pressure drop due to inertial losses is primarily associated with the acceleration and deceleration effects of the fluid as it travels through the tortuous flow path of the porous media.

Factor β , was first introduced by Cornel and Katz, and it is widely agreed that β is a property of the porous media; it is a strong function of the tortuosity of the flow path and it is usually determined from laboratory measurements and multi-rate well tests (Amao A. M., 2007). Several different empirical correlations are used in the literature to determine the beta factor.

3.4.3 Fluid flow within a Fracture

Flow within the crack can be modeled by lubrication theory, since the ratio:

$$\frac{w}{l} \ll 1 \quad (3.51)$$

Where w is the fracture width and l is the length, as seen in figure 3.10.

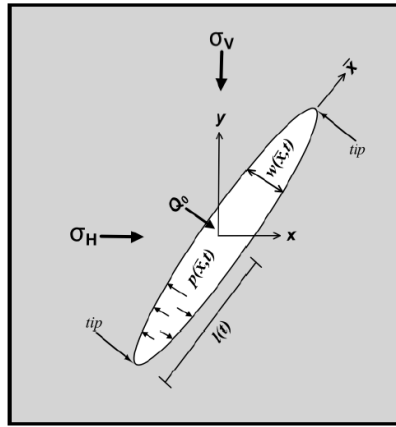


Figure 3.10 - 2D schematic Hydraulic fracture representation (Adachi, J. et al, 2007)

The other required condition is that the velocity inside the crack is sufficiently low to ensure the occurrence of laminar flow. Then, the flow can be modeled by Poiseuille's law:

$$q(x, t) = - \frac{w(x, t)^3}{12\mu} \frac{\partial p(x, t)}{\partial x} \quad (3.52)$$

Where $q(x, t)$ is the flow passing through a section normal to the x axis in a specific time t , μ is the fluid viscosity and $p(x, t)$ is the fluid pressure within the fracture.

Considering the fluid as incompressible, the dimensional continuity equation is given by the following expression:

$$\frac{\delta w(x, t)}{\delta t} + \frac{\delta q(x, t)}{\delta x} = 0 \quad (3.53)$$

Combining equations 3.52 e 3.53, the Reynolds equation can be retrieved. Reynolds equation is expressed as:

$$\frac{\delta w(x, t)}{\delta t} = \frac{\delta}{\delta x} \left(\frac{w(x, t)^3}{12\mu} \frac{\delta p(x, t)}{\delta x} \right) + \delta(x_0) Q_0 \quad (3.54)$$

Where $\delta(x_0)$ is the delta Dirac function of the injection at point x_0 and Q_0 is the fluid injection in the fracture at point x_0 .

Two boundary conditions are essential to predict the fluid flow inside fracture:

$$q(0_+, t) = \frac{Q_0}{2} \quad (3.55)$$

$$q(l, t) = 0 \quad (3.56)$$

Equation 3.55 is a consequence of fracture symmetry in relation to the injection point and Equation 3.56 imposes the impermeability of the rock mass (zero flow along the perimeter of the fracture (assuming a no fluid leak-off situation)).

To ensure the existence and uniqueness of the solution to the Reynolds equation with the specified boundary conditions, it is necessary to impose the condition of mass conservation:

$$\int_0^{l(t)} w(x, t) \delta x = \int_0^t Q(t) \delta t \quad (3.57)$$

4. Numerical methods for fracture analysis

4.1 Introduction

The finite element method (FEM) is a capable method to model fractures; however, some effects require a locally refined mesh. As a consequence, there is an increase in terms of nodal quantity and consequently in the simulation running time.

The fracture propagation analysis is a challenge for traditional FEM, since it requires the mesh discretization to be compatible with the discontinuity which requires re-meshing at each time step, increasing the computational costs, loss of accuracy and convergence.

As a solution, the extended finite element method (XFEM) allows to model discontinuities and singularities independently on the initial mesh. XFEM is based on the principles of partition of unity and enriched local shape functions for strong discontinuities.

This chapter provides a description of the two numerical methods, highlighting their capabilities and limitations.

4.2 Finite Element Method (FEM)

The aim of the finite element method is to calculate the field of stresses and deformations across the problem domain. In the FEM the analysis domain is divided into elements and the unknowns, for a traditional formulation, are the values of the displacements at the nodes, and all the remaining quantities can be expressed as a function of the nodal displacements.

The formulation of the finite element method to problems with a two dimensional geometry is obtained from a solution represented by $u(x) = (u_x, u_y)$, where $u(x)$ is the displacement field, which can be expressed by the following expression:

$$u(x) = \sum_{i=1}^n N_i(x)u_i \quad (4.1)$$

Where $N_i(x)$ are the shape functions, n the number of nodes in the element and u_i are the nodal displacement values, which are obtained by solving the following equation:

$$Mu_i = F \quad (4.2)$$

Where F is the external loads/forces vector and M is the stiffness matrix, obtained through the virtual work theorem.

For a two-dimensional problem, equation 4.1 may be represented as:

$$\begin{bmatrix} u_x \\ u_y \end{bmatrix} = \sum_{i=1}^n \begin{bmatrix} N_i & 0 \\ 0 & N_i \end{bmatrix} \begin{bmatrix} u_x^i \\ u_y^i \end{bmatrix} \quad (4.3)$$

The application of the FEM requires the domain to be discretized, through the division of the continuum into elements of known geometries (typically quadrilateral or trilateral shapes), and all these elements are interconnected through nodes. Then, interpolation shape functions should be selected (Mohammadi S., 2008).

4.2.1 Virtual work theorem

The virtual work theorem postulates that when the body is subjected to a set of forces f (Figure 4.1) the energy dissipated during the body deformation is equal to the work of the external forces, which is expressed by the following expression:

$$\int_{\Omega} (\partial \varepsilon)^T \sigma \partial \Omega = \int_{\Omega} (\partial u)^T f^t \partial \Omega + \int_{\Gamma_t} (\partial u)^T t \partial \Gamma \quad (4.4)$$

Where f^t represents the volume forces, t represents the superficial tensile forces, Ω the domain, Γ_t represents the contour line of the domain Ω , and $\partial \varepsilon$ and ∂u are the strain and displacements vectors, respectively.

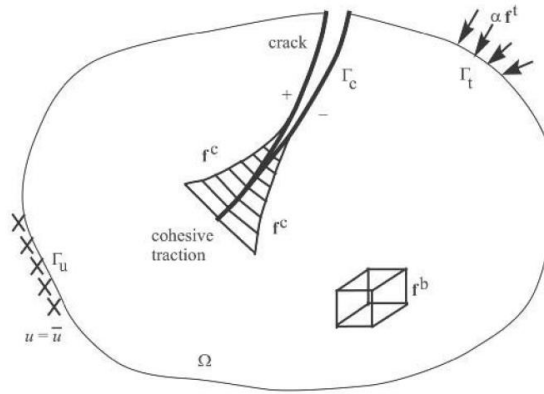


Figure 4.1 - FEM domain for application of virtual work principle (adapted from (Mohammadi S., 2008))

The expressions for the displacement, strains and their virtual components can be expressed in terms of the nodal displacements:

$$\varepsilon = \sum_{i=1}^m B_i u_i \quad (4.5)$$

$$\partial \varepsilon = \sum_{i=1}^m B_i \partial u_i \quad (4.6)$$

$$u = \sum_{i=1}^m N_i u_i \quad (4.7)$$

$$\partial u = \sum_{i=1}^m N_i \partial u_i \quad (4.8)$$

Where N_i represents the global matrix of the shape functions and B_i the global matrix of the derivatives of the shape function, expressed as:

$$B_i = \begin{bmatrix} \frac{\partial N_i}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{bmatrix} \quad (4.9)$$

When Equations 4.5 to 4.8 are substituted in Equation 4.4, then the virtual work for each element can be expressed as:

$$\int_{\Omega} (B_i)^T \sigma \delta \Omega = \int_{\Omega} (N_i)^T f^t \delta \Omega + \int_{\Gamma} (N_i)^T t \delta \Gamma \quad (4.10)$$

The displacements of the continuous medium can be calculated as a function of the nodal displacements. The contribution of the displacement of each finite element can be expressed as:

$$u^{(e)} = \sum_{i=1}^n N_i^{(e)} u_i^{(e)} \quad (4.11)$$

Due to the complexity of the elements in study, the shape functions are mapped in a local coordinate system, as seen in figure 4.2 (Caldeira L. et Cardoso R., 2013).

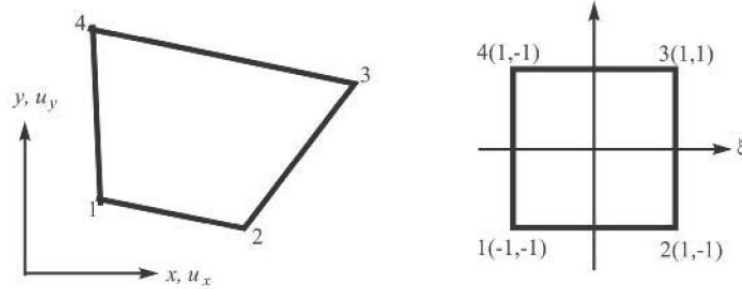


Figure 4.2 - Mapping of a Finite element in global and local coordinates (Mohammadi S., 2008)

The discrete volume of each finite element can be expressed as:

$$\delta \Omega^{(e)} = \text{Det} J^{(e)} \delta \xi \delta \eta \quad (4.12)$$

Where $\delta \xi$ and $\delta \eta$ are the local coordinates and $J^{(e)}$ is the jacobian matrix of each finite element, defined as:

$$J^{(e)} = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n \frac{\partial N_i^{(e)}}{\partial \xi} x_i^{(e)} & \sum_{i=1}^n \frac{\partial N_i^{(e)}}{\partial \xi} y_i^{(e)} \\ \sum_{i=1}^n \frac{\partial N_i^{(e)}}{\partial \eta} x_i^{(e)} & \sum_{i=1}^n \frac{\partial N_i^{(e)}}{\partial \eta} y_i^{(e)} \end{bmatrix} \quad (4.13)$$

Finally, the stiffness matrix K_e of an element with volume Ω_e can be determined in local coordinates, as:

$$K_e = \int_{-1}^{+1} \int_{-1}^{+1} B_i^{(e)T} D^{(e)} B_j^{(e)} DetJ^{(e)} \partial \xi \partial \eta \quad (4.14)$$

Where $D^{(e)}$ is the material constitutive matrix (Caldeira L. et Cardoso R., 2013). Then the element stiffness matrixes of the various elements need to be assembled to give the global stiffness matrix K , giving the final global relation between global force, displacement, and stiffness matrices (Anderson T. L., 2005):

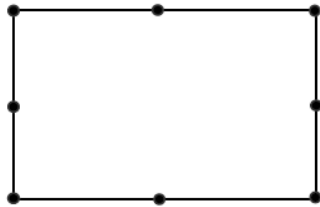
$$[K][u] = [F] \quad (4.15)$$

Where $[K]$ is the stiffness matrix, $[u]$ is the global displacement vector and $[F]$ is the global force vector.

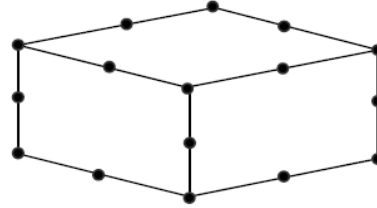
4.2.2 Discretization of the elements

As explained later in the Chapter, in the XFEM, fracture is modeled independent of mesh configuration and element type. This means that no re-meshing is required and fractures do not need to be aligned with element boundaries. However, the type of element and mesh discretization have effects on simulation convergence and results (Sepehri J., 2014).

The discretization by finite elements for the analysis of a propagating crack in the context of computational fracture mechanics, recommends biquadratic Lagrangian elements of eight nodes for two-dimensional problems and the twenty nodes element for three-dimensional problems, as seen in figure 4.3.



2 dimensional element of 8 nodes



3 dimensional element of 20 nodes

Figure 4.3 - Finite elements discretization for 2D and 3D classical fracture mechanics (Mohammadi S., 2008)

As one of the objectives of the element discretization is to analyze the stress and strain at the crack tip (by the calculation of stress intensity factors), once the stresses and strains vary in the form $\frac{1}{r}$ at the crack tip, is possible to show that triangular geometries introduces more accurate and satisfactory results than quadrilateral elements (Barsoum R. S., 1976).

To capture the behavior of the singularity, quadrilateral elements are degenerated into triangular geometry, as seen in Figure 4.4. In quadrilateral elements the singularity is only found inside the element, whereas in the triangular elements it can be located inside and on the contour of the element. This is a simple and effective way to build elements that incorporate in

their strain field the singularity $\frac{1}{r}$. In elastic problems, the nodes at the crack tip are moved to the $\frac{1}{4}$ points (quarter-point). This modification is necessary to introduce a $\frac{1}{r}$ strain singularity in the element, which brings numerical accuracy (Oliveira F., 2013).

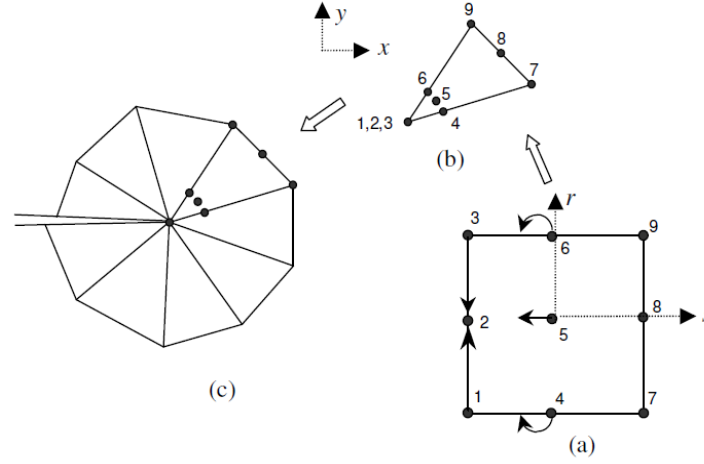


Figure 4.4 - Construction of the spider-web mesh, based on the degeneration of quadrilateral elements in triangular elements (Dassault Systèmes, 2013)

As seen in figure 4.4, the spider web configuration near the crack tip is the most efficient mesh discretization to analyze the crack tip stress intensity factors.

4.3 Partition of the Unity

The theory of partition of unity is an important tool for fracture modelling. The word unity stands for the constant shape function, η_i , equal to the unity on some given space X . These shape functions form a partition of unity, as seen in figure 4.5.

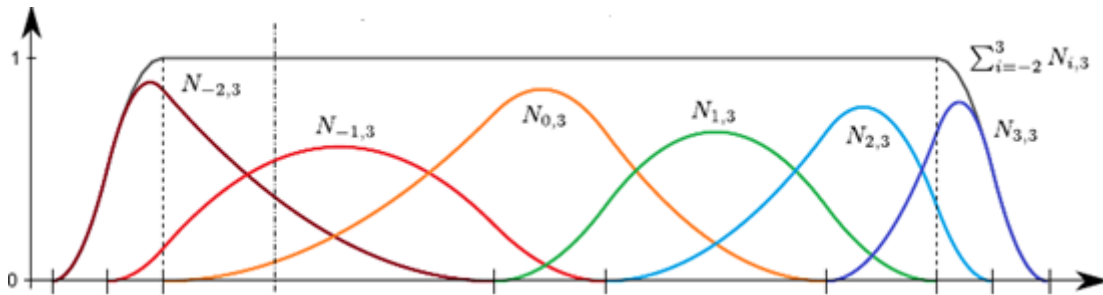


Figure 4.5 - Partition of unity concept (Wikipedia, 2015) - $N_{i,i} = \eta_i$

In a mathematical formulation, Partition of unity is a set $\mathbb{R} \in [0,1]$ of continuous functions from X such that for every point, $x \in X$ the following conditions are verified (Robbin J. W., 2014):

- $\sum_{i=1}^n \eta_i = 1, \forall x \in X$
- At any given point x all but one finite number of the functions are equal to 0

It can be shown that for any arbitrary function $\psi(x)$, the following property is automatically satisfied:

$$\sum_{i=1}^n \eta_i(x) \psi(x) = \psi(x) \quad (4.16)$$

Where η_i is a constant shape function and $\psi(x)$ is an arbitrary function.

This formalism is the basis of the XFEM. By appropriately choosing the function $\psi(x)$ for each node, the computation time may be reduced (McNary M. J., 2009).

4.3.1 Partition of Unity Finite Element Method

To improve a finite element approximation, the enrichment procedure may be applied. The methodology followed in the partition of unity finite element method (PUFEM) is defined using the classical finite element shape functions N_i . For a general point x within a finite element, the displacement field is given by:

$$u^{(e)}(x) = \sum_{i=1}^n N_i(x) u_i(x) + \sum_{i=1}^n \sum_{j=1}^m N_i(x) p_j(x) a_{ij} = \sum_{i=1}^n N_i(x) \left(u_i(x) + \sum_{k=1}^m p_k(x) a_{ik} \right) \quad (4.17)$$

Where $u^{(e)}$ is the element displacement, $p_j(x)$ are the enrichment functions, a_{ij} are the additional unknowns or degrees of freedom associated to the enriched solution and n and m are the total number of nodes of finite elements and the number of enrichment functions p_j , respectively.

The approximation is clearly a partition of unity. As a result, a compatible solution is guaranteed. Examining the approximate solution for a typical enriched node x_i , equation 4.17 can be written as:

$$u^{(e)}(x_i) = u_i + \sum_{k=1}^m p_k(x_i) a_{ik} \quad (4.18)$$

Which is not a possible solution because it does not respect the condition $(u^{(e)}(x_i) = u_i)$. A slightly modified expression is required for the enriched displacement field to satisfy the interpolation at nodes:

$$u^{(e)}(x) = \sum_{i=1}^n N_i(x) \left(u_i(x) + \sum_{k=1}^m (p_k(x) - p_k(x_i)) a_{ik} \right) \quad (4.19)$$

Which ensures $u^{(e)}(x_i) = u_i$.

4.3.2 Generalized Finite element method

In the generalized FEM, different shape functions are used for the classical and enriched parts of the approximation, consisting in an advance in relation to the partition of unity FEM and allowing a great improvement in finite element discretization. Generalized FEM can be expressed as:

$$u^{(e)}(x) = \sum_{i=1}^n N_i(x)u_i(x) + \sum_{i=1}^n \sum_{j=1}^m \bar{N}_i(x)p_j(x)a_{ij} \quad (4.20)$$

Where $\bar{N}_i(x)$ are the new set of shape functions associated with the enrichment part of the approximation. For the same reason explained before, equation 4.20 should be modified as follows:

$$u^{(e)}(x) = \sum_{i=1}^n N_i(x)u_i(x) + \sum_{i=1}^n \sum_{j=1}^m \bar{N}_i(x) \left(p_j(x) - p_j(x_i) \right) a_{ij} \quad (4.21)$$

4.4 Extended Finite Element Method (XFEM)

The XFEM method consists in a mesh of finite elements which is independent of the discontinuity. This formulation takes off the need to define a new mesh configuration (remeshing) for the simulation of crack propagation, by opposition to the classical FEM. Therefore, the same mesh is used for all time steps during crack propagation.

The XFEM is based on the introduction of additional degrees of freedom, which are established in the nodes of the elements that are intersected by the crack geometry, as seen in figure 4.6. These extra degrees of freedom in combination with special shape functions allow extra accuracy in modeling the crack.

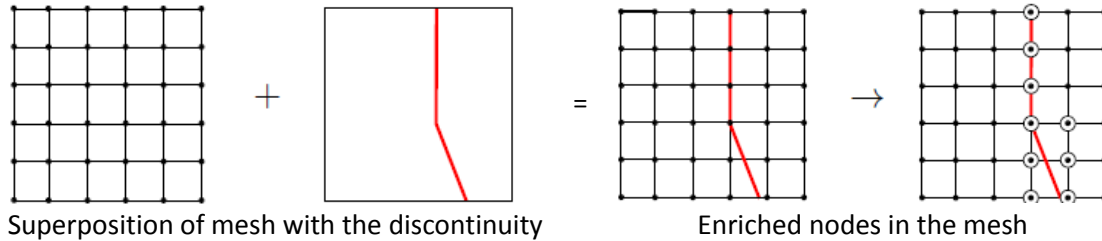


Figure 4.6 - Definition of the enriched nodes in a mesh of finite elements (Duarte A. et Simone A., 2012)

In contrast to partition of unity FEM and generalized FEM, where the enrichments are usually employed on a global level and over the entire domain, the extended finite element method adopts the same procedure but at local level.

This is a considerable computational advantage as it is equivalent to enriching only nodes close to the crack tip, an important step for the extended finite element solution.

Consider x , a point in a finite element that is intersected by a crack. To calculate the displacement at point x located within the domain, the following approximation is utilized in XFEM:

$$u^{(e)}(x) = u^{FEM} + u^{Enrichment} = \sum_{i=1}^n N_i(x)u_i(x) + \sum_{j=1}^m \bar{N}_j(x)\psi(x)a_j \quad (4.22)$$

Where u_i is the vector of nodal degrees of freedom, a_j is the set of degrees of freedom added to the standard finite element model by the introducing of the enrichment functions $\psi(x)$, defined for the set of nodes that are within the influence area of the discontinuity.

The definition of the influence area depends on the location of the discontinuity. When it is located on an edge, it consists of the elements containing that node, while for an interior node (in higher order elements) it is the element surrounding the node, as seen in figure 4.7.

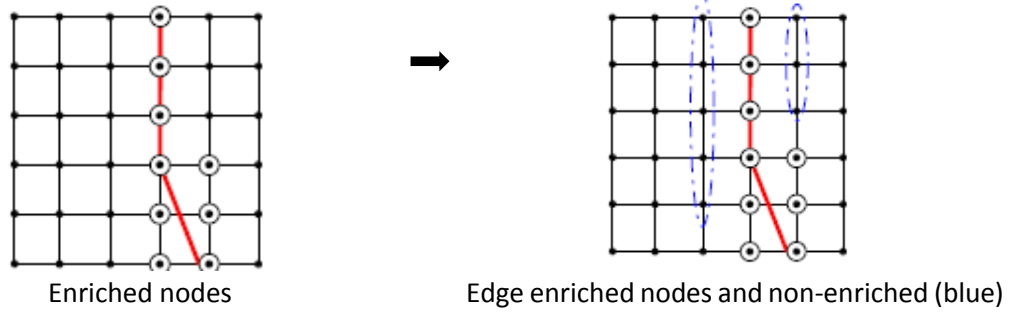


Figure 4.7 - Enriched nodes by the discontinuity contour line in the interior or on the edge of the element (Duarte A. et Simone A., 2012)

The selection of the enrichment function $\psi(x)$ should be done by applying appropriate analytical solutions according to the type of discontinuity. The main objectives for using various types of enrichment functions within an XFEM procedure are the following (Mohammadi S., 2008):

- Reproduce the displacement field around the crack tip.
- Define the compatibility displacement between adjacent finite elements.
- Reproduce different strain fields on both sides of a crack surface.

For an ordinary crack implementation problem, the approximation function for the displacement in an element node, can be defined as:

$$u^{(e)}(x) = u^{FEM} + u^{Enrichment} = \sum_{i \in J} N_i(x) u_i(x) + \sum_{j \in K_D} \bar{N}_j(x) H(x) a_j + \sum_{j \in K_T} \bar{N}_j(x) K(x) b_j \quad (4.23)$$

Where J set contains all nodes, K set contains enriched nodes, K_D set the enriched nodes associated with the discontinuity and K_T the set of enriched nodes associated with the crack tip.

The functions $H(x)$ and $K(x)$ are the enrichment functions for a strong discontinuity, as seen in figure 4.8.

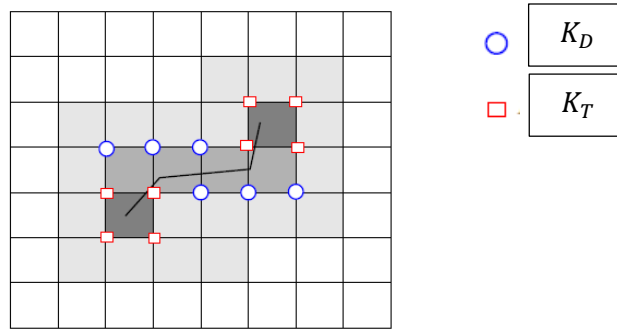


Figure 4.8 - Definition of the enriched nodes and domains in XFEM : Light grey – Heaviside function ; Heavy grey – Near-tip functions ((Thoi T. N. et al, 2015) and (Natarajan S. et al, 2011))

4.4.1 Enrichment functions

As stated before, the enrichment functions are essential for the accuracy of the computed results. Computational fracture mechanics is essentially designed to deal with strong discontinuities, where the strains and displacements are discontinuous through a crack surface, as seen in figure 4.9. But XFEM can also be used in weak discontinuities problems. The difference between strong and weak discontinuities is that in a weak discontinuity, the displacement field remains continuous across the limits of a narrow strain localization band. On the other hand, for a strong discontinuity the displacement field becomes discontinuous across the surface (Chaves E. W. et Oliver J., 2001).

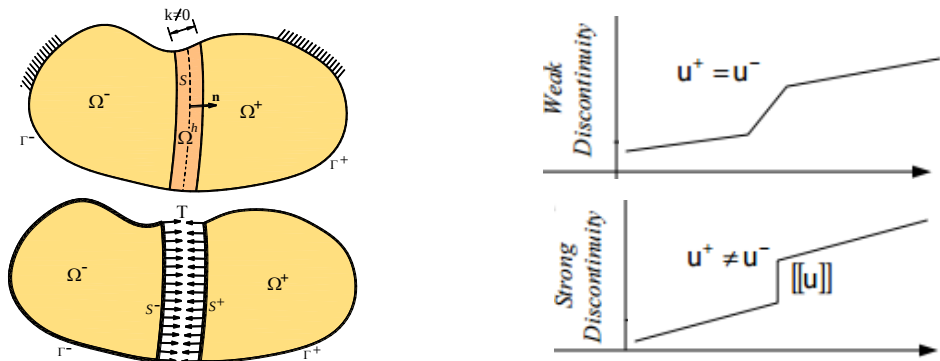


Figure 4.9 - Strong and weak discontinuity definition, adapted from (Chaves E. W. et Oliver J., 2001) and (Ayala G., 2006)

From now on, the study will be focused on the computational implementation of strong discontinuities. For a strong discontinuity, assuming linear elastic fracture mechanics, two sets of enrichment functions are mainly used:

- Heaviside function $H(x)$
- Near-tip asymptotic enrichment functions $K(x)$

Heaviside/jump functions

The elements which are completely crossed by the singularity/crack, i.e. which have a discontinuous displacement field, are enriched with the Heaviside/jump function. The Heaviside function can be defined as:

$$H(x) = \begin{cases} -1, & x < 0 \\ 1, & x \geq 0 \end{cases} \quad (4.23)$$

This function is defined in the element displacement equation, and introduces the possibility to model the crack geometry with a mesh which does not contain any discontinuity but have the ability to reproduce it, as seen in figure 4.10 ((Oliveira F., 2013) and (Chahine E. et al, 2006)).

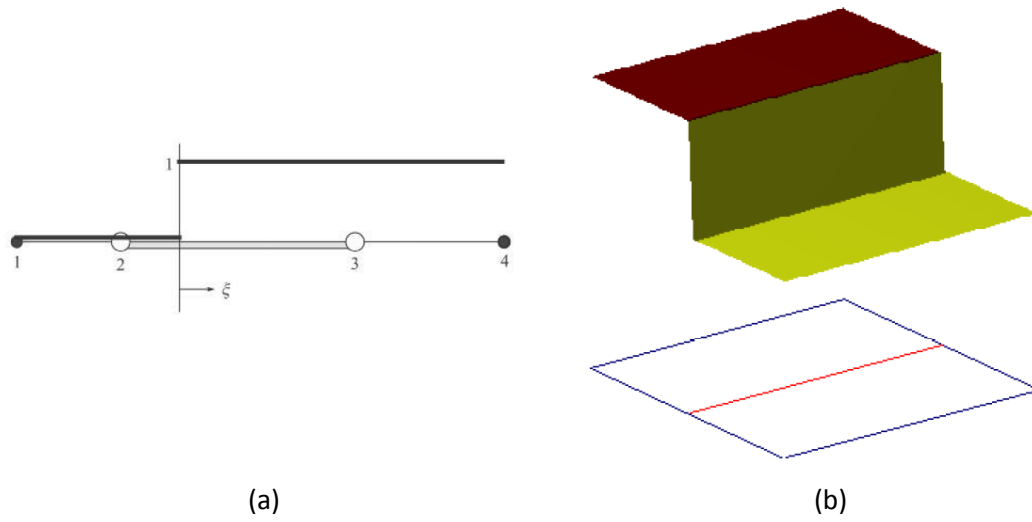


Figure 4.10 - Heaviside function (a)) and schematic representation of it in a finite element (b)) ((Mohammadi S., 2008) and (Ahmed A., 2009)).

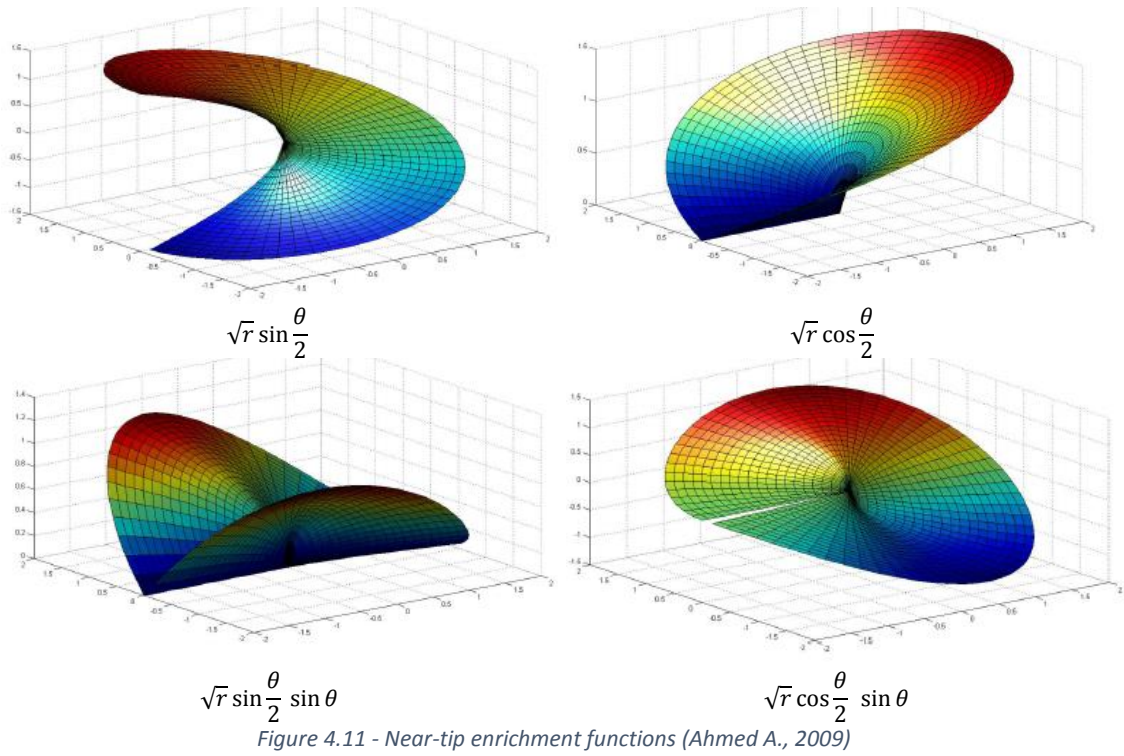
Near-tip asymptotic functions

The element that contains the crack tip is partially cut by a discontinuity. For this situation the heaviside function is not adequate to enrich the domain. In Linear elastic fracture mechanics (LEFM), the exact solution of the stress and displacement field is presented in equations 3.19 to 3.24.

The displacement field is solved at the crack tip using near-tip enrichment functions, defined in terms of the local crack tip coordinate system (r, θ) for an isotropic material as:

$$K(r, \theta) = \left\{ \sqrt{r} \sin \frac{\theta}{2}, \sqrt{r} \cos \frac{\theta}{2}, \sqrt{r} \sin \theta \sin \frac{\theta}{2}, \sqrt{r} \sin \theta \cos \frac{\theta}{2} \right\} \quad (4.24)$$

These functions form the basis of the asymptotic field $\frac{1}{r}$ around the crack tip, and introduce additional degrees of freedom in each node, improving the solution accuracy near the crack tip. The first function $\sqrt{r} \sin \frac{\theta}{2}$ is discontinuous along the crack surfaces, giving the effect of required discontinuity in the approximation along the crack, as seen in figure 4.11 and 4.12.



With the use of the above mentioned near-tip enrichment functions an element partially cut by the crack can be modeled (Ahmed A., 2009)

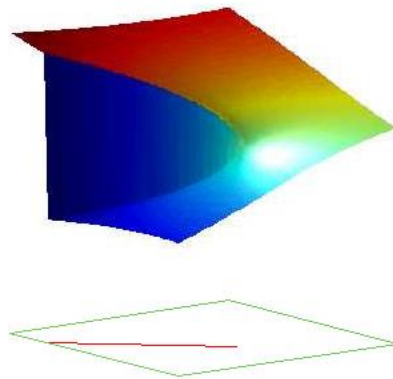


Figure 4.12 - Enrichment function (b) modelling the crack in a partially cut tip element (Ahmed A., 2009)

4.4.2 Level Set Method

In order to model complex crack configurations as the ones created by hydraulic fracturing, more powerful and convenient techniques for representing internal discontinuities are required. The level set method (LSM), allows to describe and track the motion of the crack (crack propagation). With the use of the LSM, the motion of the interface is computed on a fixed mesh, and LSM is capable of handling the topology changes (discontinuities) without any additional set function. The geometric properties of the interface can also be obtained from the level set function.

The LSM facilitates the selection and computation of the enrichment nodes as it stores all the information needed for crack growth representation.

To fully characterize a fracture, two different level set functions are defined, as seen in figure 4.13:

- A normal/distance function $\phi(x)$
- A tangential function $\psi(x)$

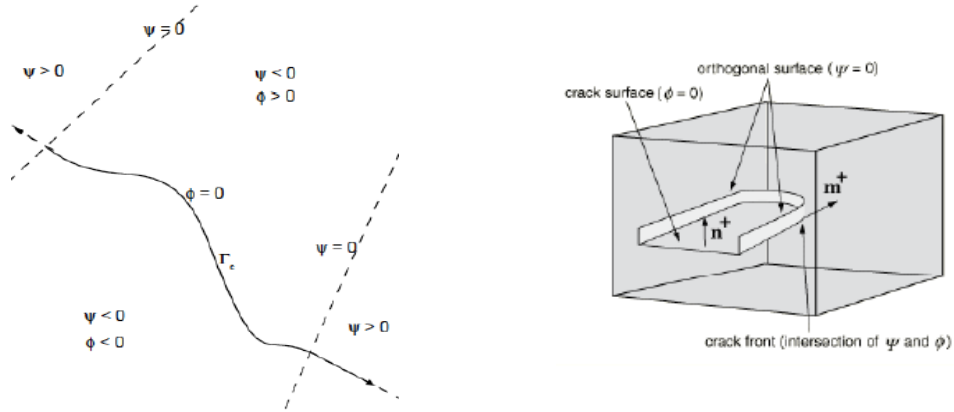


Figure 4.13 - Level set functions representation (Zhen-zhong D, 2009)

To model crack propagation, a distance/normal function needs to be defined. For the evaluation of distance function, with crack surface Γ_c , for any point, the closest point in the crack domain, x_c , should respect that the following condition is verified as possible:

$$|x - x_c| \rightarrow 0 \quad (4.25)$$

The distance/normal function is then computed as follows:

$$\phi = (X - X_\Gamma) \cdot n^+ \quad (4.26)$$

Where X is any computed model point, X_Γ is a crack surface point and n^+ is the normal outwards the fracture surface (Figure 4.13). This expression is defined considering an interior crack. In case of an interior discontinuity, two different functions can be applied, as seen in figure 4.13. However, the tangential level set function is defined based on the following criteria:

$$\psi = \max(\psi_{\Gamma_{ctip1}}, \psi_{\Gamma_{ctip2}}) \quad (4.27)$$

As said before, the main objective of the LSM is to define in a crack propagation problem the crack surface and crack tip location. Based on this, two expressions can be pointed for this purpose:

$$X \in \Gamma_c, \text{ if } \phi = 0 \text{ and } \psi \leq 0 \quad (4.28)$$

$$X \in \Gamma_{ctip}, \text{ if } \phi = 0 \text{ and } \psi = 0 \quad (4.29)$$

With this methodology, the crack propagation is controlled, as suggested by figure 4.14 and 4.15.

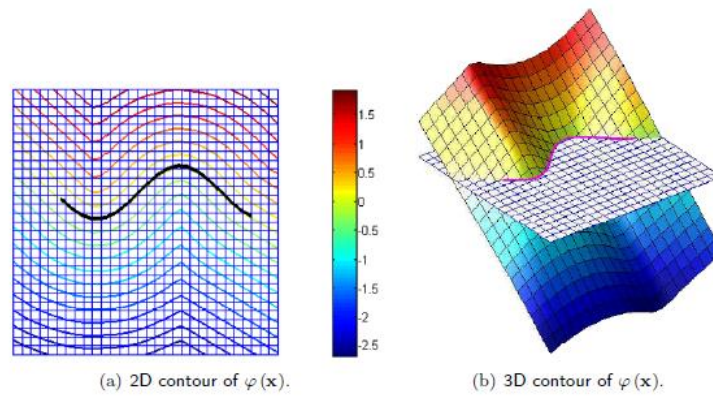


Figure 4.14 - Normal LSF for an interior crack (Gigliotti L., 2012)

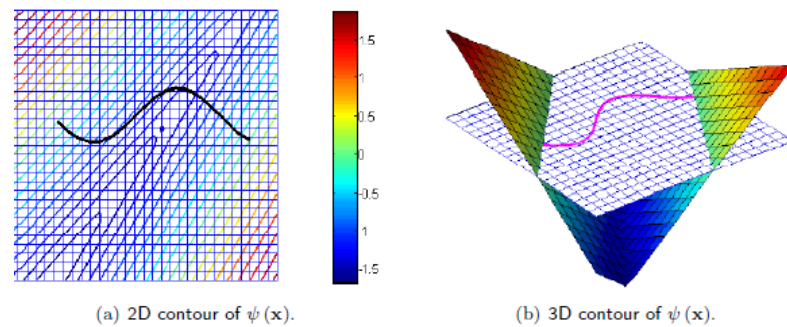


Figure 4.15 - Tangential LSF for an interior crack (Gigliotti L., 2012)

4.4.3 Fracture propagation criteria

Two approaches can be used to model fracture propagation. The virtual crack closing technique (VVCT) is based on the concept of Linear Elastic Fracture Mechanics and fracture growth is predicted when a combination of the components of the energy release rate is equal to, or greater than, a critical value.

On the other hand, the cohesive zone method (CZM) is developed based on Damage Mechanics. Then, fracture is initiated when a damage criterion reaches its maximum value. In Abaqus®, both the cohesive segment method and the virtual crack closing technique are used in combination with enriched node technique to model moving fracture.

The main difference between the two criteria is that the VCCT method is a purely energetic method; i.e. the fracture initiation and propagation depends only on the critical energy release rate. The CZM, adopted in the analyses presented in Chapters 5 and 6, defines fracture initiation based on the tensile strength of the material.

In table 4.1 the main differences between the two criteria are presented.

Table 4.1 - Differences between VCCT method and CZM - adapted from (Dassault Systèmes, 2015)

VCCT	Cohesive zone method
Is contact (surface based)	Interface elements (element based) or contact (surface based)
Assumes an existing singularity	Can model fracture initiation
Brittle fracture using LEFM	Ductile fracturing occurring over a crack front modeled with cohesive elements
Requires G_I, G_{II} and G_{III}	Requires E, σ_T and G_I, G_{II} and G_{III}
Crack initiates when strain energy release rate exceed the critical	Crack initiates when traction exceeds critical value
Crack surface are rigidly bonded when uncracked	Crack surfaces are joined elastically when uncracked

When using the VCCT criteria this is only verified for the fracture contact surface. In contrast, the cohesive zone method is applied to the entire analysis domain, which ensures that fracture initiation and propagation can occur in any point of the analysis domain.

4.4.4 XFEM limitations

Since 2009, the XFEM is available in Abaqus© software (version 6.9); however the method presents several relevant limitations, the most important being (Gigliotti L., 2012):

- XFEM has adequated conditions for stationary cracks, but needs additional functions for crack propagation problems;
- Only linear continuum elements can be used, with or without reduced integration;
- Only single or non-interacting cracks can be contained in the domain;
- A crack cannot turn more than 90 degrees within an element;

For stationary cracks the crack tip can be located inside a finite element, but in a crack propagation problem, the crack is required to go all through the element such that the tip is on the edge of an element. Once the crack propagation starts, it will keep cutting completely each of the elements and the crack tip motion cannot be analyzed within an element.

This difference between stationary and propagating cracks is mainly due to the different enrichment procedure. For propagating cracks, the asymptotic near-tip singularity functions are not included (reason for the crack tip to be forced to be in an element edge) in the enrichment scheme and only the Heaviside function is used. On the other hand, for a stationary crack both Heaviside and crack tip singularity functions are included in the XFEM discretization, as seen in figure 4.16.

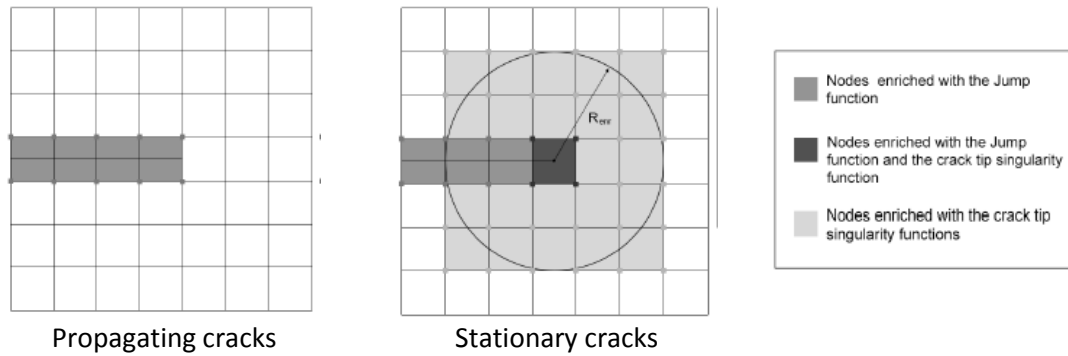


Figure 4.16 - Schematic representation of the Abaqus© enrichment functions for stationary and propagating singularities (Oliveira F., 2013)

To solve this problem, the LSM is essential to define the crack surfaces and the crack tip location in Abaqus©, and to increase to accuracy of crack propagation models.

With Abaqus 6.14 version (released in 2014) it becomes possible to carry out the XFEM analysis of fracture propagation with full hydromechanic coupling. That is software used in the study presented in this thesis.

5. Modelling

5.1 Introduction

This chapter aims to provide a correct and complete overview of the modelling capabilities, shortcomings and underlying assumptions made by the Extended Finite Element Method. The software employed to carry out the analyses is Abaqus, a multi-physics finite element based software that incorporates the XFEM approach to model fractures/singularities.

This Chapter has been divided in two main parts:

In the first part, a fracture toughness determination test on an infinite plate with a center crack is simulated numerically using the XFEM approach and the numerical results are compared with well-established theoretical solutions. To simulate the fracture toughness determination test it is necessary to model both a stationary fracture and a propagating fracture, which involves different numerical procedures and highlight the limitations of the XFEM approach. This study allows to gain sensitivity regarding the performance of the numerical tool, to assess the representativeness of the numerical simulations.

In the second part of this Chapter, it is presented a numerical study aiming to simulate a series of laboratory experiments described by Abass et al. (1994) in order to analyze the influence of oriented perforations in the development and propagation of hydraulic fractures. The definition of the numerical model is discussed in detail. The results are examined in detail, with comparison being made with analytical solutions when available. Note that in Medinas M. (2015) these tests have been previously simulated using the XFEM approach; however in that study hydraulic fracturing was simulated as a mechanical pressured applied on the walls of the borehole and it was found that the results were significantly affected by the simplification introduced by the manner in which the load was being applied. In the present study, fluid injection is explicitly considered and the tests are model by means of fully couple hydro mechanic XFEM analyses.

5.2 Numerical modelling of the fracture toughness determination test

5.2.1 Fracture Toughness determination

One of the tests for the determination of the fracture toughness is a test on an infinite plate with a center crack that is subjected to tension, as shown schematically in Figure 5.1.

Determining the value of fracture toughness requires using a sample that contains a crack of known length. Fracture toughness is the stress intensity factor at the critical load, K_{IC} i.e. when the pre-existing crack is reinitiated ((Patrício M. et Mattheij R. M., 2007) and (Economides M. J.

et al, 2000)). The stress intensity factor, and fracture toughness, is a function of the load and sample geometry, including the length of the pre-existing crack.

Though its simplicity, the test is, in practice, difficult to perform. It is of extreme interest to model this test because there are well-defined analytical solutions for this loading condition, against which the numerical results can be compared (Paris C. et Sih G., 1964).

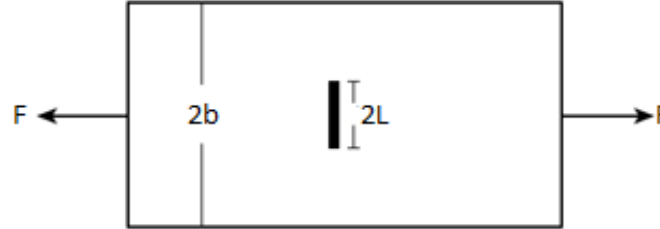


Figure 5.1 - Set up for fracture toughness determination - infinite plate with known central crack under tension (Economides M. J. et al, 2000)

For an infinite plate with a central crack under tension, the stress intensity factors, assuming the LEFM can be calculated using the following expression:

$$K_I = \sigma \sqrt{\pi L} \left(1 - \frac{L}{2b}\right) \left(1 - \frac{L}{b}\right)^{-1/2} \quad (5.1)$$

where l is half of the crack length, b is half of the plate width and σ is the stress applied to the sample, equivalent to $\sigma = \frac{F}{2b}$.

The calculation of the critical stress intensity factor is based on the following expression:

$$K_{IC} = \frac{F_c}{2b} \sqrt{\pi L} \left(1 - \frac{L}{2b}\right) \left(1 - \frac{L}{b}\right)^{-1/2} \quad (5.2)$$

where F_c is the critical load that determines the crack propagation, for which $K_I = K_{IC}$.

Assuming LEFM, the critical fracture energy G_{IC} is then estimated as:

$$G_{IC} = \frac{1 - \nu^2}{E} K_{IC}^2 \quad (5.3)$$

5.2.2 Model initialization/Pre-processing

Numerical modelling of the fracture toughness determination test requires the definition of two independent models with the same geometry: one XFEM model for the determination of critical load, when fracture propagation first occurs and a stationary fracture model to calculate the stress intensity factors.

The test was modelled by means of plane strain analyses. The same geometry and material parameters are used in the two models and these are presented in Table 5.1 and Figure 5.2.

Table 5.1 - Input parameters for fracture toughness determination test

Parameters	
$E(Kpa)$	$1.427 \times 10^7 Kpa$
ν	0,2
$\sigma_t(Kpa)$	5560 Kpa
ϕ	0,265
$2b(m)$	0,04
$2L(m)$	0,004

The boundary conditions used aim to ensure the verticality of the sample throughout the loading procedure, on top of the sample. It was chosen conservatively to use embedding/encastre on the bottom of the sample, while the reaction make the turn of the applied load to the sample bottom, as seen in Figure 5.2.

A line load was applied in the middle point of one end of the plate (see Figure 5.2); its value was set at 400 kN/m/min, increasing linearly during the time step.

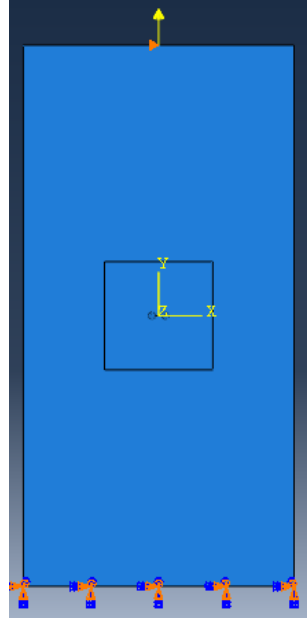


Figure 5.2 - Geometry, boundary conditions and loads for fracture toughness test

Different geometries are adopted for the two models, because of differences in the calculation procedures and respective mesh requirements. In the first model (i.e. the fracture propagation model), the objective is to determine the critical load that causes the fracture to propagate. Accordingly, to avoid convergence problems and loss of accuracy in the results, the adopted mesh should be structured, as seen in Figure 5.6a). The model is composed by 3200 plane strain linear quadrilateral elements and 3321 nodes.

Once the critical load is known, this is applied to a stationary fracture model for the calculation of the stress intensity factor. For this model, the mesh is required to have a set of characteristics to guarantee the accuracy of the numerical results, because the calculation procedure takes

advantage of the mesh geometry (Dassault Systèmes, 2015). For the calculation of the contour integral, it is desirable that the quadrilateral elements around the crack tip collapse in triangular shapes, to capture the singularity at the crack tip with accuracy (Figure 5.3).

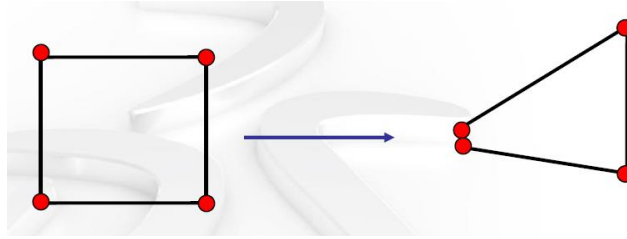


Figure 5.3 - Linear quadrilateral element degeneration (Dassault Systèmes, 2015)

As can be seen in Figure 5.4, the degeneration of the elements ensures the existence of nodes organized in a way that was not possible to obtain otherwise.

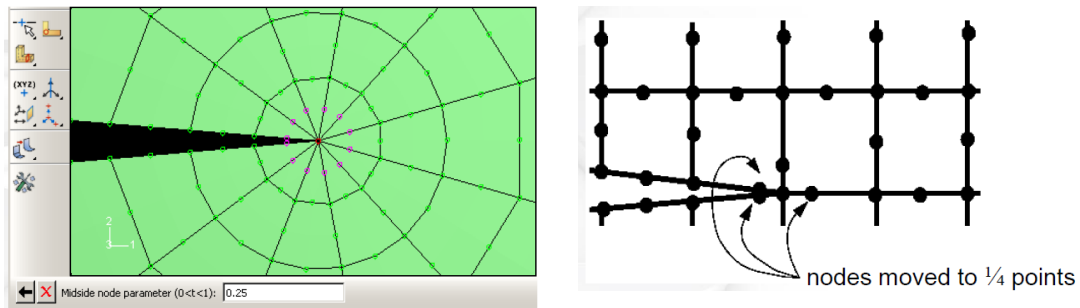


Figure 5.4 - Mesh degeneracy to $r=1/4$ (Dassault Systèmes, 2015)

As a consequence of the elements degeneration and the collapse of nodes surrounding the crack tip, the mesh geometry becomes more complex, as shown in the Figure 5.5.

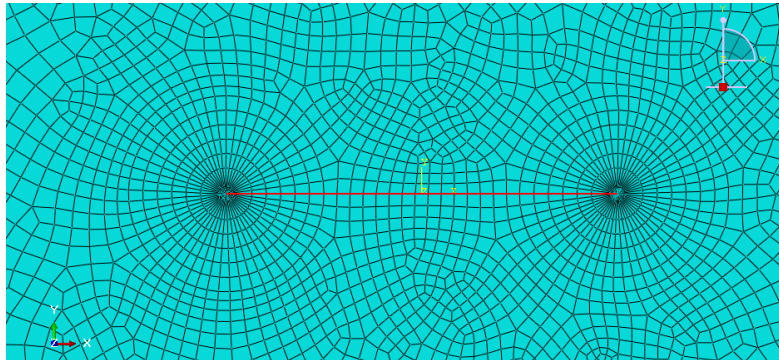


Figure 5.5 - Mesh around the crack tip/singularity

Therefore, for the stationary fracture model, the mesh is structured in most parts/regions, except around the fracture where the mesh is defined with sweep mesh configuration, characteristic of circular geometries (Dassault Systèmes, 2015), as seen in Figures 5.5 and 5.6b). The model is composed by 4833 plane strain linear quadrilateral elements and 4961 nodes.

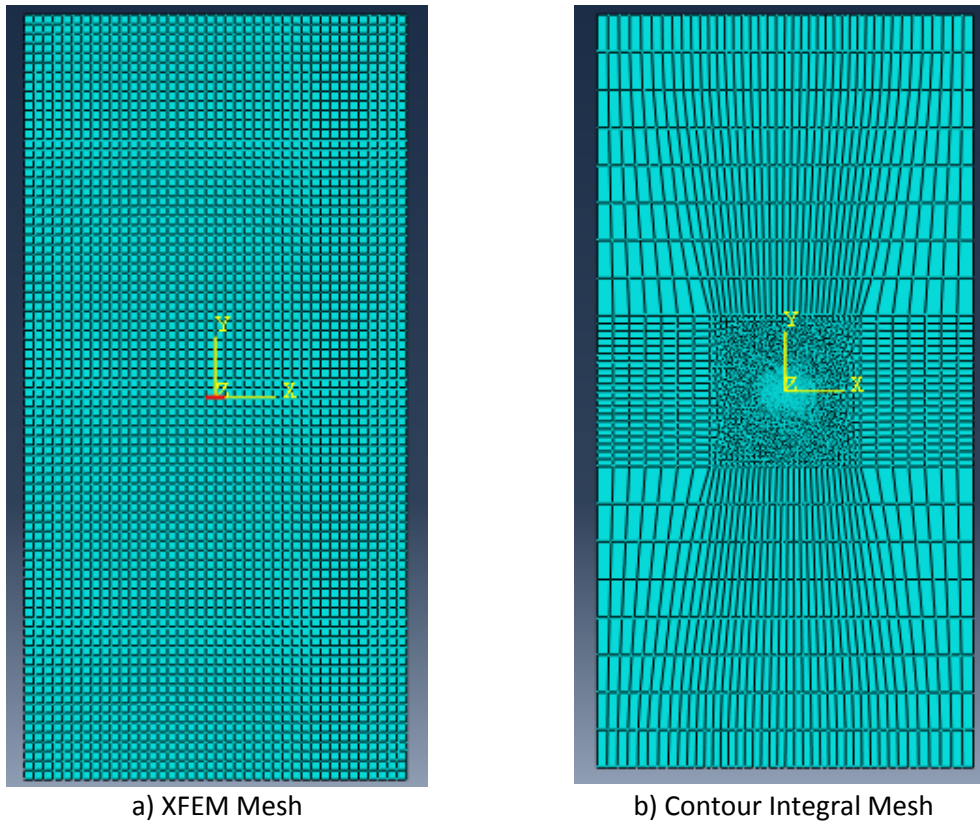


Figure 5.6 - Mesh geometry for the propagation XFEM and contour integral stationary crack

In the stationary fracture model, the number of contours for which the stress intensity factors are calculated, for reasons of calculation ease and quality of the results, should be six contours. The obtained stress intensity factor is then the arithmetic average of the five contours further away from the crack tip. This procedure is essential because the contour close to the crack tip is always too far from the solution ((Oliveira F., 2013) and (Dassault Systèmes, 2015)).

5.2.3 Results and discussion

Based on the fracture propagation XFEM model, the fracture propagation critical load is 288 KN (Figure 5.7). Figure 5.7 shows the output STATUSXFEM that is the output which shows if the fracture has propagated. When the STATUSXFEM is different from zero, the fracture has propagated, and for a value equal to one, the section has lost the cohesive behaviour.

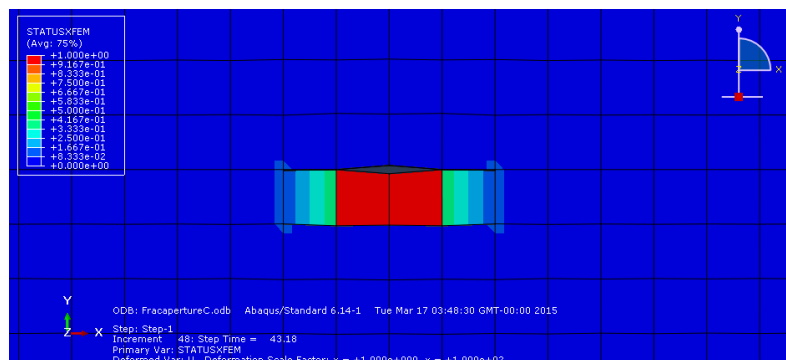


Figure 5.7 - Crack propagation initiation based on XFEM model

Once the critical load F_c is known, the critical stress intensity factor K_{IC} (or fracture toughness) can be calculated using Equation 5.2:

$$K_{IC} = \frac{288000}{2 \times 0,02} \sqrt{0,002\pi} \left(1 - \frac{0,002}{2 \times 0,02}\right) \left(1 - \frac{0,02}{2 \times 0,002}\right)^{-1/2} = 0,5715 \text{ MPa}\sqrt{m} = 571,5 \text{ KPa}\sqrt{m} \quad (5.4)$$

Based on Equation 5.3, the critical fracture energy is then:

$$G_{IC} = \frac{1 - 0,2^2}{1,427 \times 10^{10}} (0,5715 \times 10^6)^2 = 21,97 \frac{N}{m} \approx 0,022 \frac{KN}{m} \quad (5.5)$$

On the other hand, from the stationary fracture model output file, the values of the stress intensity factors calculated for the various contours can be obtained and these are shown in Figure 5.8.

CRACK NAME	CRACKFRONT NODE SET	K FACTOR ESTIMATES				
		C O N T O U R S				
		1	2	3	4	5
		6				
SIFCRACK2_CRACK-2						
-14-						
	K1:	264.0	422.3	506.2	578.0	641.8
	K2:	0.000	0.000	0.000	0.000	0.000
MERR DIRECTION (DEG):		0.000	0.000	0.000	0.000	0.000
J from Ks:		4.6902E-03	1.1999E-02	1.7237E-02	2.2474E-02	2.7712E-02
	K1:	699.8				
	K2:	0.000				
MERR DIRECTION (DEG):		0.000				
J from Ks:		3.2949E-02				

Figure 5.8 - Contour stress intensity factors for F=288KN

The stress intensity factor is then the average of the values obtained on the specified contours, ignoring the contour closest to the crack tip:

$$K_I = \frac{422,3 + 506,2 + 578,0 + 641,8 + 699,0}{5} = 569,5 \text{ KPa}\sqrt{m} \quad (5.6)$$

It is found that the difference between the numerical results and the theoretical solution is only 0.35%. The numerical models fully reproduce the test of the infinite plate with a central crack under tension.

5.3 Numerical modelling of oriented perforations

The aim of this section is to simulate numerically, using the Abaqus 6.14, the laboratory experiments described by Abass H. et al. (1994) in order to analyze the effect of oriented perforations in drilling wells when using the technique of hydraulic fracturing. This study focuses on phenomena such as the initiation and propagation of hydraulic fractures in vertical wells. As noted previously, these experimental tests were previously simulated using the XFEM in Medinas M. (2015). However in that study hydraulic fracturing was simulated as a mechanical pressured applied on the walls of the borehole and it was found that the results were significantly affected by the simplification introduced by the manner in which the load was being

applied. In the present study, fluid injection is explicitly considered and the tests are model by means of fully couple hydro mechanic XFEM analyses.

5.3.1 Experimental setup and material parameters

Abass H. et al. (1994) describe a series of laboratory test on blocks of hydrostone (Gypsum cement) in a water / hydrostone weight ratio of 32: 100, with dimensions 0.154 m x 0.154 m 0,254 m.

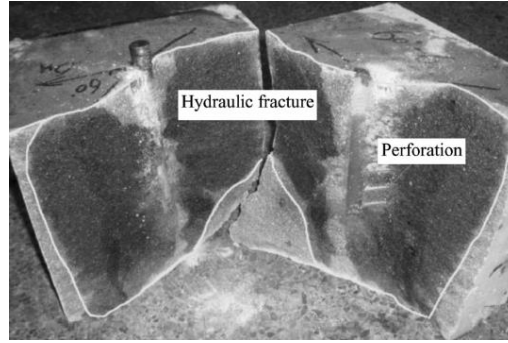


Figure 5.9 - Core sample geometry, wellbore and perforations (Abass H. et al, 1994)

The study focused on perforations at 180° phasing, having been studied the direction (θ) of 0°, 15°, 30°, 45°, 60°, 75° and 90° relative to the preferred fracture plane (PFP, that corresponds to the maximum horizontal stress direction), as shown in Figure 5.10. A total of 10 perforations were made on each block, 5 on each side. The perforations were 12.7 mm long with 3.429 mm diameter.

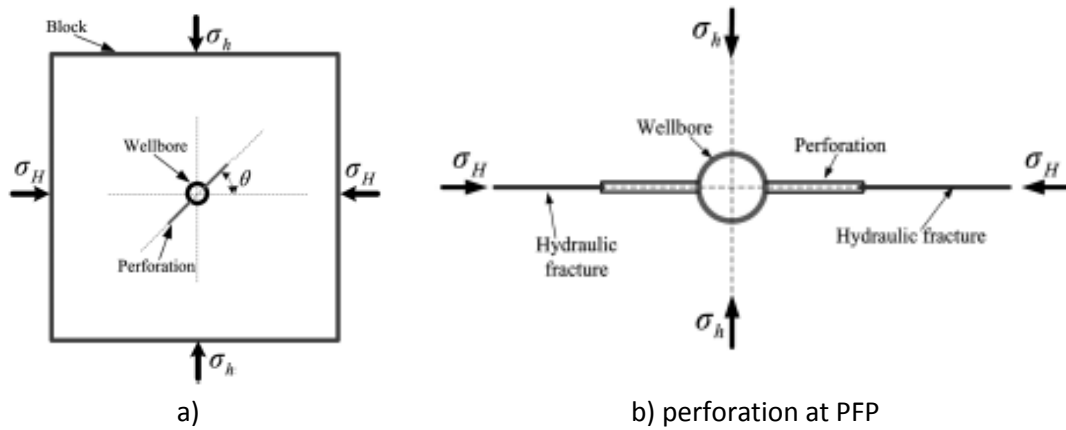


Figure 5.10-Perforations direction relative to the PFP b)

In all tests the fracturing fluid was a 90-weight gear oil with approximately 1180 cp viscosity, with an injection rate of $3 \times 10^{-5} \text{ m}^3/\text{min}$. It was assumed that the initial pore pressure Pp_i present within the samples was is zero (Abass H. et al, 1994).

The samples were confined in a true triaxial apparatus under 20670 kPa vertical stress, 17240 kPa maximum horizontal stress and 9650 kPa minimum horizontal stress. True-triaxial loading is

typically applied through a series of rigid platens, flexible bladders and passive confinement (e.g. steel casing) (Frash L. P. et al, 2014), as seen in figure 5.11.

In a typical true triaxial apparatus configuration loading is applied via one active hydraulic flat jack per principal axis. Specimen faces that contact directly with the flat jack are usually referred to as active faces while faces that are in contact with the frame where a reaction is mobilized are referred to as passive faces (Figure 5.12). Though true triaxial tests can replicate the in-situ conditions more accurately than conventional triaxial test, the test is hard to conduct due to the difficulties in equipment utilization, difficulties in sample preparation and pore pressure measurements during the test (Zoback M., 2007).

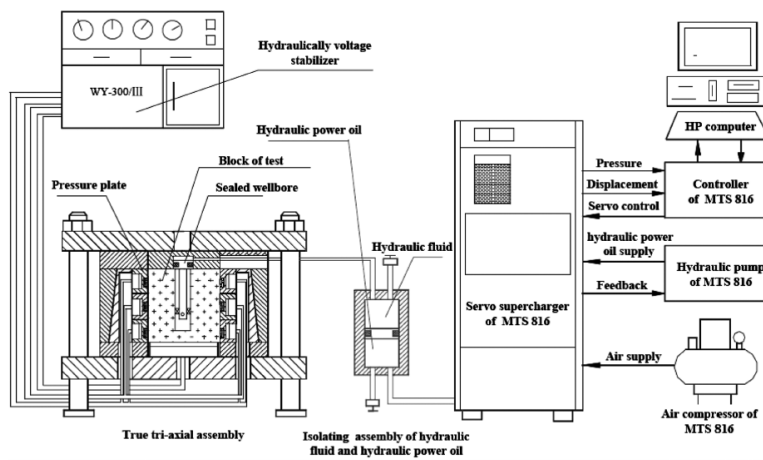


Figure 5.11 - Schematic of a true-triaxial hydraulic fracturing test system (Chen M. et al, 2010)

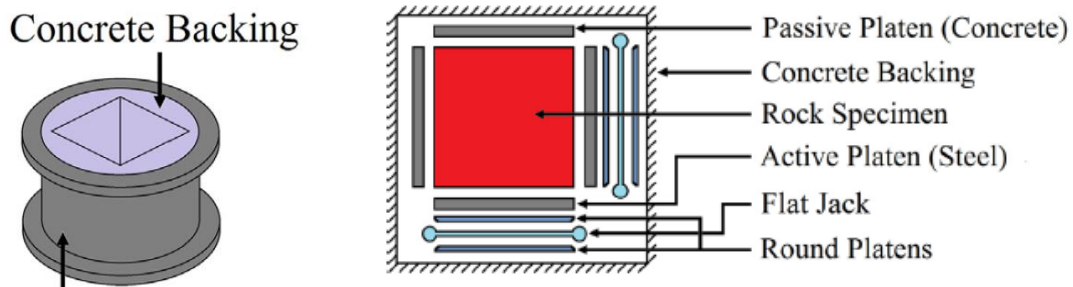


Figure 5.12 - Interior design of a true-triaxial apparatus (Frash L. P. et al, 2014)

The description above refers to typical characteristics of a true triaxial test. However Abass H. et al, (1994) do not give specific information regarding the way loads were applied, the stiffness of the equipment and materials used. It is known that in the work presented by Abass H. et al, (1994) the equipment is not servo-controlled and the minimum horizontal stress was monitored (and significant variation was recorded) during fracture propagation.

5.3.2 Material parameters

The physical and mechanical properties adopted in the numerical analysis are shown in Table 5.2.

Table 5.2 - Physical and mechanical properties of Abass H. et al. (1994) samples

Rock Sample Properties	
Dimensions	$0,1525 \times 0,1524 \times 0,254 \text{ m}$
Wellbore radius	$0,00747 \text{ m}$
Elastic Modulus	$1,714\text{e}7 \text{ KPa}$
Poisson ratio	$0,228$
Permeability	$9,5 \text{ mD}$
Porosity	$0,277$
Fracture Toughness	$2,5 \text{ MPa}\sqrt{\text{m}}$
Fracture energy	$0,341 \text{ KN/m}$
Fracturing fluid density	$0,92\rho_w$
Fracturing Fluid viscosity	1180 cP
Friction coefficient	$0,0001$
Fluid Leak-off coefficient	$1 \times 10^{-14} \text{ KPa.s}$
Tensile strength (Brazilian Test determined)	5560 KPa

Fracture toughness is not an intrinsic property of the material, but depends on the geometry of the sample under study, as illustrated in section 5.2. Because one of the input parameters required for the initialization of the numerical model (i.e. fracture energy) is function of fracture toughness, a value for this quantity had to be adopted. In the absence of any information regarding the value of fracture toughness for gypsum cement, the value used in Liu L. et al. (2014) equal to $2,5 \text{ MPa}\sqrt{\text{m}}$ was adopted, which is in the range of typical fracture toughness values for rocks (see section 3.3.6).

In Abass H. et al. (1994) no consideration is made regarding the friction on the fracture surface. To ensure the model representativeness, friction is considered in the fracture surface contact area to account for the effect of frictional energy dissipation at fracture level and its influence in fracture reorientation and propagation, and a low value of friction coefficient is initially employed. The influence of this parameter is further investigated in Chapter 6.

According to Abass H. et al. (1994) experiences were performed with a zero initial pore pressure and that was prescribed in the numerical analysis. In agreement, the initial pore saturation was also set equal to zero.

The consideration of a non-zero fluid leak-off coefficient guarantees the introduction of some important aspects of the fracturing fluid flow, since this parameter is required to compute the normal component of the fluid velocity within the fracture (Zielonka M. G. et al, 2014). No indication is given in Abass H. et al. (1994) regarding this parameter, and thus a low value equal

to $1 \times 10^{-14} \text{ KPa.s}$ is initially adopted. The influence of this parameter is subsequently analysed in Chapter 6.

Abaqus XFEM software assumes the end of cohesive behavior when $G > G_f$, where G_f is the critical fracture energy. The value of G_f corresponds to the integral of the stress-displacement diagram, as shown in Figure 5.13.

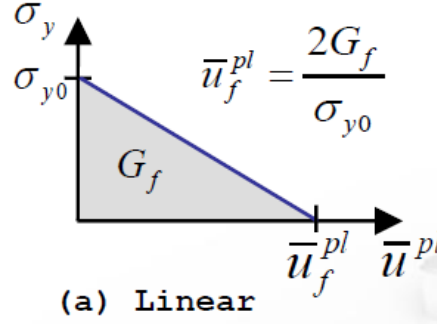


Figure 5.13 - Energy-based damage evolution for linear softening (Dassault Systèmes, 2015)

Where u_f^{pl} is the displacement to reach the end of cohesive behavior and σ_{y0} is the tensile strength of the rock.

Given the loading conditions applied in the analysis presented herein, the MAXPS (Maximum principal stress) failure criterion is adopted. This is mathematically described as:

$$MAXPS = \text{Maximum principal stress} = f = \left\{ \frac{\langle \sigma_{max} \rangle}{\sigma_{max}^0} \right\} \quad (5.7)$$

Where σ_{max} is the stress at a specific point, σ_{max}^0 is the tensile strength of the rock and the symbol $\langle \rangle$ is the Macaulay bracket that is used to ensure that a compressive stress state does not initiate damage, and a fracture is initiated or the length of an existing fracture is extended by tensile stresses, when the fracture criterion, f , reaches the value 1.0 within a specified tolerance:

$$1,0 < f < 1,0 + f_{tol} \quad (5.8)$$

Where f_{tol} is the tolerance for the initiation criterion. As the simulation of fracture propagation can be relatively instable, most authors (Dassault Systèmes, 2015) propose a value of 0,2 for the tolerance, and that is the value adopted in the analysis shown here.

The rock strength degradation/damage evolution (i.e. softening behavior) begins when the failure criterion is verified and progresses linearly until total damage and loss of the section cohesive behavior is reached.

The hydrostone was modelled as a linear elastic material with a damage law for traction separation. The *MAXPS* (Maximum principal stress) failure criterion is considered, which is given by Equation 5.7. In the absence of further information, it was considered an energy type damage evolution law, using a power law function with power equal to 1,0 and fracture energy equal to 0.341 kN/m.

5.3.3 Model geometry and finite element mesh

This set of analyses aims to simulate in an adequate manner the experiments described in the previous section, in particular the initiation and propagation mechanism of fractures. The experiments are modelled by means of a series of coupled hydro-mechanic two-dimensional plane strain analyses. This simplification (of reducing a 3D problem into a simpler 2D problem) greatly simplifies the numerical procedure and reduces computing costs and time. Assuming 2D plane strain conditions means that displacements perpendicular to the analysis plane (in this case the vertical or *z* direction) are zero; however the stresses in the *z* direction are non-zero.

The plane strain assumption is a reasonable approximation when running simplified studies of hydraulic fracturing. However it is found to lead to an overestimation of fracture width, due to the restrictive effect of the tips only in the *xy* direction, as it cannot account for similar restriction in the *z* direction (Valkó P. et Economides M. J., 1995). Given that in this study fracture width is a secondary output, the assumption is made.

To avoid any kind of scale effects the geometry adopted in the numerical analysis corresponds, as much as possible, to that of the laboratory experiments.

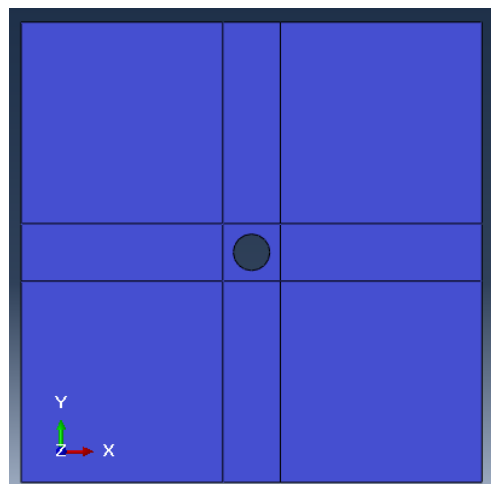


Figure 5.14 - Model geometry and partition faces

Figure 5.14 shows the model geometry and the domain partition. These partitions were introduced to optimize the mesh and the numerical calculation process, by separating the more complex areas, such as the wellbore, of the remaining areas under study. Abaqus software

allows the use of a different meshing technique for each partition. Therefore to minimize the influence of the mesh, structured meshes are used in most of the analysis domain (with different coarseness depending on the expected stress/strain changes) and in the wellbore zone is used a free mesh to overcome the complex geometry of the area (Figure 5.15).

The analyses described in this thesis are hydro-mechanic coupled analyses and thus pore pressure/ displacement finite elements are used.

Sensitivity analyses were performed to assess the influence of the mesh coarseness and configuration on the analysis results. Three structured meshed were considered: extra-fine, fine and coarse, as shown in figure 5.15. The number of nodes and elements in each mesh is summarized in Table 5.3.

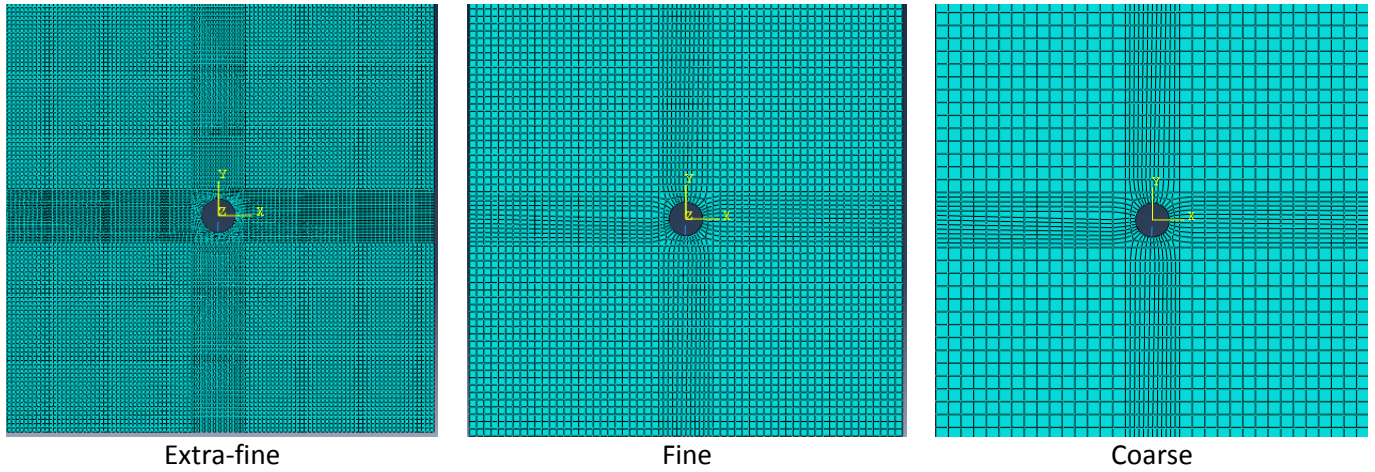


Figure 5.15 - Different Mesh configuration for XFEM oriented perforations study

Table 5.3 - 2D different mesh properties for XFEM oriented perforations study

	Extra-fine	Fine	Coarse
Elements	Linear Quadrilateral	Linear Quad.	Linear Quad.
type	CPE4P	CPE4P	CPE4P
Number of nodes	13605	4184	1808
Number of elements	13345	4036	1704

When analyzing the results obtained with the 3 meshes, it appears that, despite predicting reasonably well fracture reorientation, as a result of the large average element dimension, the coarse mesh gives breakdown pressures for perforations directions above 30° that are not consistent with Abass H. et al. (1994) experimental results.

The fine mesh provides adequate simulation of fracture initiation and propagation for most cases; However, due to the high mesh dependency of the XFEM, the study of 60° and 90° directions, is made using the extra-fine mesh, despite the significant increase in simulation time and memory consumption.

5.3.4 Boundary condition and loading procedure

Since in Abass H. et al. (1994) there is no specific information regarding the apparatus used in the experiments, the definition of loading and boundary conditions to be adopted in the analyses becomes more difficult, and a set of simplifications needs to be introduced.

Abass H. et al. (1994) states that the tests were performed in non-servo controlled conditions. In that case, and due to the fact that the stiffness of the testing equipment may be of the same order of magnitude of the sample stiffness, the stresses applied to the sample facets are not maintained constant, and Abass et al. (1994) report that significant changes in the minimum horizontal stress were recorded. Since it is not given any information regarding the true triaxial used, it is not possible to ascertain what is the stiffness of the equipment, and it is only known that the passive and active platen are often of concrete ($E = 30 \text{ GPa}$) and steel ($E = 200 \text{ GPa}$), respectively.

Following the study of Medinas M. (2015) various options for the displacements boundary conditions were considered. The adopted boundary conditions are shown in Figure 5.16 and both horizontal and vertical displacements were restrained along the samples outer boundaries. Because in the analyses describe herein the fluid is modelled explicitly, hydraulic fracturing operations, i.e. fluid injection, do not affect the stresses at the boundaries and the reactions developed at the boundaries are very small and do not affect the analysis results.

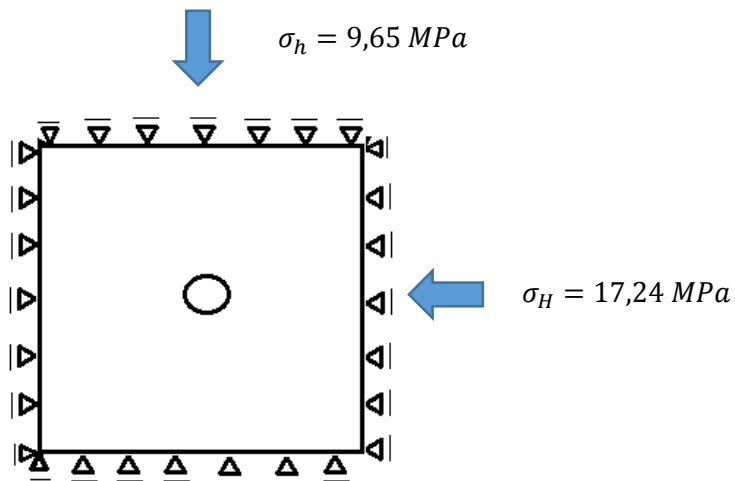


Figure 5.16 - Displacement boundary conditions for the oriented perforations experience

As noted fluid injection is simulated explicitly and it is considered that the fluid injection rate increases linearly during the first minute, up to $5 \times 10^{-7} \text{ m}^3/\text{s}$ and then remains constant throughout the fracturing operation (Figure 5.17).

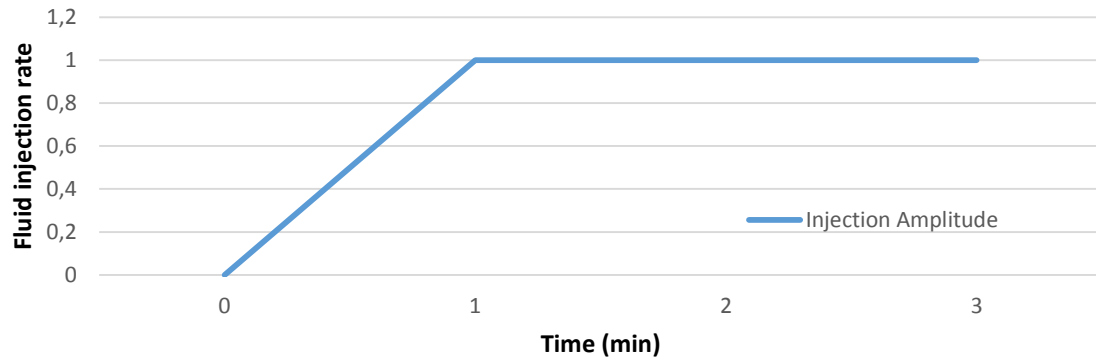


Figure 5.17 - Fluid Injection amplitude through time

The use of an increasing injection flow in the first minute aims to increase the simulation stability and avoid excessive and instantaneous pressure increments that could cause rock failure in conditions different from those expected.

During the injection and fracturing process, the typical fracture pressure profile is similar to that shown in Figure 5.18; the pressure increases until it reaches a peak that corresponds to the breakdown pressure, after which there is a pressure relief in the fracture as a result of its propagation until eventually it stabilizes (extension pressure) remaining constant until the end of the injection process (shut-in). The remaining stages correspond to the closure of the fracture in the long term as a result of overburden stresses, leading to screen outs, possibly only prevented with a re-injection operation (Soliman M. Y. et Boonen P., 2000).

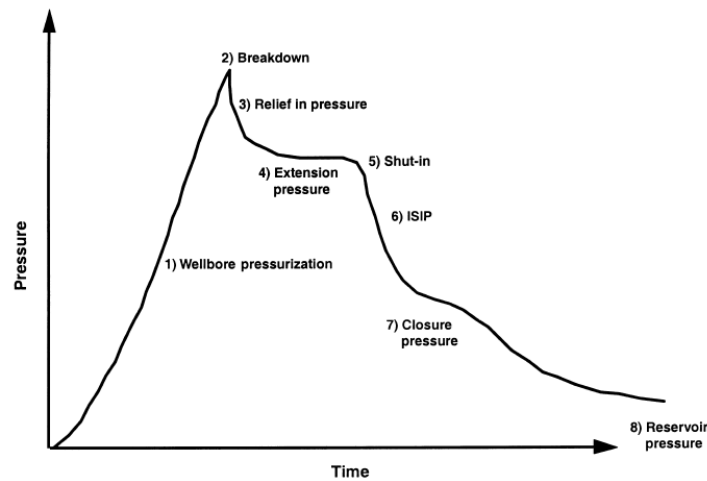


Figure 5.18- Typical fracture pressure profile during and post-injection (Soliman M. Y. et Boonen P., 2000)

From a real point of view, the injection causes the application of a pressure P_w on all the exposed surface of the wellbore. However, one of the XFEM limitations is the fact that flow injection occurs at the fracture level, through the use of phantom nodes concept.

Comparing with the analytical models presented in section 3.4.3, the assumption that the fluid is injected at fracture level is valid since the leak-off into the surrounding fluid is null or

negligible. Since in the numerical analyses it is considered a minimal fluid leak-off fluid at the fracture contact surface, this simplification is consistent.

In agreement with the information in Abass H. et al. (1994) an initial pore pressure equal to zero was adopted and a zero pressure boundary condition was prescribed on all the outer boundaries.

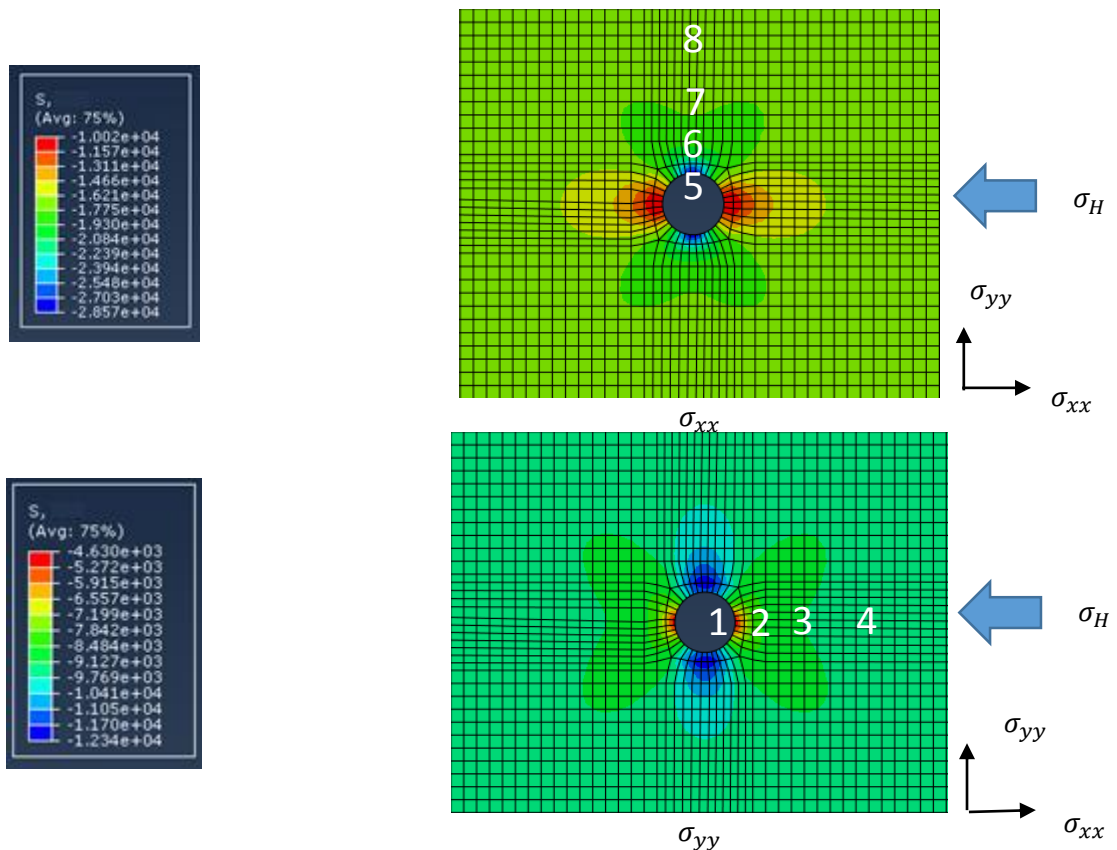
5.3.4 Results and discussion

Geostatic – Near-wellbore stresses equilibrium

According to the formulation of the software, due to the existence of an opening for the wellbore in the center of the sample and the applied boundary conditions, it is necessary to introduce an analysis step (before the simulation of hydraulic fracturing operations) to enable the stresses within the analysis domain to reach equilibrium, with the prescribed geometry, applied loads and boundary conditions. This step is referred to as a geostatic step.

Figure 5.19 shows the distribution of the total horizontal stresses and pore water pressure after the excavation of the wellbore.

Assuming that a vertical wellbore is drilled in a linearly elastic semi-infinite, homogenous and isotropic medium, the stress state around a wellbore can be expressed by equation 3.11 and 3.12 (Jaeger J. C. et Cook N. G. W., 1971).



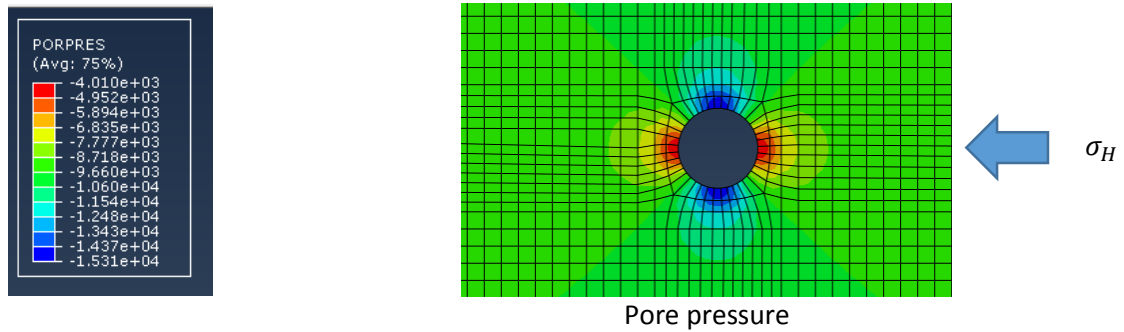


Figure 5.19 - Stress initialization due to wellbore excavation

When analysing Figure 5.19 is important to bear in mind that Abaqus convention is that positive stresses correspond to tension and negative to compression, but the opposite applies to pore pressure, such that negative pore pressures correspond to suctions. Despite the fact that at the start of the analysis a zero pore pressure and zero saturation has been specified to all the analysis domain, at the start of the geostatic step the software assumes 100% saturation. Given that a small time step has been assigned to the geostatic step and the sample has a very small permeability, the analysis predict an undrained response to the formation of the wellbore and negative pore pressures (suctions) are predicted. The fluid injection is modelled immediately after this geostatic step. It is acknowledge that stress conditions at the start of the fluid injection predicted by the numerical analysis differ significantly from those existing in the laboratory tests carried out by Abass et al. (1994).

Table 5.4 compares the value of the total tangential stress obtained in the numerical analyses at the end of the geostatic step at a set of points (see figure 5.19), with the values obtained through the application of Equation 3.12.

Table 5.4 - Comparison between measured initial tangential stresses between analytical and numerical solutions

Point	$\sigma_{numerical}$	$\sigma_{analitical}$	error(%)
1	-4,63	-11,71	60,5
2	-7,79	-13,62	42,8
3	-8,864	-12,3	28,0
4	-9,46	-10,4	9,0
5	-28,57	-42,07	32,1
6	-23,45	-29,4	20,2
7	-18,03	-19,89	9,4
8	-17,56	-17,8	1,3

It is found that the difference between the tangential stresses obtained in the numerical analyses and those given by equation 4.12, reduces with the distance to the wellbore, presenting an average error of 25,4% for this set of points. However some studies (Tie Y. et al, 2011) have found that the analytical expression can lead to errors on the order of 25% when these are applied to laboratory test samples with non-zero pore pressures. Another reason for the

difference may be related with the influence area of the analytical equation, usually defined for distances 2-3 times de wellbore radius (Sepehri J., 2014).

It appears that towards the preferred fracture plane (Points 1 and 2) the tangential stresses near the wellbore are underestimated, as well as in the direction of σ_h (Points 5 and 6).

Figure 5.20 shows the distribution of the pore pressure, after reequilibrium of the pore pressures with the beginning of the fluid injection, showing that these stabilize at values around 8-10 MPa. Notice this equilibrium condition is independent of the geostatic step.

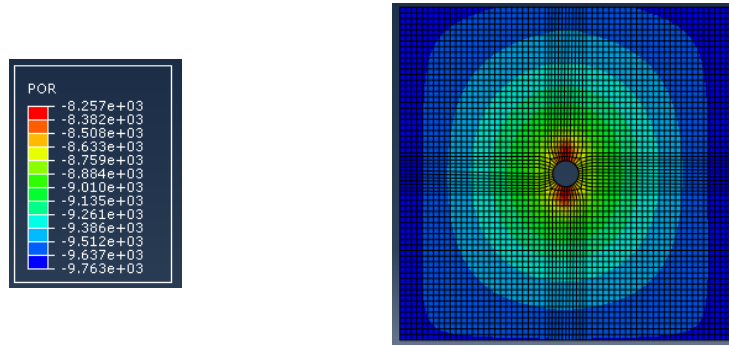


Figure 5.20 - Pore pressure distribution in the sample with the start of fluid injection

Breakdown pressure without perforation

The XFEM functionality enables the simulation of fracture initiation and propagation without having to initially set its location. For the analysis without perforation, fracture initiated in the direction of the maximum horizontal stress (i.e. the PFP) associated with an injected flow at the element edge of $2,5e^{-7} m^3/s$ and a breakdown pressure is obtained from the output PORPRES and equal to 13,58 MPa. Theoretically that value should correspond to:

$$P_{Bk} = \sigma'_{00} + \sigma_T \quad (5.9)$$

Where P_{Bk} is the breakdown pressure, σ'_{00} is the effective tangential stress in the near-wellbore expected fracture location and σ_T is the rock tensile strength.

Computing the values using the various output available, the following value is obtained:

$$P_{Bk} = 7,033 + 5,560 = 12,593 \text{ Mpa} \quad (5.10)$$

The value 7,033 MPa is a tangential stress and was obtained from the numerical analysis at the rock breakdown instant. It is thought that this difference of about 8% is related to the friction effect on the fracture contact surface, as well as some difficulties in the numerical calculation of the stresses around the fracture tip. According to Equation 5.10 the breakdown pressure depends only of the tangential stresses at the fracture tip and the tensile strength of the material.

Assuming the validity of equations 3.11 and 3.12 to describe the effective stresses around the wellbore region, and invoking the material tensile failure criteria, the breakdown pressure is then:

$$P_{Bk} = 3\sigma_h - \sigma_H - P_p + \sigma_T \quad (5.11)$$

Where σ_h is the minimum horizontal stress, σ_H is the maximum horizontal stress and P_p is the pore pressure in that location. Based on the integration of the numerical (pore pressure) and theoretical values (stresses and tensile strength), the following expression is achieved:

$$P_{Bk} = 3 \times 9,65 - 17,24 - (-1,7) + 5,56 = 18,97 \text{ Mpa} \quad (5.12)$$

A 28,4% difference in relation to the value obtained in the numerical analysis is obtained. This difference may be related to the re-equilibrium of the near-wellbore stress made on geostatic step, since it was found that in the direction of the PFP, the stress values were underestimated.

It is verified that in this situation the injected flow rate to cause tensile failure of the sample without perforation is 3187% the flow rate required to cause the failure when considering the existence of a perforation in the same direction. This is extremely important when designing an operation to minimize costs and ensure the project feasibility.

Breakdown pressure with oriented perforations

The breakdown pressure is the pressure required to cause the tensile failure of the rock at the perforation tip. The breakdown pressure was calculated for the simulated perforations in the angles $\theta = 0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ$ and 90° , as seen in table 5.5.

Table 5.5 - Breakdown pressure for direction 0-90° for studied model

Breakdown pressure (MPa)	
Direction	
0°	12,36
15°	15,85
30°	20,09
45°	24,77
60°	25,8
75°	30,44
90°	32,18

As expected, there is an increase in the breakdown pressure as θ increases, since for higher θ , near-wellbore tangential stresses are greater. It is also observed a slight reduction in breakdown pressure in the direction 0° compared with the situation without initial perforation.

The results obtained for the breakdown pressure are compared with the results obtained in laboratory (Abass H. et al, 1994), as seen in figure 5.21.

In directions lower than 45° , as result of the initial stress equilibrium, the model underestimates the breakdown pressure values, and in the direction 45° there is a perfect match between laboratory and numerical values which remain reasonably adjusted for the remaining directions. Notice that for directions 45° - 90° the average error between the obtained values is 7% and the breakdown pressure profiles are approximately parallel.

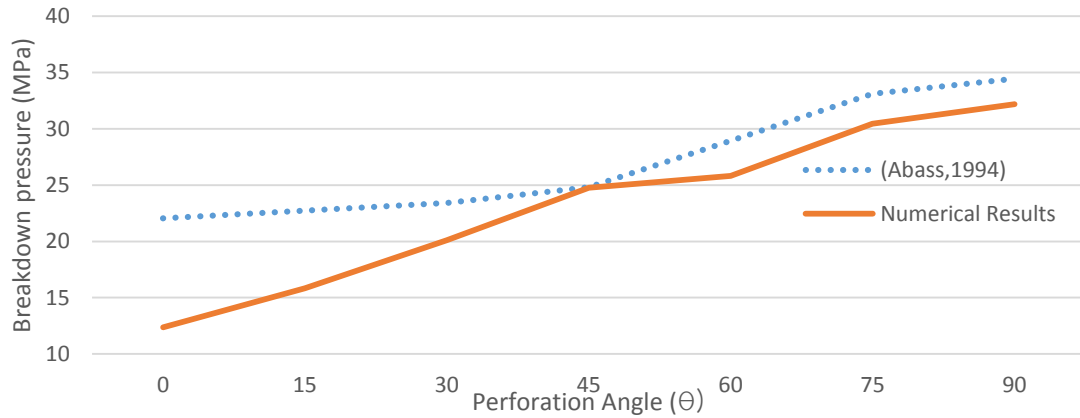


Figure 5.21 - Breakdown pressure comparison between (Abass, 1994) and the numerical simulation

Given that fracture initiation is mainly controlled by the tangential stresses in the crack tip region (Economides M. J. et al, 2000), to explain the differences in breakdown pressure shown in Figure 5.21, the evolution of the tangential stress with the fluid injection is analysed.

Figure 5.22 shows the tangential stresses at the crack tip at the end of the geostatic step following the stress re-equilibrium due to excavation of the wellbore and at the time of fracture initiation.

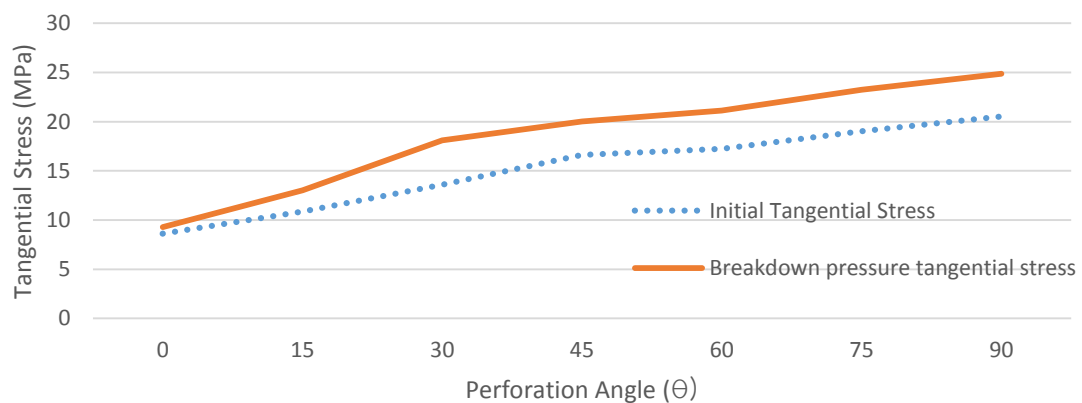


Figure 5.22 - Tangential stresses in the initial geostatic equilibrium and through the tensile failure in the crack tip

The difference between these two data sets should provide an estimate of the changes in tangential stresses at the crack tip caused by fluid injection. This effect is estimated at about 4 MPa for all perforation direction, except for 0° and 15° , where it is much smaller.

Values of tangential stresses at perforation tip at the moment of fracture initiation occurring in the laboratory samples are estimated, rewriting Equation 5.9 as:

$$\sigma'_{00} = P_{Bk} - \sigma_T \quad (5.13)$$

Figure 5.23 compared the values of tangential stresses at the crack tip obtained by application of Equation 5.13 to the laboratory testing data with the numerical simulation results.

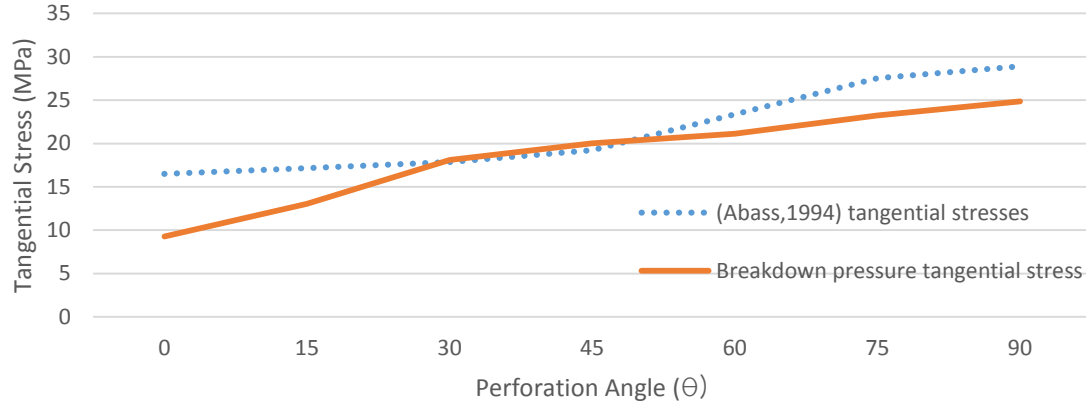


Figure 5.23 - Tangential stresses in crack tip for numerical and laboratorial results

There is still a difference between the Abass et al. (1994) experiments and the simulations results. This difference, whose cause is not known, may be related to the friction coefficients and leak-off coefficients used in the model.

Fracture reorientation

Fracture tends to propagate in a plane of least resistance, which is usually called preferred fracture plane (PFP) and is perpendicular to the minimum horizontal stress. If the perforation orientation is out of the direction of PFP, induced fracture may initiate along the perforation and as it propagates away from the near wellbore toward the unaltered in-situ state of stress, will reorient itself to be perpendicular to the minimum horizontal stress.

A visual comparison between Abass H. et al. (1994) results and the numerical simulations is shown in Figures 5.24.

in section 5.2, Abaqus only allows the examination of stress intensity factors when simulating stationary fractures.

The analysis can be made based on the stress anisotropy, with the fracture to reorient to the direction of greatest principal stress (perpendicular to the direction of least principal stress), if the stress anisotropy is enough to ensure fracture reorientation.

To analyse in a quantitative manner fracture reorientation, the fracture was digitalized using software Data GetGraph Digitalizer, and the data treated to obtain the variation of the fracture direction along its length (Figure 5.25).

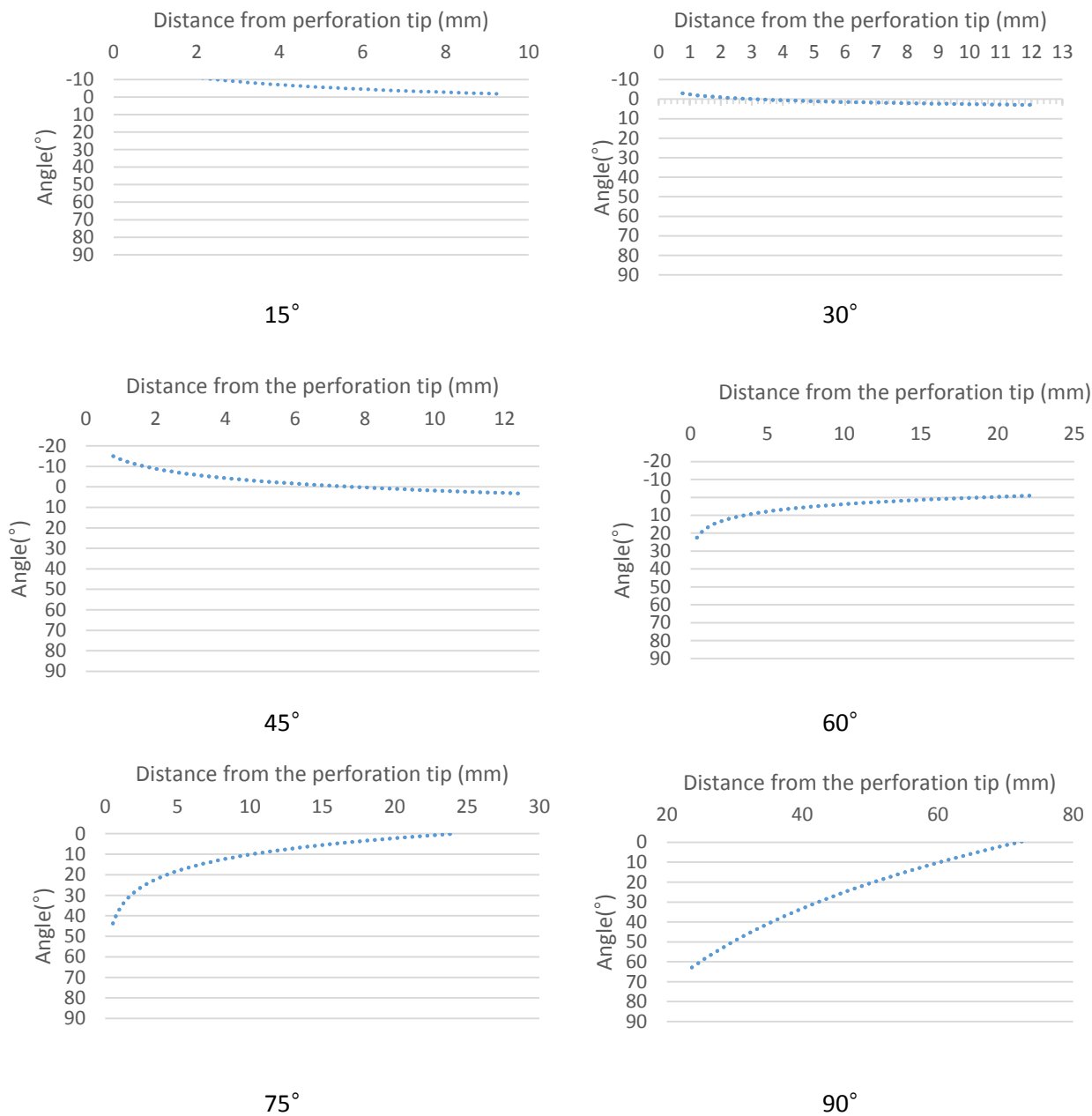


Figure 5.25 - Fracture reorientation for all perforation directions

In the directions 15° to 45° there is the so-called pinch effect, which makes the fracture initial reorientation to be negative. In the other directions this effect is not felt, as studied in more detail later in this chapter.

Another way to analyze fracture reorientation is using the concept of reorientation radius, which is the distance from the wellbore to the point where fracture is completely aligned with PFP, as shown graphically in Figure 5.26.

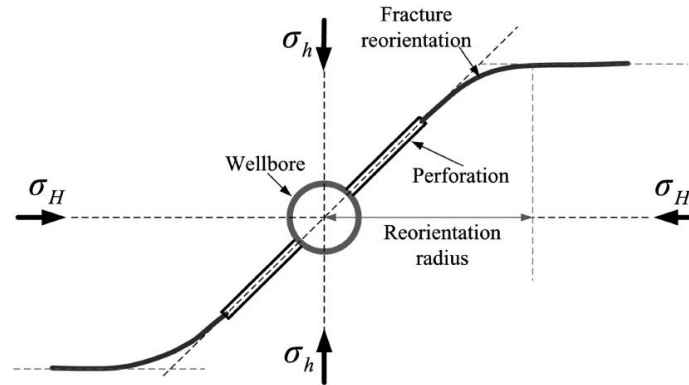


Figure 5.26 - Schematic representation of reorientation radius (Chen M. et al, 2010)

Table 5.6 shows the reorientation radius for different perforation angles. This distance should be considered with caution as a result of the pinch effect which occurs in directions 15° - 45° .

Table 5.6 - Reorientation radius for different perforation angle

Angle	Reorientation radius
15	$2,62r_w$
30	$3,07r_w$
45	$2,86r_w$
60	$3,28r_w$
75	$4,13r_w$
90	$6,27r_w$

Based on the results presented in the Figure 5.24 and Table 5.6 it appears that fracture reorientation towards the PFP is achieved in all directions. The larger the perforation angle, the greater the reorientation angle and thus the distance required for the fracture to re-orientate. The slight reduction in the reorientation radius from perforation angle 30° to 45° is related with pressure peaks at the fracture tip immediately after fracture initiation, which generates an instantaneous stress anisotropy, giving rise to a larger reorientation than expected. According to Abass H. et al. (1994), during injection is verified the existence of peak pressures at the fracture tip, which goes with the numerical results. However this effect may be due to the instability of XFEM associated with mesh dependency in directions near 45° .

In the directions 15° to 45° there is the so-called pinch effect, which makes fracture reorientation analysis more difficult. According to Zhang X. et al. (2011) and Cherny S. et al. (2009) the pinch effect is a consequence of the adopted friction coefficient, which is an input parameter of the analysis. This effect can cause a slight increase in the reorientation radius and difficulties related with the fluid injection and subsequent loss of productive capacity as a consequence of the fracture width reduction. This effect will be studied later in this chapter in more detail.

As said before, fracture reorientation seems to be dictated by the stress anisotropy at the crack tip. The tendency for the fracture to propagate toward the initial perforation direction increases as the anisotropy ratio tends to 1 (Solyman M. Y. et Boonen P., 1999).

Therefore the evolution of the anisotropy at the crack tip while fracture propagates and re-orientates has been analysed, and is shown in Figure 5.27 and table 5.7, for perforation angle equal to 90° . Notice that herein the anisotropy ratio is the ratio between the vertical (yy) stresses and the horizontal (xx) stresses at the fracture tip.

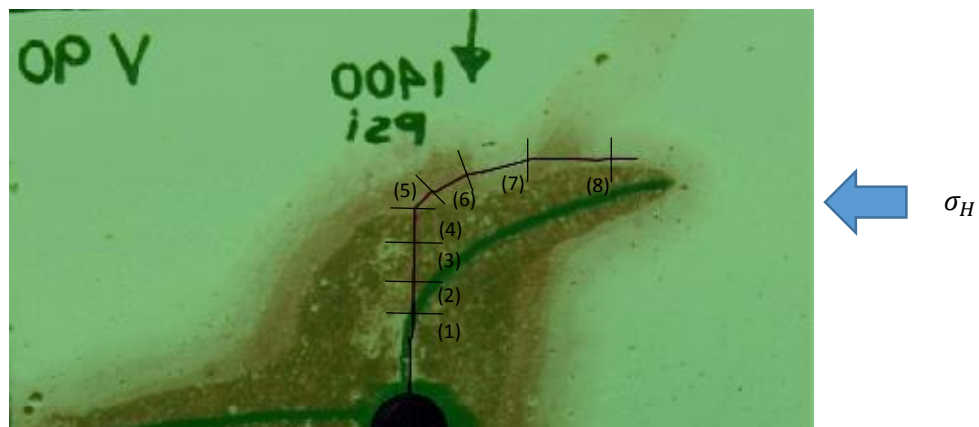


Figure 5.27 - Stress Anisotropy ratios for different moments in the crack propagation for 90° perforation angle
Table 5.7 - Stress Anisotropy ratios for different moments in the crack propagation (at the element level) for 90° perforation angle

Points	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Anisotropy ratio (σ_{yy}/σ_{xx})	1,31	1,49	1,69	2,89	1,97	2,04	1,73	1,88

Initially fracture reorientation to the PFP does not occur, since the stress anisotropy ratios are just above 1, with a tendency to grow. After (4), where the stress anisotropy ratio is equal to 2,89 there is a clear shift in fracture orientation and then to a lesser degree following (5).

With the initiation of fracture reorientation there is a tendency for the fracture to maintain the previous propagation direction if there is no change in anisotropy. A further increase in the ratio (6) , allows the reorientation to the PFP direction in which eventually remain, since the subsequent local anisotropy ratios are not generating any reorientation in the fracture.

Abass H. et al. (1994) note that, in practice, the use of perforation angles above 30° - 45° is very complicated, as result of tortuosity introduced by the reorientation, which reduces the communication between the fluids and the fracture, with possible fracture width reductions and followed by fluids screen out. This effect is studied below.

Fracture Width

Fracture reorientation can cause tortuosity that can lead to reductions in fracture width for a certain applied pressure. The loss of communication between the perforation and fracture is defined based on a geometry factor defined as (Abass H. et al, 1994):

$$GF = \frac{A_{com}}{A_{per}} \quad (5.14)$$

Where A_{com} is the communication area between the fracture and the perforation tunnel and A_{per} is the perforation cross-sectional area.

The fracture width (W_f) is calculated as:

$$W_f = \frac{2D}{E} (P_f - \sigma_h) \quad (5.15)$$

Where D is half-length of the sample in a principal direction and E is the Young's modulus.

Differentiating Equation 5.15 with respect to time:

$$\frac{W_f}{dt} = \frac{2D}{E} \frac{d}{dt} (P_f - \sigma_h) \quad (5.16)$$

The width function is expressed as:

$$Width\ function = \frac{W_f}{dt} \frac{E}{2D} = \frac{d}{dt} (P_f - \sigma_h) \quad (5.17)$$

Analyzing the average opening of the fracture at the crack tip level (with the output PFOPEMXFEM) for the various directions for a range of different accumulated injected flows, is possible to verify that the fracture opening is similar for a injected flow of $3,75 \times 10^{-6} m^3$, as can be seen in table 5.8.

Table 5.8 - Fracture opening (mm) for different perforation direction to different injected flows

Direction	Injected flow = $3,75 \times 10^{-6} m^3$	Injected flow = $5 \times 10^{-6} m^3$
0	2,0395	2,0395
15	2,0340	2,0330
30	2,0359	2,0320
45	2,0350	2,0314
60	2,0437	2,0335
75	2,0417	2,0332
90	2,0362	2,0289

It is found that this initial opening to directions different of zero (PFP) is rapidly reduced to lower values of injected flow. Once this is the magnitude of the injected flow for fracture propagation, this effect shows the effect of the tortuosity for all directions less the non deviated perforation.

However, this effect should be more accentuated with the perforation direction deviation increase with the PFP. The openings in the directions 60 ° and 75 ° for the two analyses stages considered in Table 5.8, do not comply with the stated above. This can be related to pressure peaks due to numerical calculation difficulties, since the data does not follow the trend observed at smaller perforation angles.

According to Abass H. et al. (1994), it appears that for directions less than 45, there is a clear increase in fracture width with fluids injection. However, Abass H. et al. (1994) found that for directions above 45°, as fracture propagation progresses the fracture width function tends to zero, becoming even negative to 75-90° directions, which indicates fracture width reduction and possible consequent the fluid screen-out. This effect is resultant of fracture tortuosity.

Table 5.9 shows the values of the fluid pressure in fracture and σ_h , at 5 instances of the analysis procedure when admitting a perforation angle equal to 90°, in order to assess their effect in the fracture opening. Figure 5.28 shows the propagation of the fracture at the same instances.

Table 5.9 - Fracture opening rate (Width function) verification to direction 90 °

	Injected Flow - 10s	Injected Flow - 20s	Injected Flow - 23,99	Injected Flow - 31s	Injected Flow - 60s
P_f (MPa)	32,06	23,10	14,43	4,74	7,26
σ_h (MPa)	18,57	16,88	11,71	9,21	6,42
$P_f - \sigma_h$ (MPa)	13,49	6,22	2,72	-4,47	0,84
Fracture opening	2	2,0288	2,0288	2,0210	2,0943

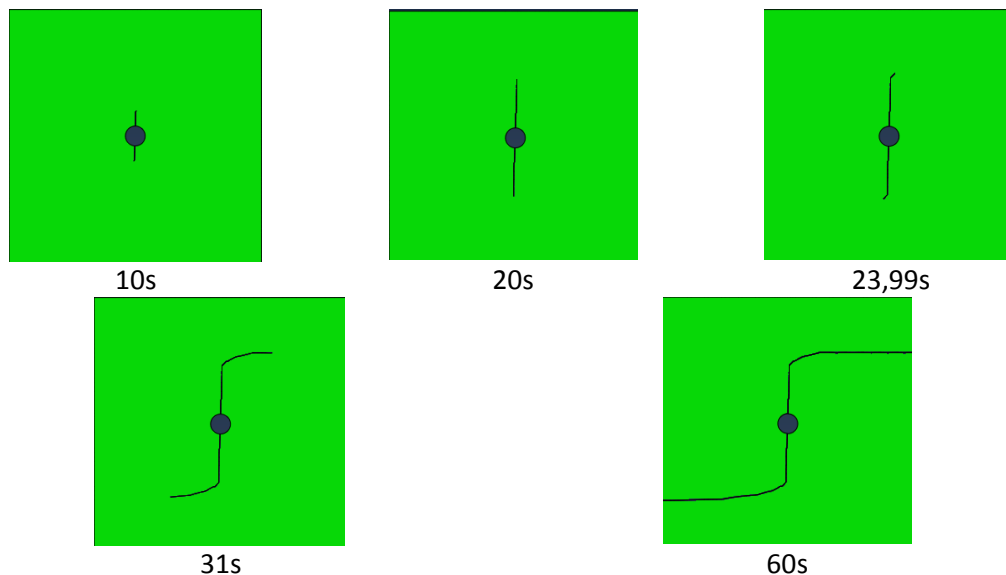


Figure 5.28 - Fracture propagation with fluid injection

It is found that initially and as a result of the existent stress anisotropy, the fracture propagates in the vertical direction, and an increase in the fracture opening is found. Since the tangential stresses in the surrounding of the fracture are very high, the extension of the fracture is small.

With the continued fluid injection, the fracture propagates and it reorientates towards the PFP. It is found, at this stage, a reduction of fracture opening, with the fracture function width to take even negative values, which can generate screenout effects. This decrease in fracture pressure may be related to the tortuosity and loss of communication between the perforation and the fracture, which hinders the access of the fluid to the crack tip zone (fluid lag effect).

Nevertheless it appears that with a major increase in the injected flow, there is a significant increase in fracture opening, either due to a reduction in the tortuosity effect or as a result of the reduction in tangential stresses and loss of fracture cohesive behaviour. The reduced permeability of the sample is one of the decisive factors that contribute to the increase in fracture width, since the fluids are located in the fracture and the amount of fluid lost (fluid leak-off) into the surrounding area is minor.

Fracture Pressure profile

Figure 5.29 shows the evolution of the fluid pressure in the fracture with time, obtained in the numerical simulations for a perforation angle equal to 0°.

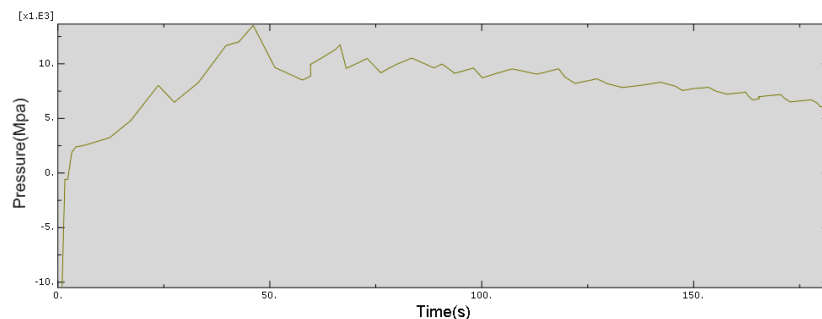


Figure 5.29 - Fracture Pressure profile for perforation near-crack tip for direction 0

Comparing Figure 5.29 with Figure 5.18 (that shows a typical pressure profile in the fracture during HF operations) it is found that the various stages are represented in the numerical curve.

Table 5.10 shows the average fracture pressure at equilibrium (i.e. the extension pressure) for the various perforation angles, obtained from an Abaqus output file

Table 5.10 - Equilibrium fracture pressure for different perforation direction after reorientation to the PFP

Direction	Without perforation	0	15	30	45	60	75	90
Equilibrium pressure (MPa)	5,09	4,73	5,31	5,46	6,08	6,41	6,43	7,30

It is found an increase in the equilibrium pressure for higher directions. This can be explained by the increase of the tangential stresses, as θ increases. It is found that the equilibrium pressure for the situation without initial perforation is similar to the direction 0° with perforation.

It should be noted that as indicated in Abass H. et al. (1994), some peak pressures are felt in the fracture during the injection phase. These can explain the pressure increments before fracture reorientation. In the numerical simulations the pressure peaks may as well result of numerical calculation difficulties instead.

The XFEM functionality is heavily dependent on mesh and the location of the fracture tip. The adopted procedures aimed to ensure that the results presented herein were not affected by these aspects or numerical instabilities. However their influence cannot be ruled out.

6. Validation - Parametric study

The studies described in the previous chapter, i.e., the simulation of a fracture toughness determination test and the laboratory tests described by Abass H. et al. (1994) on induced fractures using oriented perforations, showed that Abaqus XFEM software is able to analyze with confidence the mechanical behaviour of rocks, fracture initiation and propagation by fluid injection.

Based on the model set up used for the simulation of the laboratory tests described by Abass et al. (1994), in this chapter, it is presented a numerical study on the effect of a series of parameters on the outcome of hydraulic fracturing operations, in particular the breakdown pressure and fracture re-orientation. The parameters considered in this study are porosity, permeability, friction, anisotropy, fluid viscosity, fluid leak-off, flow rate, perforation phasing and phasing miss-alignment. For simplicity it will only be considered perforations oriented at 0°, 45° and 90° with the PFP.

6.1 Porosity

Porosity is one a rock physical properties and different porosities were studied: 13,25%; 6,625% and 3,3125% in addition to the base case of 27,7%. This effect is not documented in the literature.

It was found that the porosity does not affect the fracture initiation and propagation, since no changes were identified in the breakdown pressure, fracture reorientation profile or fracture width.

6.2 Permeability

Permeability is a rock physical property which affects the fluid displacement velocity within the sample. Different permeabilities were used: 50mD, 30 mD, 5mD, 2,5 mD and 0,5mD, in addition to the base case of 9,5 mD permeability.

The breakdown pressure for the different permeability values and perforation directions was calculated, and is presented in Figure 6.1.

It is possible to verify that with an increase in permeability, the breakdown pressure also increases. The increase in permeability increases the capacity of the fluid to move within the rock pore space, which increases the effective stresses around the perforation, and thus the tangential stresses.

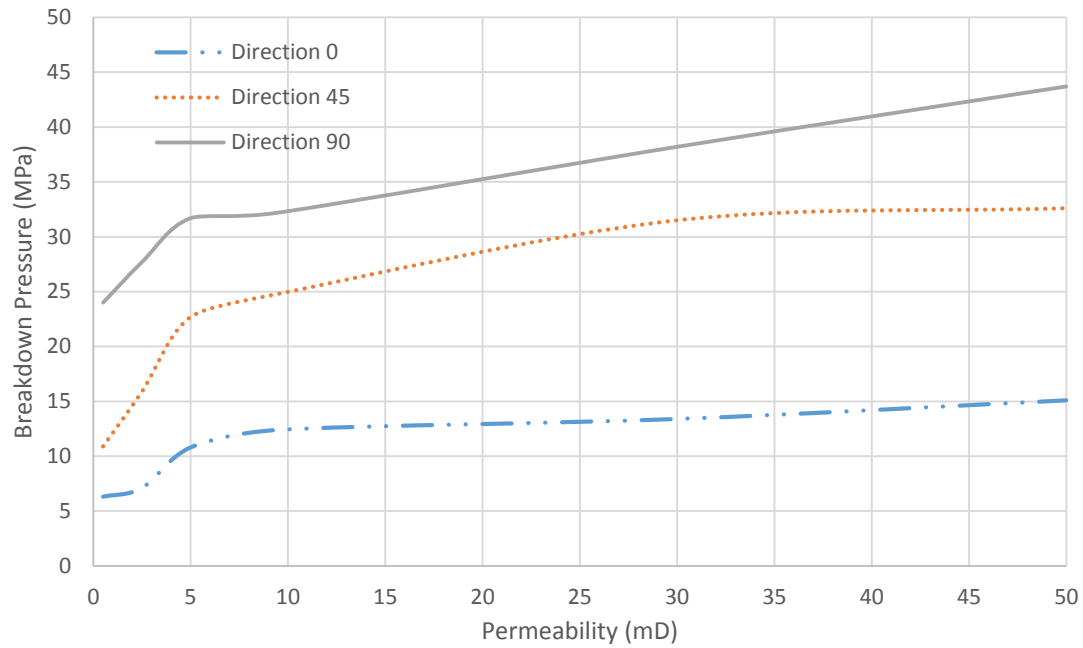


Figure 6.1 - Breakdown pressure for different directions and different permeabilities

The breakdown pressure variation with the permeability for the different directions follows the same evolution path, which indicates that the effect of the perforation direction has a low impact in the variation of the breakdown pressure when changing the rock permeability.

While the permeability increases, the injected flow that induces the breakdown pressure also increases for the different directions, as seen in figure 6.2. This was anticipated, once that for higher permeability it is easier for the fluid to move through the rock mass and it is necessary sufficient fluid material to sustain the fracture opening and avoid the screen out effect. For higher permeabilities the injected flow is similar, and follow the same trend of the breakdown pressure for high permeabilities (see Figure 6.1 and 6.2).

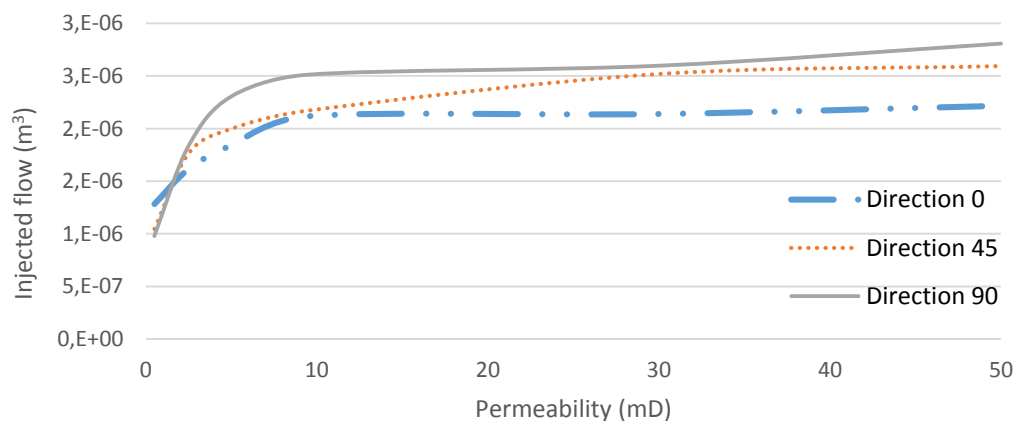


Figure 6.2 - Evolution of the injected flow with permeability for different perforation directions

The fracture reorientation path is also affected by the adopted permeability, as can be seen in figure 6.3 and figure 6.4.

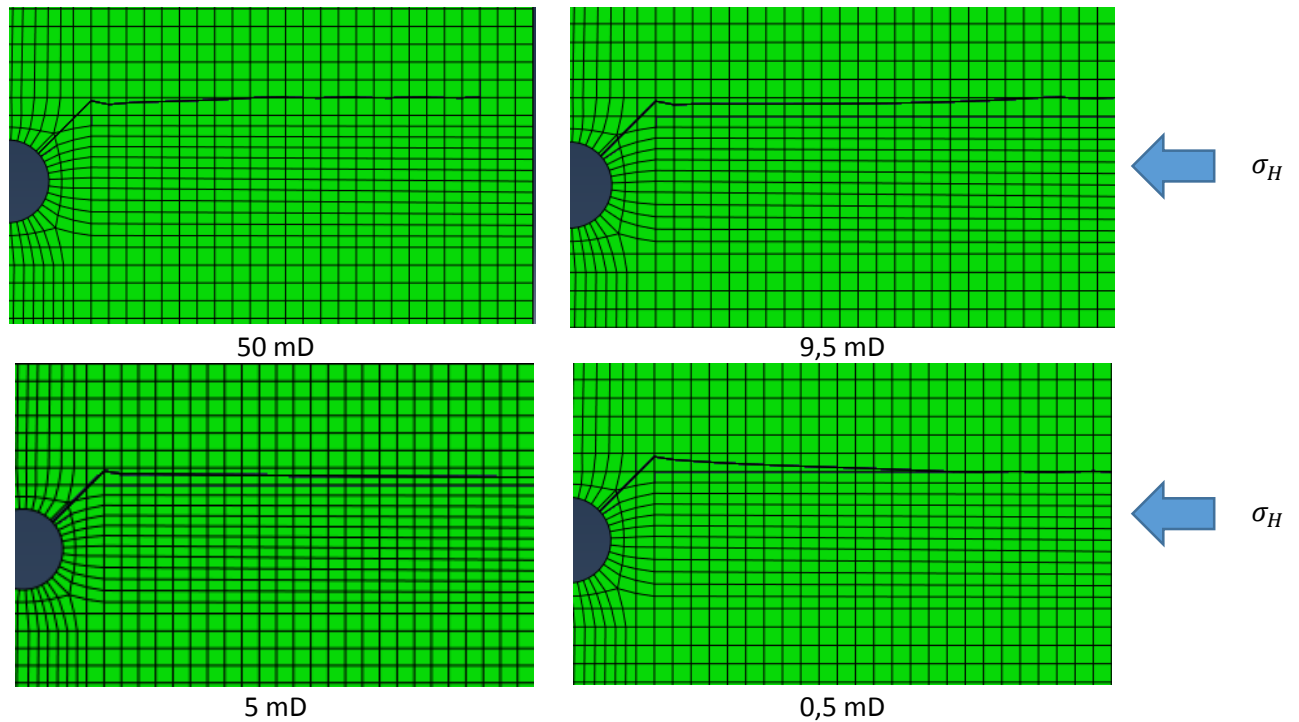


Figure 6.3 - Fracture propagation for different permeabilities to a perforation in direction 45

Figure 6.3 suggests that the pinch effect reduces with increasing permeability increases, due to the stress changes introduced by the fluid injection.

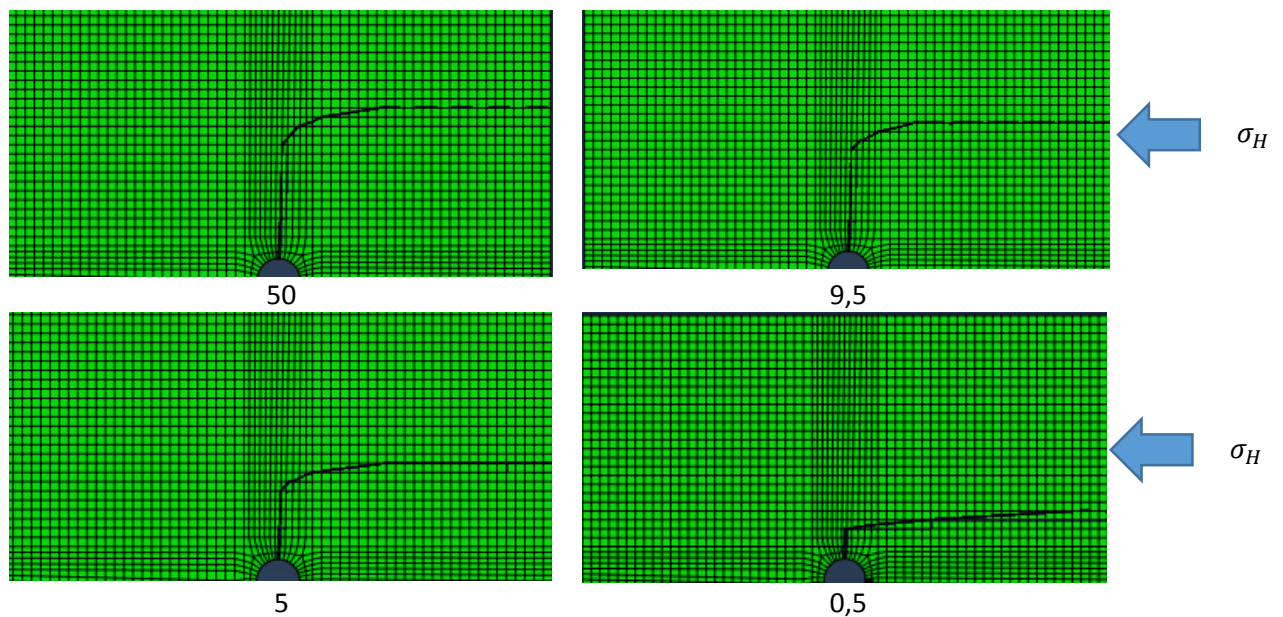


Figure 6.4 - Fracture propagation for different permeabilities to a perforation in direction 90

Although there is a tendency of the complete reorientation to the PFP to happen for bigger distances (i.e higher reorientation radius), the perforation in direction 90° reorientation to the PFP happens earlier (in terms of Cartesian distance) as lower the permeability.

The flow injected to achieve complete reorientation of the fracture to the PFP is found to increase with permeability, as seen in figure 6.5.

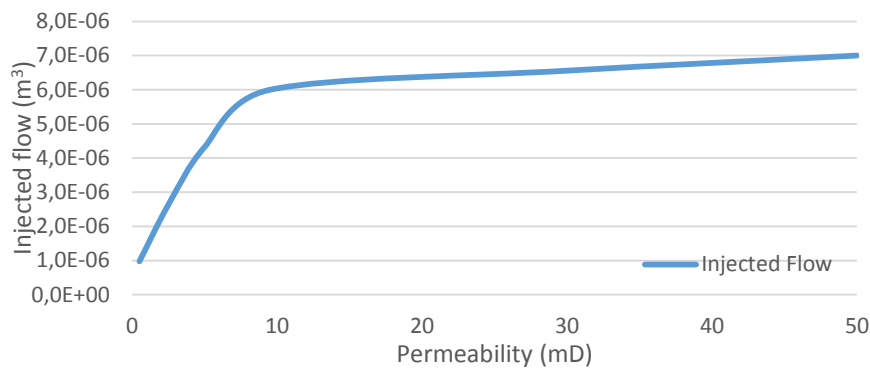


Figure 6.5 - Injected flow for the complete reorientation to the PFP of 90 ° perforations

The reduction in fluid injection to achieve complete fracture reorientation for lower permeabilities has consequences on the evolution of fracture width.

For lower permeabilities, the fluid displacement (i.e. percolation) is lower to the surrounding media and the fluid within the fracture is higher (due to less fluid loss) inducing greater fracture openings that in a high permeability rock, increasing the productivity of the fracture and the economic feasibility of the operation.

6.3 Friction coefficient

The fracture surface friction coefficient influences the fracture reorientation path, and the so-called pinch effect is a direct consequence of the friction coefficient. More information regarding the principles and effects of the friction coefficient in the fracture reorientation can be found in the literature (Zhang X. et al, 2011). Different coefficients were used: 0,000001; 0,0001 (base case); 0,01; 0,1.

This effect may be felt in all the perforation direction, however, due to the higher tangential stresses in perforations 60° or more from the PFP, the effect of the friction coefficient becomes residual when compared with the in-situ stress state (Zhang X. et al, 2011).

The direction of the model that shows more clearly the pinch effect is the direction 45°, for which the influence of the fracture surface friction on the fracture propagation is shown in Figure 6.6.

The pinch effect is noticed for friction coefficients lower than 0,1 (case d). Higher friction coefficients were not studied because the software was providing numerically unstable and having difficulties in reaching convergence.

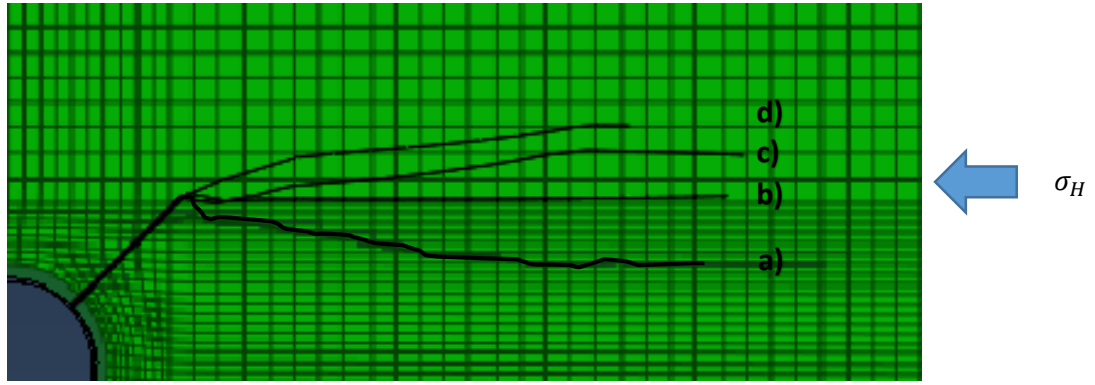


Figure 6.6 - Fracture reorientation for different friction coefficients: a) 0,000001 ; b) 0,0001 ; c) 0,001 ; d) 0,1 to a perforation 45°

The breakdown pressure is not affected by the friction coefficient.

6.4 Stress Anisotropy

The stress anisotropy is one of the most important factors in the design of an hydraulic fracturing operation. The stress anisotropy is often quantified in terms of the ratio of the principal horizontal stresses and Table 6.1 presents the studied options. In all the scenarios considered the value of σ_H remained constant and equal to 17240 KPa.

Table 6.1 - Stress anisotropies

(KPa)	$\sigma_H/\sigma_h = 1,79$	$\sigma_H/\sigma_h = 2$	$\sigma_H/\sigma_h = 3$	$\sigma_H/\sigma_h = 4$	$\sigma_H/\sigma_h = 5$
σ_h	9650	8620	5747	4310	3448

Stress ratios greater than 5 are not realistic and therefore were not considered in this study. Table 6.2 shows the breakdown pressure for all the situations considered.

Table 6.2 - Breakdown pressure for different stress ration for perforation directions 0°, 45° and 90°

	Direction 0	Direction 45	Direction 90
$\sigma_H/\sigma_h = 1,79$	12,36	24,77	32,18
$\sigma_H/\sigma_h = 2$	11,18	22,34	36,26
$\sigma_H/\sigma_h = 3$	7,46	17,77	39,13
$\sigma_H/\sigma_h = 4$	6,54	-	40,75
$\sigma_H/\sigma_h = 5$	3,64	-	40,86

It is verified that in the 0° direction there is a clear reduction in the breakdown pressure with increasing anisotropy. In the direction 45° the reduction is less significant, but still happens, contrary to what happens in the direction 90°, where the breakdown pressure increases.

According to equation 3.13, which defines breakdown pressure for the case when there is no perforation, the breakdown pressure is a function of the principal stresses and the material tensile strength only.

For perforation direction 0° , a reduction in the breakdown pressure would be expected, because σ_h controls the tangential stresses generated in the direction normal to the perforation direction, and subsequently the reduction in the breakdown pressure.

Although, if this result is purely compared with the analytical expression (equation 3.13), the result is coherent, because the near-wellbore stress re-equilibrium for high stress anisotropy requires even negative breakdown pressures for direction 0° , as seen in table 6.3. This negative value means that the stresses generated in the near-wellbore region are tensile stresses.

Table 6.3 - Analytical breakdown pressure values for different stress ratios to direction 0° and 90° (without perforation) – Based on equation 3.12

(MPa)	$\sigma_H/\sigma_h = 1,79$	$\sigma_H/\sigma_h = 2$	$\sigma_H/\sigma_h = 3$	$\sigma_H/\sigma_h = 4$	$\sigma_H/\sigma_h = 5$
Direction 90	47,63	48,66	51,53	52,97	53,83
Direction 0	17,27	14,18	4,74	1,25	-1,34

For perforation direction 90° , is verified an increase in the breakdown pressure for higher stress anisotropy ratios. This effect is also verified in the experiments of (Chen M. et al, 2010) and in table 6.3 results. This effect is a consequence of the different stress re-equilibrium in the near-wellbore region for different stress anisotropy ratios. A different stress state generate the higher or lower tangential stresses at the perforation tip, and consequently different breakdown pressures, even with a constant maximum principal stress.

In the direction 45° there is an intermediate change between the two cases explained; this is because it is not a principal stress direction, resulting in an intermediate solution. Notice however that for the three studied directions, the injected fluid at fracture initiation (at breakdown pressure) decreases for higher stress anisotropy ratios, as can be seen in figure 6.7.

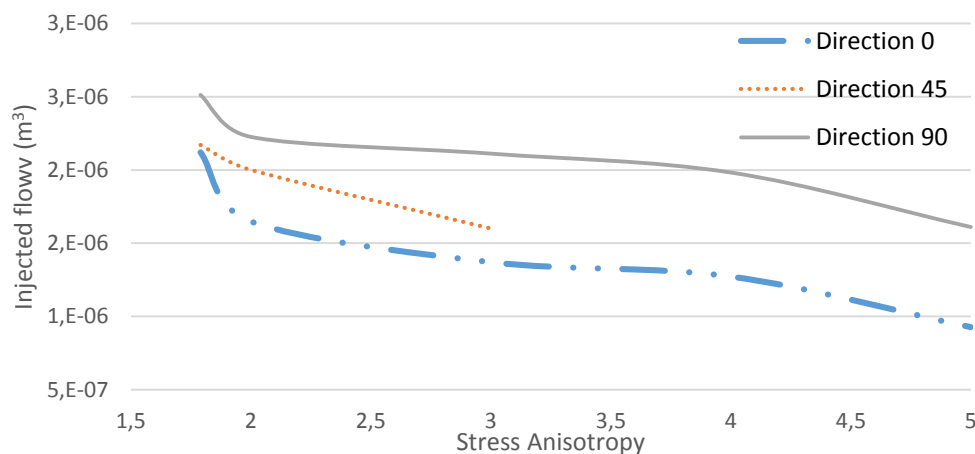


Figure 6.7 - Injected flow to cause the rock tensile failure for different directions and anisotropy ratios

The stress anisotropy has an effect in fracture reorientation, as seen in Figure 6.8 and 6.9.

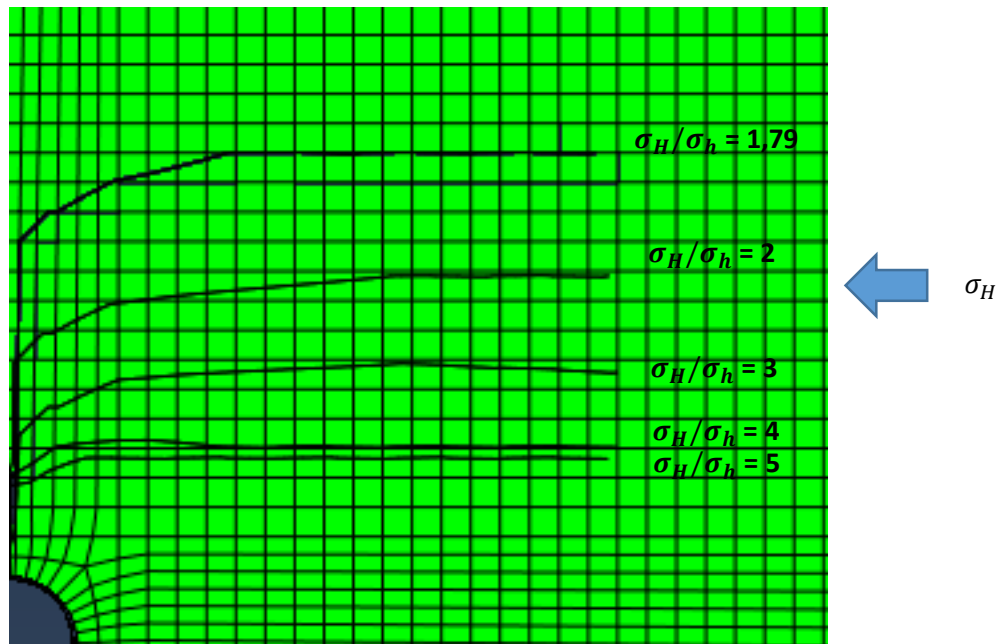


Figure 6.8 - Fracture propagation for different stress ratio in direction 90 °

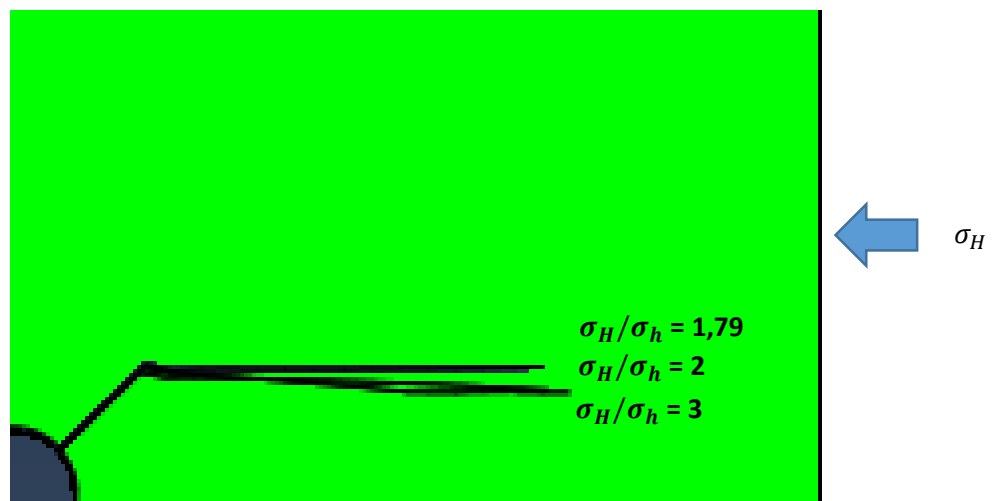


Figure 6.9 - Fracture propagation for different stress ratio in direction 45 °

The effect of the reorientation with the stress anisotropy is more apparent in the 90° direction than in 45°. This occurs because the stress equilibrium in the proximity of the fracture in 90° direction ensures the existence of significant reductions in normal stresses, allowing the fracture to reorientate toward PFP, i.e. the fracture tip stress state for direction 90° is more influenced than the direction 45°, once in this is not a principal direction

In direction 45°, a small increase in the pinch effect is felt at high anisotropy ratios. Notice that for higher stress anisotropies the simulations for the 45° direction perforations were not completed due to numerical difficulties.

One of the premises of fracture propagation is the necessity of a relatively high stress anisotropy to induce the fracture reorientation from the initial perforation direction towards the PFP. A ratio of one was used to simulate the fracture behaviour for direction 45° and 90° , as seen in figure 6.10. This simulation aims to control if Abaqus is simulating well the fracture reorientation expected behaviour if the principal stresses directions were changed.

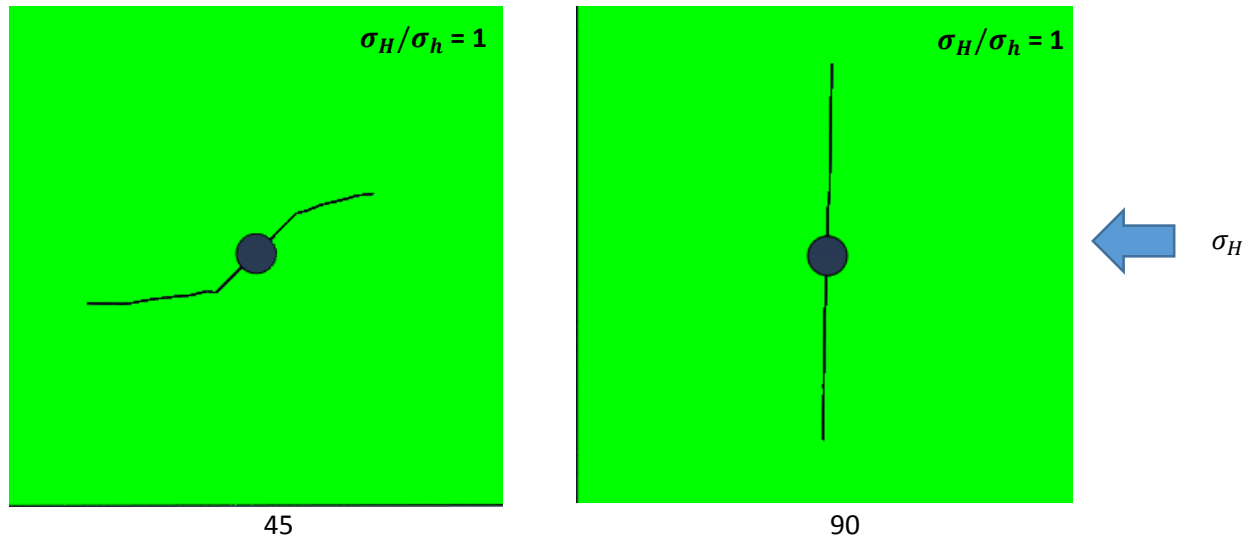


Figure 6.10 - Fracture reorientation for 45 and 90 perforations with a stress anisotropy ratio = 1

For direction 90° , and according with the previously studies, for this anisotropy ratio the fracture is unable to reorientate. For direction 45° the reorientation towards the PFP happens, however the reorientation radius increases when compared with a high anisotropy ratio. The pinch effect in direction 45° is minorly felt for this anisotropy ratio. The differences found for a ratio of one in direction 45 are related with stress changes induced by the fluid injection.

Also to verify if the model is simulating correctly the fracture propagation behaviour, a anisotropy ratio of 0,5 was simulated, to control if the fracture is able to reorientate towards the new PFP (vertical direction), as can be seen in figure 6.11.

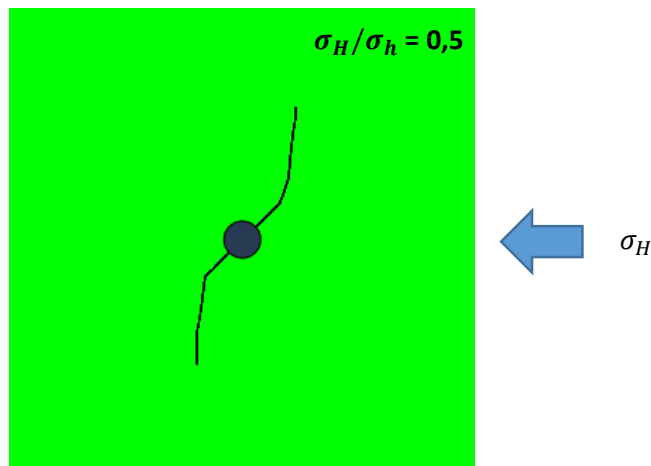


Figure 6.11 - Fracture reorientation for a 45° perforation with a stress anisotropy ratio = 0,5

The fracture reorientates to the new PFP. These results (figure 6.10 and 6.11) shows the consistency of the fracture modelling with the anisotropy.

6.5 Fluid viscosity

The fluid viscosity is one of the parameters controlling the fracture initiation and propagation patterns ((Zielonka M. G. et al, 2014), (Chen M. et al, 2010) and (Guo T. et al, 2014)).

In experimental laboratory studies, usually, the fracturing fluid viscosity is very high, in order to reduce the influence of the toughness (factor that controls the cohesive behaviour of the rock) and to compensate for the low injection rates that can be employed (Chen M. et al, 2010).

Fluid viscosity is an important variable of the common energy dissipation mechanisms in the numerical modelling. The energy dissipation in a fracture modelling problem can be conceptualized in a two-dimensional diagram, while different regimes controls the energy dissipation, as seen in figure 6.12. The parameter storage is a measure of the fluid that stays within the fracture (fluid accumulation), by the opposite of the parameter leak-off (fluid loss for the surrounding media).

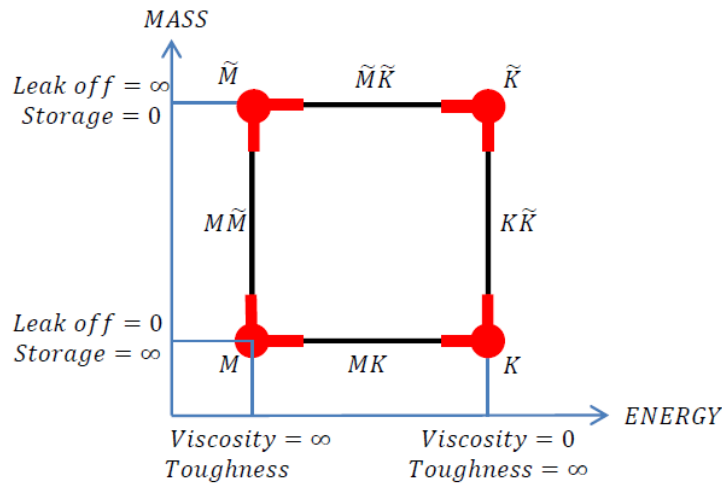


Figure 6.12 - Parametric diagram representing the four limiting propagation regimes of hydraulically induced fractures (Zielonka M. G. et al, 2014)

As the fluid leak-off considered in this study is very low, and the fluid viscosity is very high, the work is being carried under the near-M regime (viscosity and storage dominated propagation regime).

Based on the above, it is expected that the viscosity has a significant influence in fracture initiation and propagation behaviour, and the following fluid viscosity values were considered: 787 cP, 590 cP, 393 cP, 236 cP 118 cP and 1cP.

The analyses for the various perforation directions suggest that the fluid viscosity does not affect fracture initiation (breakdown pressure) and propagation (fracture reorientation). However, this result needs to be taken with caution.

The injection rates employed in this study (and in Abass et al. 1994) are very high, when compared with the usual injection rates for small-scale laboratory models ((Tie Y. et al, 2011) and (Chen M. et al, 2010)). This reduces the effect of the viscosity in the fracture initiation and propagation, as fracture behaviour becomes controlled by the injection rate.

This effect is emphasized by the fact that fracturing is being modelled using 2D plane strain analyses, and in this situation the ratio of fracture volume and injected volume is even lower than used in the Abass H. et al. (1994) experiments.

To determine if the fluid viscosity has in fact any influence in fracturing in small scale models two limiting situations were analysed: a situation of very high viscosity (1^6 cP) and very low viscosity (0,000001 cP). The case of very low viscosity did not predict fracture initiation. The fracture propagation for the case of very high viscosity is shown in Figure 6.13.

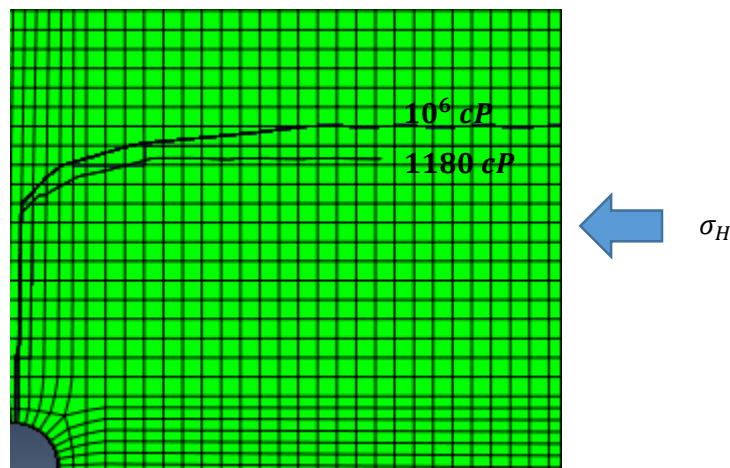


Figure 6.13 - Fracture reorientation for a direction 90° perforation for different viscosity.

When comparing the fracture propagation for the cases of viscosity equal to 1180 cP and 1×10^6 cP, it is found that fracture reorientation happen for similar injected flows, however the reorientation to the PFP happen first for the case of 1180 cP viscosity. The fracture width profile is different for both situation, with a 500% increase in the fracture width for a 1^6 cP viscosity (2,11 mm) when compared with a 1180 cP viscosity (2,022 mm) for a injected flow of $7,5 \times 10^{-6} m^3$. This was expected because of the higher induced pressures on the fracture surfaces by the high viscosity fluids. The increase in fracture width is extremely important to enhance the fracture productivity and the operation optimization.

The breakdown pressure increases from 32,18 MPa to 36,63 MPa, when the viscosity increases from 1180 cP to 1×10^6 cP, due to the low fluid displacement velocities, which increase the fluid storage within the fracture.

It can be concluded that the initial numerical simulations were unable to predict the influence of the fluid viscosity in the fracture initiation and propagation due to the high injection rates, and for this reason, a large scale model (both experimental and numerical) should be set to control the influence of this parameter (reducing the flow rate influence due to a larger relation between fracture volume/injected volume).

6.6 Fluid leak-off

A base fluid leak-off coefficient equal to 1×10^{-14} KPa.s was used in order to model the complete fracture behavior, since this coefficient is essential to compute the normal fracturing fluid velocity within the fracture, as seen in figure 6.14.

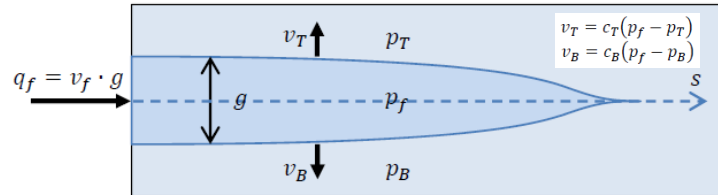


Figure 6.14 - Fluid leak-off coefficients (c_T and c_B) to the fracture computation, where v_T and v_B are the top and bottom fluid displacement velocities and p_f , p_B , and p_T are the fracture, bottom and top pressures respectively (Zielonka M. G. et al, 2014)

Based on this concept it is expected that when increasing the fluid leak-off coefficient, the fracturing fluid velocity is going to increase and induce stress changes in the near fracture region.

As said before, the value initially adopted for the fluid leak-off coefficient was introduced to account for its effect, but not to introduce major changes in the obtained results. Higher fluid leak-off coefficients are now studied: 1×10^{-10} ; 1×10^{-7} and 1×10^{-4} KPa.s. According with (Dassault Systèmes, 2013), typical value of the fluid leak-off coefficient are superior to 1×10^{-10} KPa.s.

The influence of the fracture leak-off coefficient was analysed for a perforation direction equal to 0° only, and the results in terms of breakdown pressure are shown in table 6.4. These results show that the breakdown pressure increases for smaller values of leak-off coefficient.

Table 6.4 – Breakdown pressure for different leak-off coefficients for perforation direction 0° .

1^{-4}	1^{-7}	1^{-10}	1^{-14}
7,6	8,0	10,43	12,36

However, the total volume of fluid trapped within the fracture reduces with the increase of the fluid leak-off coefficient, which influence the fracture width, as seen in figure 6.15.

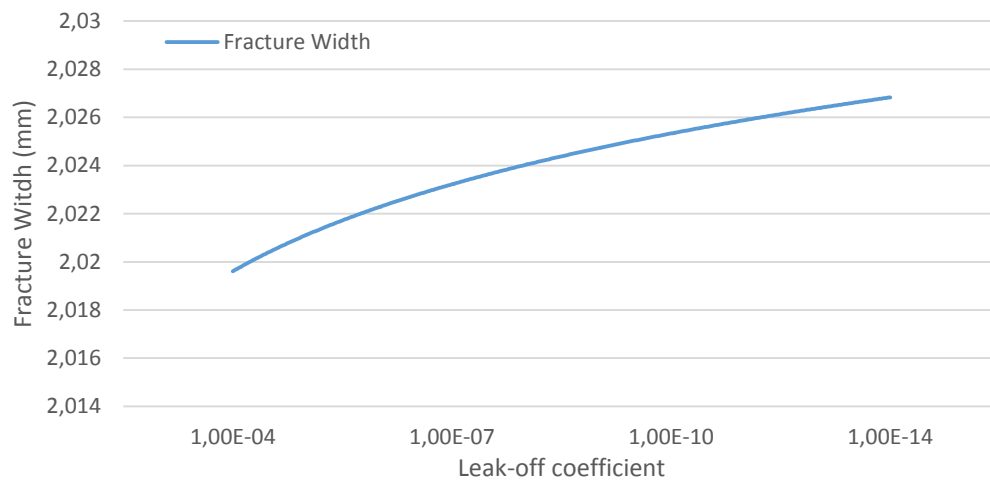


Figure 6.15 - Fracture width by leak-off coefficients to a direction 0° perforation for an injected flow = $2,5 \times 10^{-6} \text{ m}^3$

The fracture propagation direction is not affected by the fluid leak-off coefficient.

This parameter is only a numerical procedure/formalism to compute the normal fracturing velocity, and it was studied with the purpose of obtaining a better and complete understanding of the simulation procedure.

6.7 Flow rates

The flow rate is one of the main parameters controlling the initiation and propagation of the fracture. It may lead to different fracture reorientations patterns and breakdown pressures, as result of fluid pressure peaks (propagation controlled by the injection rate) or a viscosity-dominated regime (for low injection rates).

The flow rates studied are: $2,5 \times 10^{-7} \text{ m}^3/\text{s}$; $1 \times 10^{-7} \text{ m}^3/\text{s}$; $5 \times 10^{-8} \text{ m}^3/\text{s}$; $1 \times 10^{-8} \text{ m}^3/\text{s}$ and $1 \times 10^{-9} \text{ m}^3/\text{s}$. Lower injection rates are not realistic, as well as higher ones, once in the viscosity analysis, it was concluded that for the base flow rate, the viscosity effect was not measurable.

The breakdown pressure for different flow rates are presented at table 6.5.

Table 6.5 - Breakdown pressure by flow rate

Flow rate	Direction 0	Direction 45	Direction 90
$2,5 \times 10^{-7} \text{ m}^3/\text{s}$	12,36	24,77	32,18
$1 \times 10^{-7} \text{ m}^3/\text{s}$	11,29	23,416	32,18
$5 \times 10^{-8} \text{ m}^3/\text{s}$	10,86	21,76	32,18
$1 \times 10^{-8} \text{ m}^3/\text{s}$	9,623	20,09	32,18
$1 \times 10^{-9} \text{ m}^3/\text{s}$	8,428	19,07	-

For direction 0° and 45° a similar reduction in the breakdown pressure is found when the injection rate is reduced. As explained in figure 6.12, for lower flow rate, the viscosity effect is

higher, and the breakdown pressure is reduced due to the inertial effect of the high fluid viscosity.

For direction 90° , once the in-situ tangential stresses are higher than for the other direction, the fluid viscosity has a smaller influence in the near-tip stress state, and the breakdown pressure is not affected for intermediate injection rates. For very low injection rates, the fracture is not initiated, because the generated pressures are insufficient to exceed the tensile strength of the rock.

The total injected flow to cause the breakdown shows a clear reduction in all the directions (Figure 6.16) for lower flow rates. If the effect of viscosity was negligible, the injected volume necessary to cause the tensile failure would be independent of the injection rate; however that is not the case. As lower the injection rate, higher is the influence of viscosity in the initiation and propagation of the fracture.

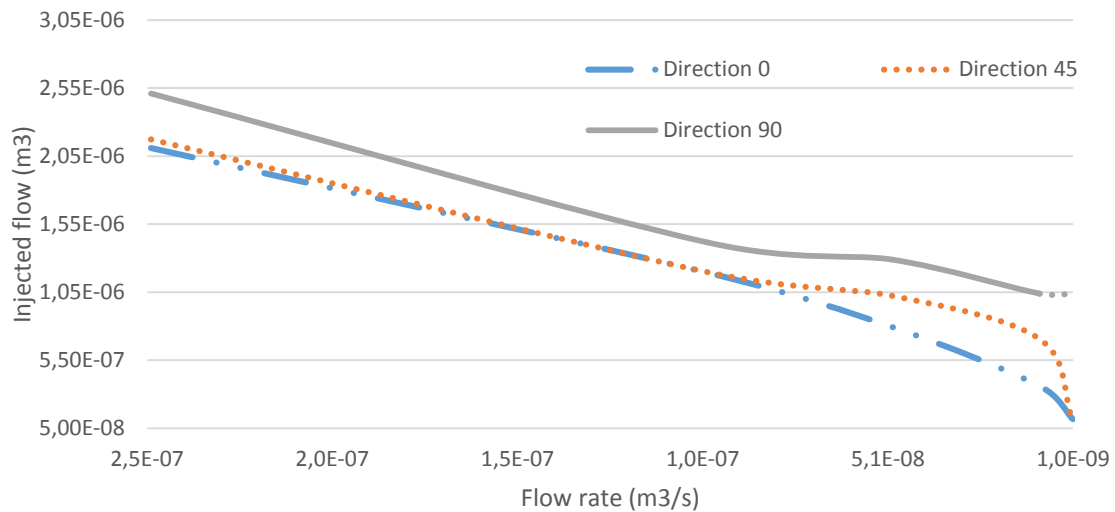


Figure 6.16 - Total injected fluid to cause for fracture initiation

The fracture reorientation is also dependent on the injection rate, as seen in figure 6.17.

For direction 90° the reorientation distance reduces with decreasing injection rate, the effect being noticeable even for very low injection rates (see fracture re-orientation for injection rate equal to $5 \times 10^{-8} \text{ m}^3/\text{s}$ and $1 \times 10^{-8} \text{ m}^3/\text{s}$). These results are in agreement with the work by Guo T. et al. (2014).

Direction 45° show a different fracture propagation behaviour. However, as the pinch effect is more pronounced in this direction, the fracture reorientation profile was expected to be affected. The results show an increase in the pinch effect for lower injection rates.

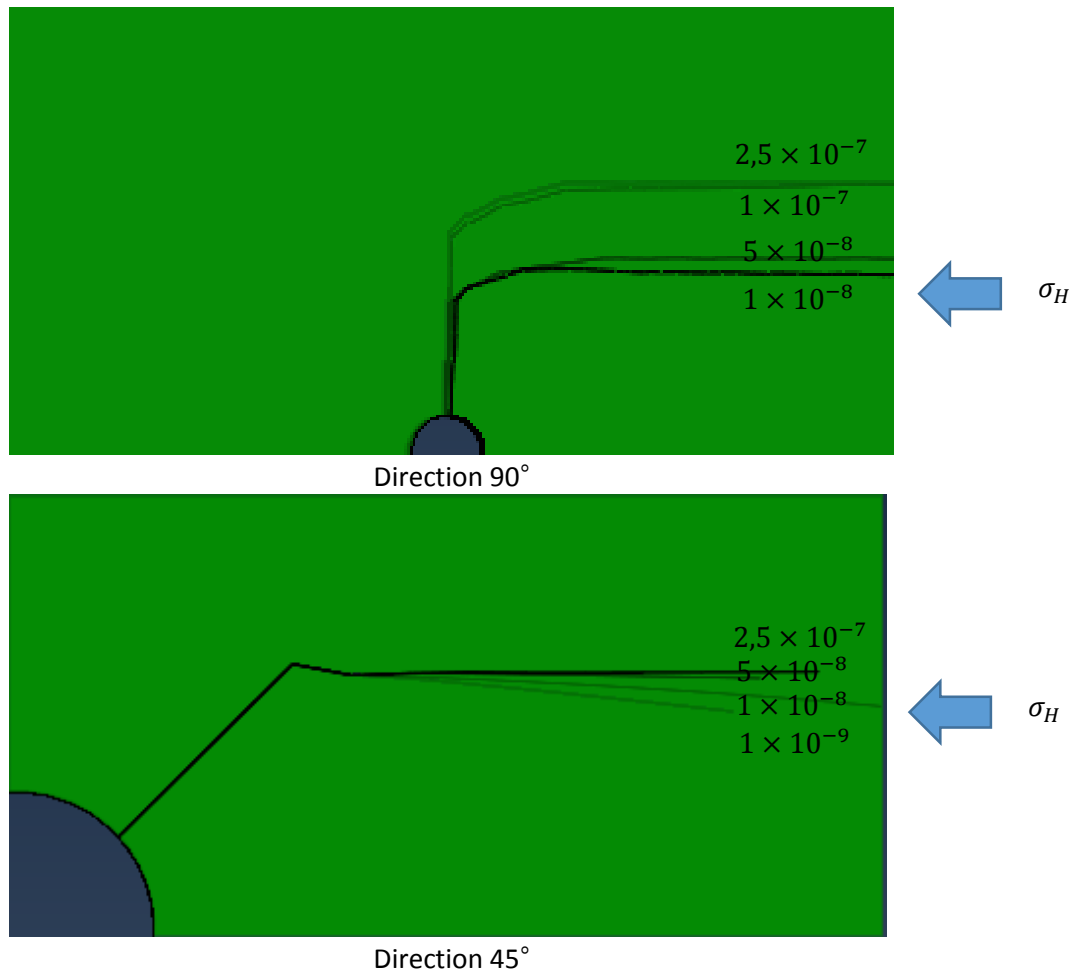


Figure 6.17 - Fracture reorientation for perforation direction 45 ° and 90 ° to different injection rates (m³/s)

As seen above, the injection rate influences the fracture initiation and reorientation. It also influences the fracture propagation velocity. As higher the injection rate, less time is required to propagate the fracture, as seen in figure 6.18.

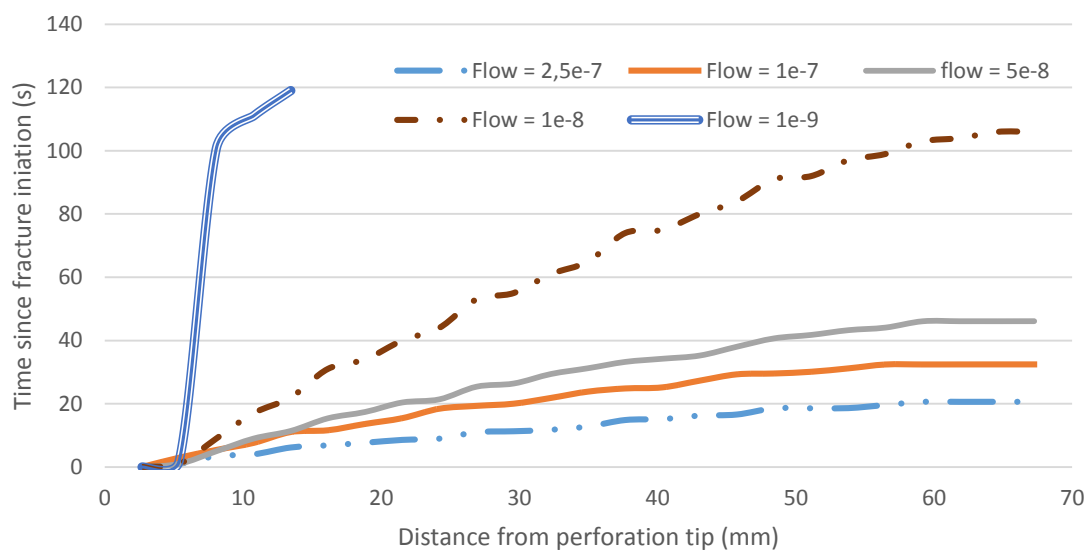


Figure 6.18 - Fracture propagation distance vs. time for a direction 0 ° perforation

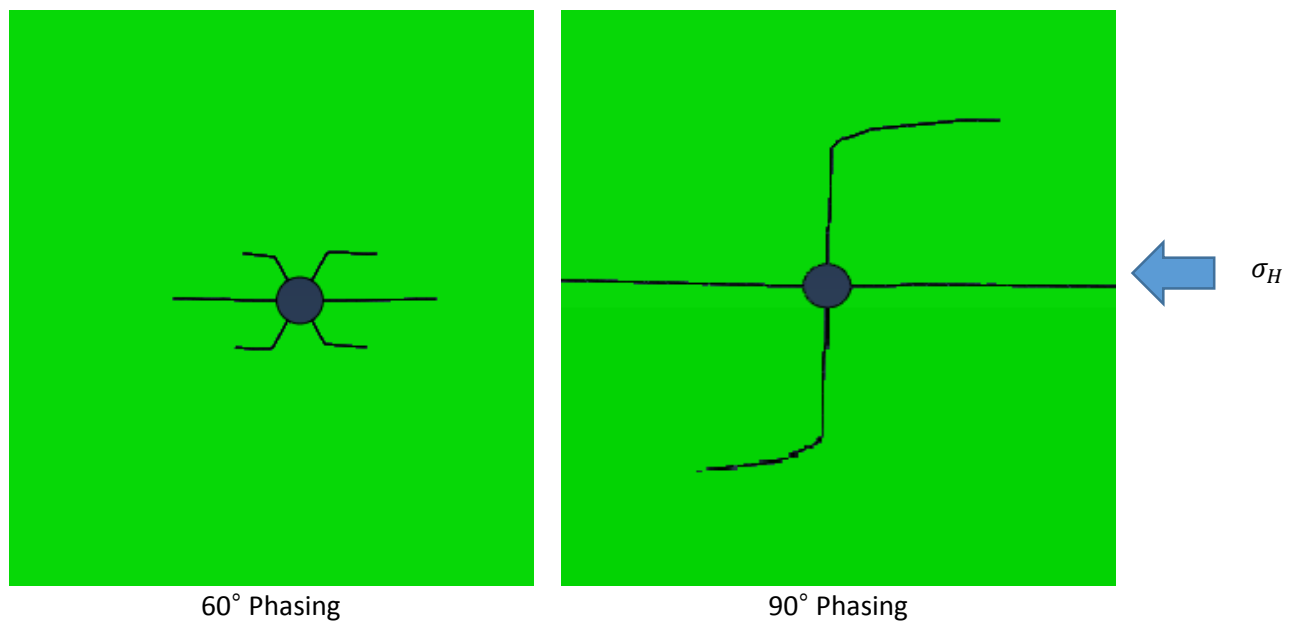
As seen in figure 6.18, even for direction 0° , for the lower studied flow rate, the fracture was able to initiate, however it was unable to propagate in distance. This happens because the generated pressures at the fracture tip are inferior to the tensile strength of the rock and the failure criteria is not satisfied.

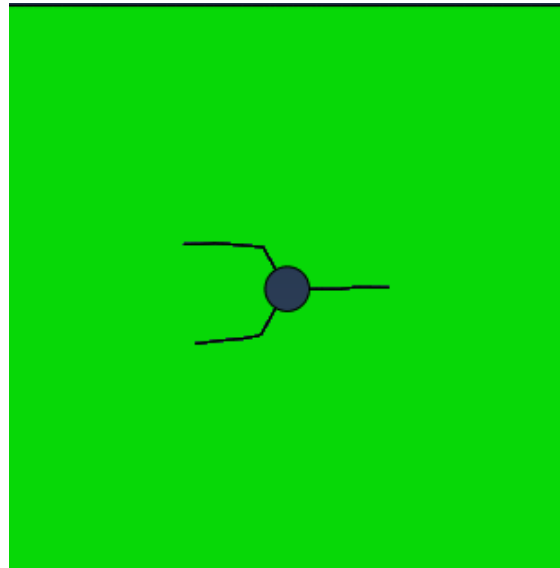
6.8 Different phasing

As already stated in this document, the perforation phasing is selected to ensure that with few degrees difference, there is a perforation in the direction of greatest principal stress σ_H , and usually the use of a higher perforation density means a higher uncertainty regarding the in-situ principal stress directions.

In this sense the following perforations phasing are analysed: 60° , 90° , 120° and 180° (i.e. the base case). According to equation 3.55, in the presence of various perforations, the flow is divided equally by the number of perforations. Therefore, the perforation phasing influences the results, once the flow rate was found to be one of the most important parameters for oriented perforation initiation and propagation.

Figure 6.19 shows the fracture reorientation for the different phasing.





120° Phasing

Figure 6.19 - Fracture propagation for different perforation phasing

The fracture reorientation is not affected by phasing. However, phasing influences the breakdown pressure of the first and second initiated fractures (table 6.6).

Table 6.6 - First and second breakdown pressure (MPa) for different phasing

Phasing	180°	120°	90°	60°
Breakdown Pressure of the first fracture	12,36	10,978	13,187	13,69
Breakdown Pressure of the second fracture	-	41,84	47,63	43,18
Direction of the second fracture		60°	90°	60°

The first breakdown pressure suffers a slight reduction for 120° phasing perforations, and an increase for the other phasing. The increase for the 90° and 60° phasing is a consequence of the perforation density. It seems that as higher the density, higher the breakdown pressure, due to the induced changes in the near-wellbore stresses.

The 120° phasing seems to provide a sufficient distance between perforations for the interaction between perforations to have a negligible effect on the first breakdown pressure. The reduction in the breakdown pressure from 180° to 120° phasing is explained by the reduction in flow rate per perforation, as examined in the previous section. This effect was also expected in the other phasing, but it seems this is masked by the effect of perforation density.

The second fracture to initiate shows a higher variation in the breakdown pressure when comparing with table 5.5, for instance. This is explained by the stress changes introduced by the fluid pressure exerted at the first propagating fracture.

Comparing the breakdown pressure of the second fracture for 120° and 60° phasing, the 60° phasing breakdown pressure is higher than the 120° phasing. Once again, this may be explained by the perforations density and its influence in the near-wellbore stress state.

Another possibility and closer to the reality is to use these same phasing slightly misaligned with the PFP.

6.9 Phasing misalignment

This study was executed with a misalignment of 15° to the preferred fracture plane, with the results presented in figure 6.20.

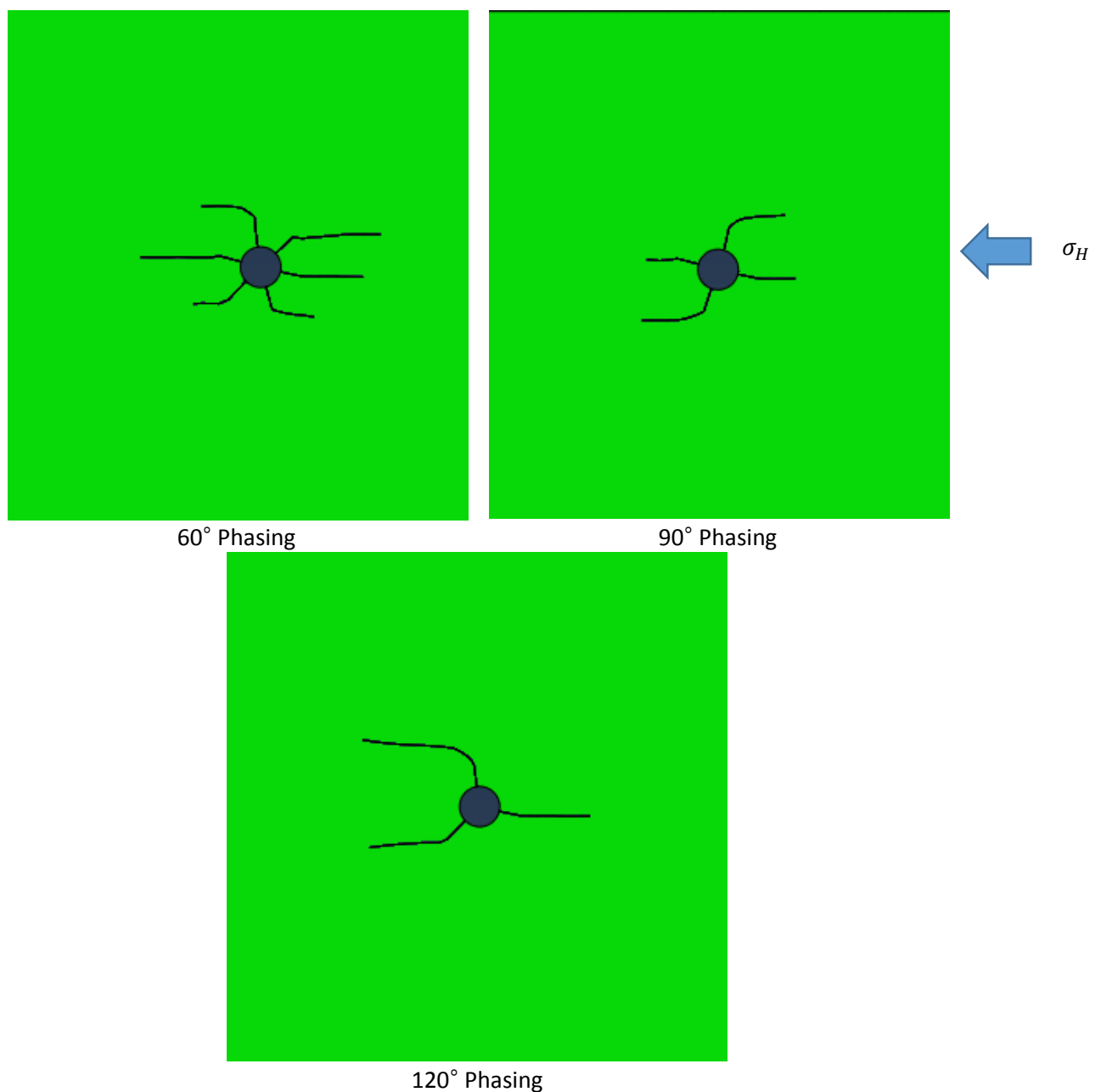


Figure 6.20 - Fracture propagation for different phasing with perforation miss alignment

The first and second fracture breakdown pressures are presented in table 6.7.

Table 6.7 - First and second breakdown pressure (MPa) for different phasing

Phasing	180°	120°	90°	60°
Breakdown Pressure of the first fracture	15,85	14,853	17,714	20,118
Breakdown Pressure of the second fracture	-	46,97	52,15	40,06
Direction of the second fracture		75°	75°	45°

The results are similar to those seen in section 6.8 for perforation phasing aligned with PFP phasing; The first breakdown pressure for 180° phasing is higher than that for 120° phasing, but lower than for 90° and 60° phasing.

This corroborates the results obtained in section 6.8. The influence area of the perforation is insufficient for the 120° phasing to affect significantly the stress state in the surrounding of the other perforation.

The second breakdown pressure also suffers a higher increase in comparison with the first. Comparing the second breakdown pressure for 90° and 120° phasing, the higher perforation density causes an increase in the pressure required to generate the rock tensile failure.

The increase in the second fracture breakdown pressure is coherent with the reality, once the oriented perforation objective is to create a single and wide fracture, to increase the productivity of the well. The second fractures at more deviated from the PFP and thus are expected to have smaller widths, at least for small injected volumes, which may reduce the cost-benefit of the operation.

7. Conclusions and future work

Due to the high cost of a Hydraulic fracturing operation, pre-design is essential to maximize the economic potential of the reservoirs. The determination of the in-situ principal stresses has a high degree of uncertainty, and the use of oriented perforations with different phasing ensures the initiation and propagation of single fractures in controlled conditions.

To understand the fracture initiation and propagation mechanism is essential to study the mechanics of the materials involved: rock, fluid and fracture. This information together with the understanding of the basis of the XFEM is essential to better understand and analyze the simulation results.

Initially a numerical study to validate and explore the capabilities of the software and the XFEM to model propagating fractures was performed. This consisted in the simulation of a fracture toughness determination test for an infinite plate with a central crack under tension, and a good match between the numerical results and the theoretical solution has been obtained. This exercise has allowed the integration of the fracture mechanics concepts and demonstrated that Abaqus software, and in particular the XFEM functionality, mimics well fracture initiation. It also allowed to understand some of the XFEM limitations, e.g. stress intensity factors can only be accessed in a stationary fracture model.

Subsequently, a series of coupled hydro-mechanical numerical analyses were carried out to simulate the laboratory tests described by Abass H. et al. (1994). Overall, the numerical analyses were found to reproduce well the observed experimental results. Based on this set of analyses some conclusion can be drawn:

- The wellbore excavation changes the near-wellbore stress state. This situation increases the stresses in the proximity of the wellbore, with the effect being negligible close to the boundaries, where stresses are similar to the confining applied pressure by the true-triaxial equipment.
- The breakdown pressure increases as the direction of the perforation deviates from the PFP. This is due to the increase of the local tangential stresses at the crack tip, with increasing the perforation angle.
- Perforations along non-preferred directions affect significantly the fracture breakdown pressure and the reorientation procedure. Perforations more deviated from the PFP perforations tend to have more reorientation difficulties and the breakdown pressure increase as a consequence of the higher tangential stresses at the crack tip. Fracture

initiation for the situation without initial perforation is well reproduced by the numerical simulations, which corroborates the model representativeness and the capabilities of the numerical tools.

- The fluid injection generates changes in the tangential stresses around the perforation tip region of 1-4 MPa, between the excavation of the wellbore and the fracture initiation.
- Fracture reorientation towards the PFP is dependent on the anisotropy ratio; it is found that the reorientation radius (i.e. reorientation distance) increases as the direction of the perforation deviates from the PFP.
- The fracture width reduces with the tortuosity. However, for the conditions examined in the simulations the tortuosity effect reduces when the injected flow increases, because the dimensions and properties of the model are not sufficient to accommodate the total injected volume.
- The equilibrium pressure (i.e. extension pressure) increases as the direction of the perforation deviates from the PFP. This is explained by the higher tangential stresses at directions more deviated from PFP. During the simulations some pressure peaks were felt, which is according with Abass H. et al. (1994) but may also be result of numerical convergence difficulties.

The results obtained from this set of analyses support the decision to use the same numerical tools to conduct a representative study on the effect of a set of parameters on hydraulic fractures initiation and propagation that make use of oriented perforations. From this study some conclusions can be drawn:

- The permeability affects fracture initiation and the breakdown pressure is found to increase with higher permeability, independently of the perforation direction. The increase in permeability facilitates the movement of the fluid within the rock mass, which increases the tangential stresses at the fracture tip.
- The fracture surface friction coefficient does not affect the breakdown pressure; however, it affects the fracture reorientation. For perforation directions higher than 45° with the PFP, the effect of the friction coefficient becomes residual when compared with the in-situ stress state. For perforation direction lower or equal to 45° with the PFP, increasing the friction coefficient reduces the pinch effect in the fracture path, which may cause an increase in the fracture productivity.

- Higher stress anisotropy ratios increase the capacity of fracture reorientation, with the reorientation to the PFP occurring over a smaller distance. The breakdown pressure is also affected by the stress anisotropy; an increase in stress anisotropy ratio causes a reduction of breakdown pressure for direction 0° and 45° , and a slight increase in direction 90° , which is coherent with the near-wellbore equilibrium conditions. The injected flow that causes fracture initiation is found to decrease for higher values of stress anisotropy ratios, following a similar trend for directions 0° , 45° and 90° .
- The fracture propagation regime presented herein was a viscosity and storage dominated propagation regime, and it was expected that the fluid viscosity would affect significantly the fracture initiation and propagation. However, the injection flow rates considered in Abass H. et al. (1994) are very high, and thus this becomes the controlling parameter of the fracture behavior. To verify the viscosity effect an infinite viscosity fluid was considered and the results showed a great increment in the fracture width, and small differences in the fracture initiation and propagation.
- For the conditions considered, a reduction in the flow rate leads to a reduction in the breakdown pressure for direction 0° and 45° , while for direction 90° , as the in-situ tangential stresses at the perforation tip are higher than for the other directions, the fluid viscosity has a smaller influence in the near-tip stress state, and the breakdown pressure is not affected for intermediate injection rates.
- As lower the injection rate, higher is the influence of viscosity in the initiation and propagation of the fracture. If the effect of viscosity was negligible, the volume of injected fluid necessary to cause the tensile failure of the rock mass would be independent of the injection rate; however, it is found that the total injected volume is reduced when reducing the flow rate, increasing the influence of viscosity in the initiation and propagation of the fracture. On the other hand, the velocity of fracture propagation reduces when considering lower injection rates, and for very low injection rates the fracture is unable to propagate for a significant distance.
- Using a perforation higher density increases the probability of a perforation being oriented close to the PFP, optimizing the success of a HF operation. The breakdown pressure of the first fracture (subjected to lower tangential stresses) is not significantly affected by phasing misalignment; however, the initiation of a second fracture tends to suffer an increase in the breakdown pressure, which is explained by the stress changes introduced by the fluid pressure exerted at the first propagating fracture.

As explained in the document, the analyses assume that the formation of the wellbore and perforations was instantaneous and during these operations the rock responded in undrained conditions, and the injection associated with hydraulic fracturing operations occurred immediately after, i.e. the excess pore pressures generated during the formation of the wellbore were not allowed to dissipate. It is acknowledged that this may have influenced the obtained results. For this reason, the other limit condition, i.e. drained conditions at the start of the injection operations, should be tested through the consideration of a consolidation period of say, at least one day, to understand and quantify the effect of the pore pressure regime on the obtained results.

Due to the sample dimension and the Abass H. et al. (1994) test conditions (i.e. high flow rates), the analysis of some of the parameters is not representative (e.g. viscosity). To overcome this, a large-scale numerical model should be considered. This model would also be important to have the real magnitude of the fracture widths during fracture propagation, once this model is only able to provide trends.

The use of three-dimensional simulations may provide information on the vertical propagation of the fracture, and would enable to quantify the consequences of the plain strain assumption made through this study.

The data retrieved from the parametric study can be used together with other operation optimization software (as input file), to study the production phase of a hydraulically fractured well.

This study has improved the understanding of the factors controlling the behavior of oriented perforation as well as the XFEM numerical tool ability to model hydraulically induced fractures.

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Annex - A.1 Execution of an Abaqus© XFEM analysis

The XFEM functionality introduces a new and easier way to simulate crack initiation and propagation. Several different simplifications are essential for fracture implementation. For this propose, a multi-physics software distributed by Dassault Systèmes, Abaqus is selected for the study carried once it provides the XFEM functionality and all the tools required for an accurate fracture modelling.

In the following pages are presented the fundamentals for the execution of an analysis of a boundary value problem with Abaqus XFEM, with higher focus to specific subjects related to the crack analysis.

A 1.1 Abaqus Software structure

The processing of each analysis performed with the aforementioned software is divided into three distinct phases: pre-processing, simulation, and post-processing. These three processing stages are inter-connected, wherein in each stage, are produced data files which serves as the basis for the sequential stage in the processing chain , as seen in Figure A.1. Then are presented each of these phases:

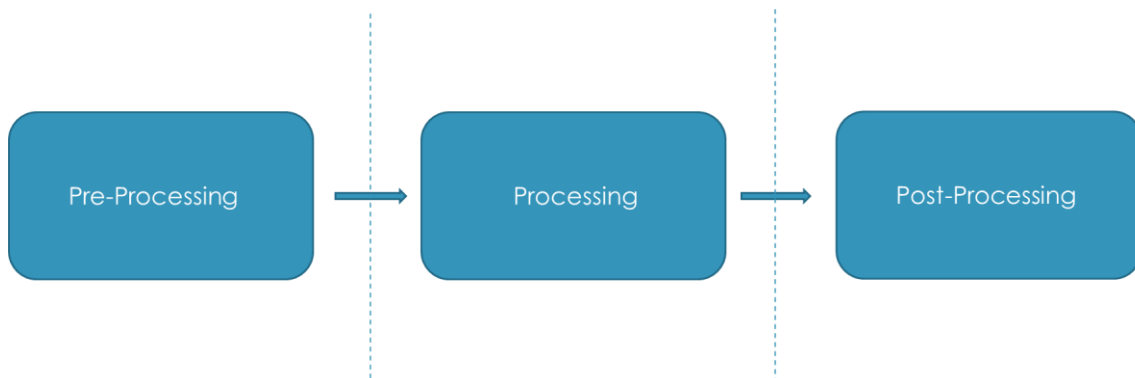


Figure A.0.1 - Scheme of the interactive processing stages (Author)

Pre-processing (ABAQUS/CAE): At this stage the numerical model is defined conceptually and formally, as an ABAQUS input file. The model is usually created graphically using the ABAQUS / CAE (Computer-Aided Engineering / Complete Abaqus Environment), however, depending on the complexity of the problem/model this may be created using the Keyword edition. Once the study takes advantage of new features introduced recently, the graphical interface does not provide all functionalities, so the Keyword edition has used at several stages of the analysis.

Simulation/processing (ABAQUS/Standard): The simulation is the stage where ABAQUS/Standard solves the numerical model. For instance, the output files of a stress analysis includes the displacements and stresses that are stored in binary files ready for post-processing.

Depending on the complexity of the problem, the input parameters and computer capability, a simulation can have an ease numerical convergence, produce more or less accurate results and a diverse computational data consumption. As example to the aforementioned, for the production of this study, more than 400 GB of CPU memory were used.

Post-processing (ABAQUS/CAE): After the end of the simulation, it is possible to evaluate the results of displacements, stresses, or other key variables whose calculation has been requested. The evaluation is usually performed interactively using the visualization module of ABAQUS / CAE. The visualization module, which reads the binary output database has a variety of options to represent the results. It is also possible to extract the produced files to Excel, enabling a broader spectrum of processing and data analysis.

A1.2 Components of the Abaqus Pre-processing phase

An ABAQUS model is composed by different components that together/assembled describe the physical problem. Depending on the complexity, the number of inputs should vary, however, it is essential that the model has the following information: geometry, properties of element section, material data, loads, boundary conditions, analysis type/procedure, and data to be requested as output.

Geometry and Material properties

The finite elements and nodes define the basic geometry of the physical structure to be modeled in ABAQUS. Each element in the model represents a discrete portion of the physical structure, which by its turn, is represented by many interconnected elements. The elements are connected to each other by shared nodes. The coordinates of the nodes and the connection elements comprise the model geometry, and constitute a mesh.

Generally, the mesh is only an approximation of the real geometry of the structure. The type, shape, and position of the element, as well as the total number of elements used in the mesh, affect the simulation results. The higher the mesh density, i.e., higher number of elements in the mesh, the more accurate the results. As the mesh density increases, the analysis results converge into a single solution, increasing the time used by the computer for analysis. However, the aim is to ensure an approximation to the analytical solution with the lowest possible space consumption and analysis time. Note that depending on the simplifications made, in the geometry, material behavior, boundary conditions and loads, is determined how the simulation approaches the physical problem.

Material properties should be defined for all elements. The ABAQUS has a wide range of element types. Some limitations are introduced by using XFEM, as explained in the previous chapter.

Loads and Boundary Conditions

The loads give rise to distortion and displacement in the physical structure, subsequently computed as stresses in the body. The Load types used throughout this document include:

- Concentrated forces;
- Surface pressures;
- Fluid concentrated pressure

The boundary conditions are used to constrain the moving parts of the model to remain stationary or to have a pre-defined displacement. From the analysis point of view, if possible, to apply a controlled displacement rather than a load/pressure allow to retain better the material behavior (Hudson J. A. et Harrison J. P., 1997) , as can be seen in figure A.2.

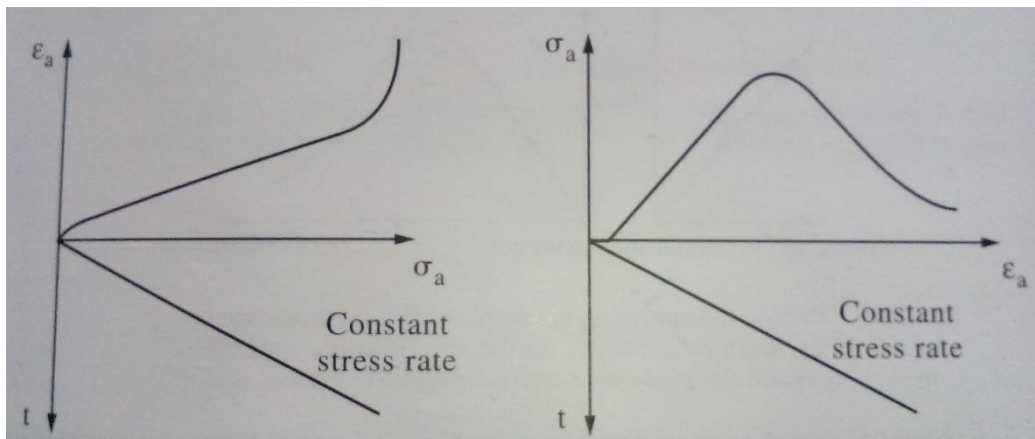


Figure A.0.2 – Stress (left) and strain (right) controlled stress-strain curves (Hudson J. A. et Harrison J. P., 1997)

In a static analysis the boundary conditions should be defined in order to constrain the rigid body displacement of the model, reducing the convergence difficulties.

Output Data

An ABAQUS simulation generates a large amount of output files. To avoid using excessive disk space, only output data essential to achieve the desired results should be required, reducing the space consumption. The XFEM is accompanied by a specific set of outputs that should be requested to make possible to analyze the fracture accurately (Dassault Systèmes, 2015).

A1.3 Abaqus/CAE modules

The ABAQUS / CAE is divided into modules, where each module defines a logical point of the modeling process. As it progresses from module to module, the model is being created. When the model is complete, the ABAQUS / CAE creates an input file that will be reviewed. After the simulation and numerical analysis, the information is returned to ABAQUS/CAE to allow monitoring of work in progress, generating a file with the output data, where it is possible to

verify the convergence process at the elements. Finally, using the visualization module it is possible to read the output files and graphically analyze the simulation results.

The ABAQUS / CAE is divided into functional units known as modules, as seen in figure A.3. Each module has only the relevant tools to a specific portion of the task, such as mesh module only has the necessary tools to create finite elements meshes.

The order of the modules in the menu corresponds to a logical sequence to establish the model. However, it is possible the selection of any module at any development time.

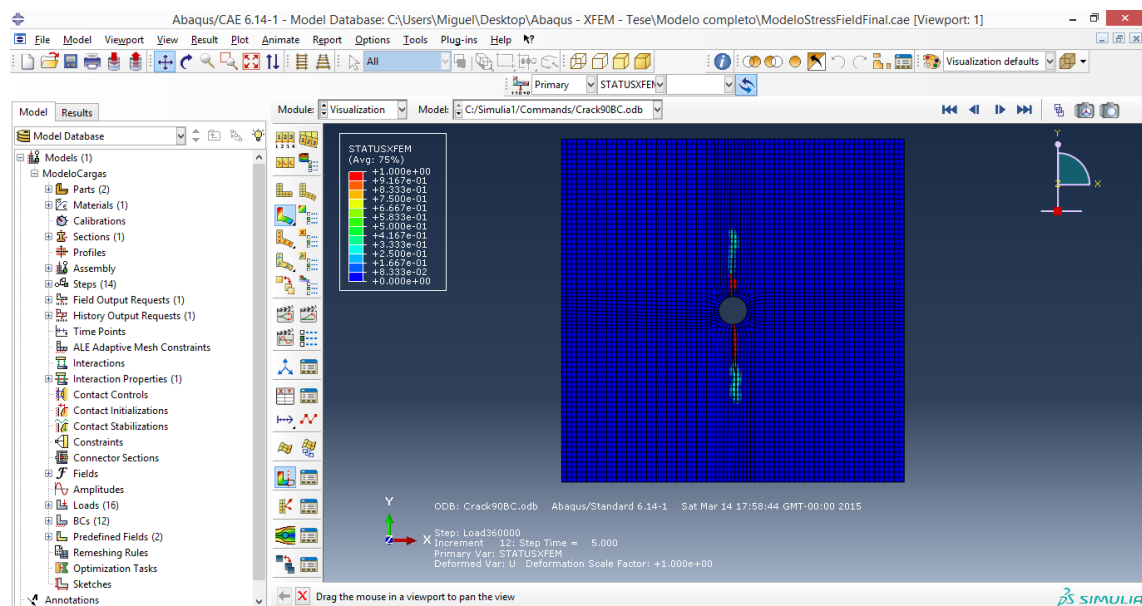


Figure A.0.3 - Abaqus/CAE interface

Following are described summarily the modeling tasks performed on each module Abaqus / CAE:

Part - The module part allows creating individual parts directly drawing the geometry. Is essential to use the correct geometry set to be reproduced, i.e. Modelling space, shape, solid type or approximate size of the introduced geometry, since ABAQUS is a non-defined-units software, which only requires unit consistency throughout the analysis (Dassault Systèmes, 2013).

Property - The section contains information about the properties of a part or a part region, as material and the cross section geometry. In the Property module sections are created, and the material behavior is assigned to the correspondent regions. The information about the used materials shall be specified in this module. Cohesive fracture behavior is assumed in the analyses presented in this thesis, and thus a damage law for traction separation needs to be defined, which describes the fracture strength loss during the loading procedure (Zielonka M. G. et al, 2014).

Abaqus XFEM software assumes the end of cohesive behavior when $G > G_f$, where G_f is the critical fracture energy. The value of G_f corresponds to the integral of the stress-displacement diagram, as shown in Figure A.4.

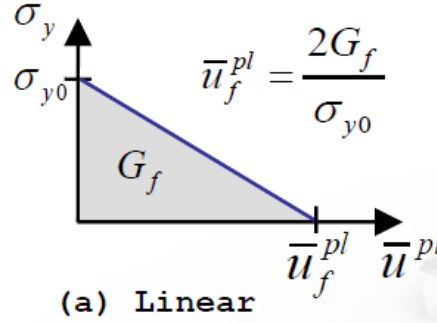


Figure A.0.4 - Energy-based damage evolution for linear softening (Dassault Systèmes, 2015)

Where u_f^{pl} is the displacement to reach the end of cohesive behavior and σ_{y0} is the tensile strength of the rock.

Given the loading conditions applied in the analysis presented herein, the *MAXPS* (Maximum principal stress) failure criterion is adopted. This is mathematically described as:

$$MAXPS = \text{Maximum principal stress} = f = \left\{ \frac{\langle \sigma_{max} \rangle}{\sigma_{max}^0} \right\} \quad (A.1)$$

Where σ_{max} is the stress at a specific point, σ_{max}^0 is the tensile strength of the rock and the symbol $\langle \rangle$ is the Macaulay bracket that is used to ensure that a compressive stress state does not initiate damage, and a fracture is initiated or the length of an existing fracture is extended by tensile stresses, when the fracture criterion, f , reaches the value 1,0 within a specified tolerance:

$$1,0 < f < 1,0 + f_{tol} \quad (A.2)$$

Where f_{tol} is the tolerance for the initiation criterion. As the simulation of fracture propagation can be relatively instable, most authors (Dassault Systèmes, 2015) propose a value of 0,2 for the tolerance, and that is the value adopted in the analysis shown here.

The rock strength degradation/damage evolution (i.e. softening behavior) begins when the failure criterion is verified and progresses linearly until total damage and loss of the section cohesive behavior is reached.

Assembly - When creating a part, there is a different coordinate system independent of the other parts of the model. The assembly module is used to assemble the parts and position them in relation to a global coordinate system, thus creating a set. If the fracture is defined as an

individual entity, it should be assembled. If it is modeled as a seam, may be modeled as a portion of the model geometry. The seam representation is more suitable to use when modeling stationary fractures.

Step - The step module is used to create and configure the analysis stages/steps, as well as the requested output data. It provides a convenient way of monitoring changes made to a model (such as the loading condition and boundary conditions through different steps). To ensure the results analysis by the user in a more systematically way, various steps should be defined in stress or deformation increments and shorter periodizations. This facilitates the numerical calculation process and the results convergence.

Interaction - In this module are specified the mechanical interactions between different regions of a model or between a region of a model and its neighbors, i.e. between the fracture and the surrounding region. ABAQUS/CAE does not recognize the mechanical contact between parts or Assembly regions, unless it is specified in the contact interaction module, since the physical proximity of two surfaces in the assembly is not sufficient to indicate some kind of interaction between the surfaces. Interactions are step-dependent objects, which means that should be specified the analysis steps in which they are active. This module allows the introduction of cohesive fracture behavior, damage laws, fracture initiation criterion through the technique of VCCT or surface friction. Note however that when considering a cohesive behavior and traction separation law for all geometry in the study, the behavior is closer to reality then considering only a fracture cohesive behavior.

Loads - Load module allows to specify the load, the boundary conditions, and the predefined fields. The directions of the loads should be checked throughout the process and based on the program's specifications; it takes tractions as positives and compressions as negatives, by following the solid mechanics and not the conventions of soil and rocks mechanics. When using fluids loads on surfaces, this feature is not yet available for XFEM analysis, so its edition must be made using Keyword edition, taking advantage of the concept of phantom nodes.

In this implementation, each enriched pressure diffusion/stress element (CPE4P) is internally duplicated with the addition of corner phantom nodes, as seen in figure A.5. Prior to damage initiation only one copy of the element is active. Upon damage initiation the displacement and pore pressure degrees of freedom associated with the corner phantom nodes are activated and both copies of the element are allowed to deform independently, pore pressures are allowed to diffuse independently, and the created interface behavior is enforced with a traction separation cohesive law.

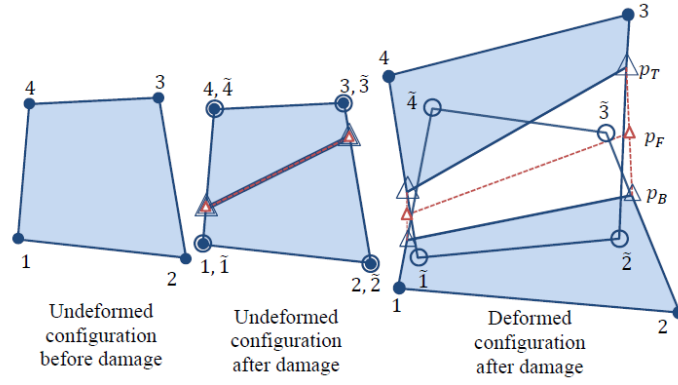


Figure A.0.5 - Phantom nodes due to pore pressure extra degrees of freedom (original nodes are represented with full circles and corner phantom nodes with hollow circles) (Zielonka M. G. et al, 2014)

For the XFEM model, the corner phantom nodes on the symmetry surfaces and boundary surfaces are constrained to move within these surfaces, as seen in figure A.6.

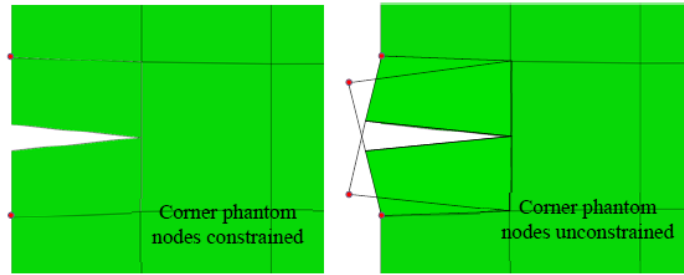


Figure A.0.6- Displacement boundary conditions for the XFEM modelling (Zielonka M. G. et al, 2014)

The fracturing fluid pressure degrees of freedom are associated with the mid-edge nodes of the cohesive elements and the edge phantom nodes of the enriched elements. Therefore, concentrated fracturing fluid flow must be applied directly to these mid-edge and phantom edge nodes, as seen in Figure A.7.

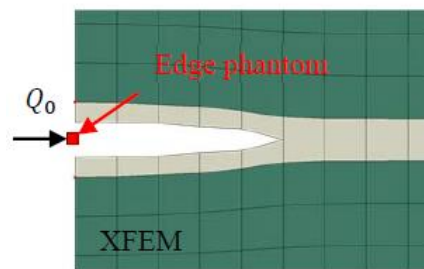


Figure A.0.7 - Concentrated flow injection in the phantom nodes/edge (Zielonka M. G. et al, 2014)

Comparing the analytical models presented previously (see chapter fracture mechanics), the assumption of a flow injected at fracture level is valid since the leak-off into the surrounding fluid is null or negligible. Since it is only taken into account a numerical fluid leak-off fluid at the fracture contact surface, this simplification is consistent.

Mesh - The mesh is an important process in creating a finite elements model, because depending on the type of mesh and the density of the elements used, the results may vary as well as the analysis time. As the XFEM is a method where the mesh dependency is high, any study should be followed with a study of the mesh influence.

The mesh module contains the tools for the generation of finite element meshes of independent parts in the assembly section. Various levels of automation and control are available in order to create a mesh that meets the analysis requirements.

As succeeded with the creation of Parts and Assembly, the process of meshing (mesh definition) of the model, the meshing techniques, and the element type are dependent on the geometry. Consequently, is possible to modify the parameters that define a part or an assembly, and the specified mesh attributes within the mesh module are automatically regenerated. The mesh module has the following characteristics: 1) Tools to set the mesh density locally and globally (Seed Part and Seed Mesh); 2) The coloring of assembled model that shows visually the used meshing technique for the model; 3) Different mesh controls, as: i) Element shape; ii) Meshing technique; iii) Mesh algorithm.

The ABAQUS / CAE can use a variety of mesh techniques with different topologies and it is possible to use one technique to work a template region/partition part and another mesh technique for a more complex region. The use of structured mesh ensures a uniform mesh, which in most cases facilitates the convergence process. However, this is only likely to be used for relatively simple geometries; i.e. uniform quadrilateral parts, so it is not applicable in more complex geometries, such as the near wellbore region, where free meshes are required.

Another limitation of XFEM is the required linear quadrilateral elements analysis, namely, four nodes, as shown in Figure A.8.

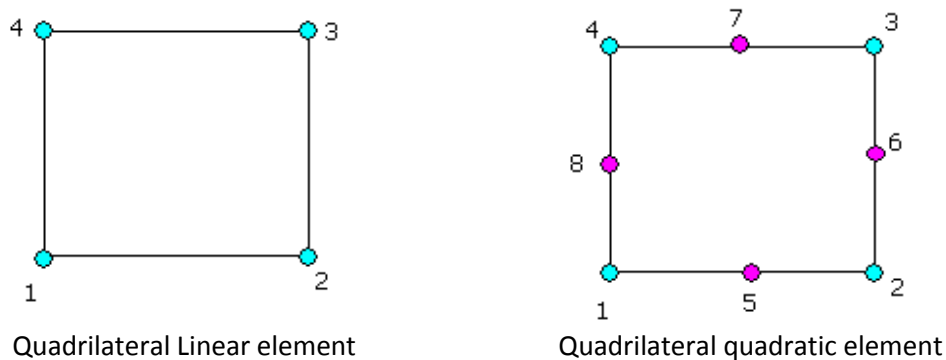


Figure A.0.8 - Quadrilateral element types (Forum 8, 2015)

Obviously, this limitation has an impact on the results, the calculation process becomes less accurate, since the interpolation function must necessarily be simpler for linear elements than quadratic elements, where polynomial interpolation functions are used.

By his turn, the use of the reduced integration process has also consequences from the results accuracy point of view. This procedure consists of using only four gauss points instead of the traditional nine, which introduces greater instability in the process, although in specific cases can be beneficial, as it allows the reduction of the element overstiffness (Varma A. H., 2013).

It is further noted that the use of certain procedure types can introduce limitations in other analysis stages. For instance, the geostatic step (essential in some of the models further analyzed to ensure the equilibrium state), does not allow the procedure of fracture propagation to be calculated at crack tip, being made instead at the centroid level.

These limitations are overcome by the use of sufficiently fine mesh to guarantee the proximity of the centroid and the crack tip, as seen in Figure A.9.

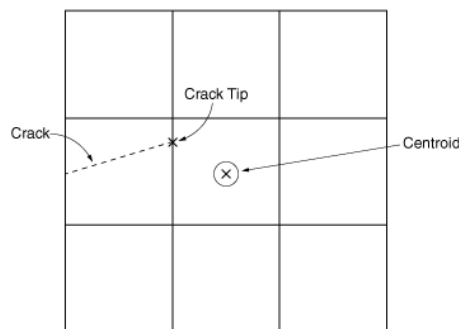


Figure A.0.9- Centroid vs Crack tip fracture propagation determination (Dassault Systèmes, 2013)

Job and visualization – When the model is completely created, the job module is used to analyze the model. The job module allows to submit interactively a job and monitoring the progress. The visualization module provides graphical presentation of models and results of the numerical finite element analysis, obtained from the output files.